Full Data - Pinellas County Model Capstone

Housing Prices, Property Characteristics, and Spatial Dependence

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Table of contents

1 Project overview **3**

2 0. Packages and data import **3**

3 1. Clean data and construct predictors **8**

4 2. Prepare Bayesian/INLA data **16**

5 3. Observation-count summary **17**

6 4. Exploratory data analysis **18**
6.1 4.1 Distribution of log adjusted price . . . 18
6.2 4.2 Distribution of adjusted price . . . 19
6.3 4.3 Sales volume by year . . . 20
6.4 4.4 Log price versus sale year . . . 21
6.5 4.5 Price by frontage . . . 22
6.6 4.6 Price by quality . . . 23
6.7 4.7 Price by seawall . . . 24
6.8 4.8 Price by stories . . . 25
6.9 4.9 Price by foundation . . . 26
6.10 4.10 Price by barrier municipality . . . 27
6.11 4.11 Categorical predictor frequencies . . . 28
6.12 4.12 Continuous predictor distributions . . . 29
6.13 4.13 Heated area versus price . . . 31
6.14 4.14 Mean price by municipality . . . 31
6.15 4.15 Geographic distribution . . . 33

7 5. Model 1 — OLS hedonic baseline **34**

8 6. Model 2 — INLA hierarchical municipality model **36**
8.1 6.1 Municipality fixed effects . . . 38
8.2 6.2 Municipality random effects . . . 39
8.3 6.3 Municipality model fit statistics . . . 40

9 7. Prepare projected spatial coordinates **40**

10 8. Construct computationally efficient INLA SPDE mesh **45**
10.1 8.1 Plot mesh 47

11 9. Define Matern SPDE model **47**

12 10. Create SPDE projection matrix **48**

13 11. Create INLA spatial index **48**

14 12. Create INLA model stack **49**

15 13. Primary INLA spatial model **50**
15.1 Mathematical Specification of the Bayesian Spatial Hedonic Model 52

16 14. Primary model fixed effects **424**

17 15. Municipality random effects **426**

18 16. Spatial random effects **427**

19 17. Spatial hyperparameters **428**

20 18. Model fit statistics **428**

21 19. Compare OLS, municipality, and spatial models **429**

22 20. Extract fitted values **429**

23 21. INLA prediction accuracy **430**

24 22. Residual distribution **432**

25 23. Residuals versus fitted values **433**

26 24. Spatial residual map **434**

27 25. Spatial field map **436**

28 26. Spatial uncertainty map **438**

29 27. Observed versus fitted values **439**

30 28. Municipality effects plot **441**

31 29. Save fitted models and objects **444**

32 30. Final model summary table **444**

1 Project overview

This project models Pinellas County housing prices using a full complete-case analytic dataset.

Three modeling tiers are estimated:

1. OLS Hedonic Baseline
2. Bayesian Hierarchical Municipality Model using INLA
3. Bayesian Spatial SPDE Model using INLA

The primary spatial model uses an INLA Matern SPDE spatial field rather than constructing a CAR adjacency matrix across every individual property coordinate. This preserves the full property-level dataset while keeping the latent spatial model sparse and computationally tractable.

2 0. Packages and data import

```
# SECTION 0: PACKAGES AND DATA IMPORT

required_packages <- c(
  "tidyverse",
  "janitor",
  "lubridate",
  "sf",
  "sp",
  "INLA",
  "ggplot2",
  "scales",
  "patchwork",
  "broom",
  "knitr",
  "viridis",
  "viridisLite"
)

# Install packages from CRAN
cran_packages <- setdiff(
  required_packages,
  rownames(installed.packages())
)

cran_packages <- setdiff(cran_packages, "INLA")

if (length(cran_packages) > 0) {
  install.packages(
    cran_packages,
    dependencies = TRUE
```

```
  )
}

# ---- 0.2 Install INLA if missing ----
# This uses the Windows binary repository for R 4.5.

if (!requireNamespace("INLA", quietly = TRUE)) {

  install.packages(
    "INLA",
    repos = c(
      getOption("repos"),
      INLA = "https://inla.r-inla-download.org/R/stable"
    ),
    dependencies = TRUE
  )
}

library(tidyverse)
library(janitor)
library(lubridate)
library(sf)
library(sp)
library(INLA)
library(ggplot2)
library(scales)
library(patchwork)
library(broom)
library(knitr)
library(viridis)
library(viridisLite)


analysis_version <- "pinellas_full_data_capstone_inla_spde_v1"


data_path <- paste0(
  "G:/",
  ".shortcut-targets-by-id/1IwJJwifiMkxN1ZtZ-lx4TooG6WGbO18w/",
  "Alderfer and Evans/1-data/pinellasdata.csv"
)

# Confirm that the file exists before importing
if (!file.exists(data_path)) {
  stop(
    paste0(
      "DATA FILE NOT FOUND.\n",
```

```
      "Check that the Google Drive location is available.\n",
      "Current path: ",
      data_path
    )
  )
}

required_17_columns <- c(
  "uid",
  "parcelid",
  "sale_year",
  "adj_price",
  "heated_area",
  "acreage",
  "stories",
  "quality",
  "foundation",
  "seawall",
  "cccl_distance",
  "tb_distance",
  "icw_distance",
  "frontage",
  "x_coord",
  "y_coord",
  "site_city"
)

# ---- 0.7 Import the full dataset ----
# Import as character first to avoid accidental type conversion.
# Numeric conversion will happen during the cleaning section.

raw_data <- readr::read_csv(
  file = data_path,
  col_select = tidyselect::all_of(required_17_columns),
  col_types = readr::cols(
    .default = readr::col_character()
  ),
  na = c(
    "",
    "NA",
    "N/A",
    ".",
    "null",
    "NULL",
    "*"
  ),
  progress = TRUE,
```

```
  show_col_types = FALSE
)
```

```
indexing pinellasdata.csv [---------------------------------] 2.15GB/s, eta:  0s
indexing pinellasdata.csv [=================================] 234.42MB/s, eta:  0s
indexing pinellasdata.csv [================---------------] 320.02MB/s, eta:  0s
indexing pinellasdata.csv [================---------------] 232.66MB/s, eta:  0s
indexing pinellasdata.csv [====================-----------] 285.91MB/s, eta:  0s
indexing pinellasdata.csv [=====================----------] 222.67MB/s, eta:  0s
indexing pinellasdata.csv [========================-------] 258.50MB/s, eta:  0s
indexing pinellasdata.csv [=========================------] 211.30MB/s, eta:  0s
indexing pinellasdata.csv [============================---] 228.46MB/s, eta:  0s
indexing pinellasdata.csv [=============================--] 238.24MB/s, eta:  0s
indexing pinellasdata.csv [==============================] 241.49MB/s, eta:  0s
indexing pinellasdata.csv [==============================] 243.55MB/s, eta:  0s
```

```
# ---- 0.8 Verify imported dataset ----

# Import verification

import_summary <- tibble(
  Check = c(
    "File exists",
    "Data object class",
    "Observations imported",
    "Variables imported"
  ),
  Result = c(
    file.exists(data_path),
    paste(class(raw_data), collapse = ", "),
    format(nrow(raw_data), big.mark = ","),
    ncol(raw_data)
  )
)

knitr::kable(import_summary, caption = "Data Import Verification")
```

Table 1: Data Import Verification

Check	Result
File exists	TRUE
Data object class	spec_tbl_df, tbl_df, tbl, data.frame
Observations imported	267,074

Check	Result
Variables imported	17

```
# Missing values by column

missing_summary <- tibble(Variable = names(raw_data), Missing = colSums(is.na(raw_data))) %>%
  arrange(desc(Missing))

knitr::kable(missing_summary, caption = "Missing Values by Variable")
```

Table 2: Missing Values by Variable

Variable	Missing
site_city	3461
uid	0
parcelid	0
sale_year	0
adj_price	0
heated_area	0
acreage	0
stories	0
quality	0
foundation	0
seawall	0
cccl_distance	0
tb_distance	0
icw_distance	0
frontage	0
x_coord	0
y_coord	0

```
# ---- 0.9 Stop if the import does not match expectations ----

if (!is.data.frame(raw_data)) {
  stop("Import failed: raw_data is not a data frame.")
}

if (ncol(raw_data) != 17) {
  stop(
    paste(
      "Expected 17 columns but imported",
      ncol(raw_data)
    )
  )
}
```

```r
if (!all(required_17_columns %in% names(raw_data))) {
  stop("One or more required columns are missing.")
}

if (nrow(raw_data) == 0) {
  stop("The imported dataset contains zero rows.")
}
```

3 1. Clean data and construct predictors

```r
# 1. CLEANING AND VARIABLE CONSTRUCTION


df_clean <- raw_data %>%
  mutate(
    sale_year =
      parse_number(sale_year),

    sale_year_c =
      sale_year - 2000,

    adj_price =
      parse_number(adj_price),

    heated_area =
      parse_number(heated_area),

    acreage =
      parse_number(acreage),

    cccl_distance =
      parse_number(cccl_distance),

    tb_distance =
      parse_number(tb_distance),

    icw_distance =
      parse_number(icw_distance),

    x_coord =
      parse_number(x_coord),

    y_coord =
      parse_number(y_coord),
```

```
    frontage_raw =
      str_squish(
        str_to_upper(frontage)
      ),

    seawall_raw =
      str_squish(
        str_to_upper(seawall)
      ),

    quality_raw =
      str_squish(
        str_to_upper(quality)
      ),

    foundation_raw =
      str_squish(
        str_to_upper(foundation)
      ),

    site_city =
      str_squish(site_city),
    frontage_group = case_when(

      is.na(frontage_raw) ~
        NA_character_,

      frontage_raw %in% c(
        "NONE",
        "NO",
        "NO FRONTAGE",
        "NO FRONT"
      ) ~
        "None",

      str_detect(
        frontage_raw,
        "BAY|TAMPA BAY"
      ) ~
        "Bay",

      str_detect(
        frontage_raw,
        "LAKE|POND"
      ) ~
        "Lake",
```

```
    TRUE ~
      "Other"
  ),
  seawall_group = case_when(

    is.na(seawall_raw) ~
      NA_character_,

    str_detect(
      seawall_raw,
      "NO SEAWALL|^NO|N|FALSE|^0|NONE"
    ) ~
      "No",

    str_detect(
      seawall_raw,
      "^YES|Y|TRUE|^1$|HAS SEAWALL|SEAWALL"
    ) ~
      "Yes",

    TRUE ~
      NA_character_
  ),
  quality_group = case_when(

    is.na(quality_raw) ~
      NA_character_,

    str_detect(
      quality_raw,
      "ABOVE\\s*AVERAGE|SUPERIOR|EXCELLENT"
    ) ~
      "Above Average",

    str_detect(
      quality_raw,
      "MINIMAL|FAIR"
    ) ~
      "Fair",

    str_detect(
      quality_raw,
      "AVERAGE"
    ) ~
      "Average",

    TRUE ~
```

```
      NA_character_
  ),
  foundation_group = case_when(

    is.na(foundation_raw) ~
      NA_character_,

    str_detect(
      foundation_raw,
      "CONTINUOUS"
    ) ~
      "Yes",

    TRUE ~
      "No"
  ),
  stories_num =
    parse_number(stories),

  stories_group = case_when(

    is.na(stories_num) ~
      NA_character_,

    stories_num == 1 ~
      "1",

    stories_num == 2 ~
      "2",

    stories_num >= 3 ~
      "3+",

    TRUE ~
      NA_character_
  ),
  barrier_municipality = if_else(

    str_to_upper(site_city) %in% c(
      "ST PETE BEACH",
      "TREASURE ISLAND",
      "MADEIRA BEACH",
      "INDIAN ROCKS BEACH",
      "REDINGTON SHORES",
      "BELLEAIR BEACH",
      "REDINGTON BEACH",
      "NORTH REDINGTON BEACH",
```

```
        "INDIAN SHORES",
        "CLEARWATER BEACH"
      ),

      "Barrier",

      "Non-Barrier",

      missing = NA_character_
    ),
    log_price = if_else(
      !is.na(adj_price) &
        adj_price > 0,
      log(adj_price),
      NA_real_
    )
  ) %>%
  filter(

    sale_year >= 2000,

    sale_year <= 2015,

    !is.na(log_price),

    heated_area > 200,

    !is.na(x_coord),

    !is.na(y_coord),

    !is.na(site_city)
  )


# Variables required for complete-case modeling


model_vars <- c(
  "log_price",
  "sale_year_c",
  "frontage_group",
  "seawall_group",
  "cccl_distance",
  "tb_distance",
  "quality_group",
  "heated_area",
```

```
  "acreage",
  "site_city",
  "barrier_municipality",
  "stories_group",
  "foundation_group",
  "x_coord",
  "y_coord"
)


# Complete-case analytic dataset


model_data <- df_clean %>%

  filter(
    if_all(
      all_of(model_vars),
      ~ !is.na(.x)
    )
  ) %>%

  mutate(

    frontage_group =
      factor(
        frontage_group,
        levels = c(
          "None",
          "Lake",
          "Bay",
          "Other"
        )
      ),

    seawall_group =
      factor(
        seawall_group,
        levels = c(
          "No",
          "Yes"
        )
      ),

    quality_group =
      factor(
        quality_group,
```

```
      levels = c(
        "Average",
        "Fair",
        "Above Average"
      )
    ),

  foundation_group =
    factor(
      foundation_group,
      levels = c(
        "No",
        "Yes"
      )
    ),

  stories_group =
    factor(
      stories_group,
      levels = c(
        "1",
        "2",
        "3+"
      )
    ),

  barrier_municipality =
    factor(
      barrier_municipality,
      levels = c(
        "Non-Barrier",
        "Barrier"
      )
    ),

  site_city =
    factor(site_city)
) %>%

droplevels() %>%

mutate(

  frontage_group =
    relevel(
      frontage_group,
      ref = "None"
```

```
      ),

    seawall_group =
      relevel(
        seawall_group,
        ref = "No"
      ),

    quality_group =
      relevel(
        quality_group,
        ref = "Average"
      ),

    foundation_group =
      relevel(
        foundation_group,
        ref = "No"
      ),

    stories_group =
      relevel(
        stories_group,
        ref = "1"
      ),

    barrier_municipality =
      relevel(
        barrier_municipality,
        ref = "Non-Barrier"
      )
  )

rownames(model_data) <- NULL

cleaning_summary <- tibble(
  Stage = c(
    "After initial filters",
    "Final complete-case dataset"
  ),
  Observations = c(
    nrow(df_clean),
    nrow(model_data)
  ),
  Municipalities = c(
    NA_integer_,
    nlevels(model_data$site_city)
```

```
  )
)

knitr::kable(
  cleaning_summary,
  caption = "Data Cleaning and Sample Selection"
)
```

Table 3: Data Cleaning and Sample Selection

Stage	Observations	Municipalities
After initial filters	245477	NA
Final complete-case dataset	245477	27

4 2. Prepare Bayesian/INLA data

```
# 2. BAYESIAN / INLA DATA


bayes_data <- model_data %>%

  mutate(

    # ----------------------------------------------------------
    # Standardized continuous predictors
    # ----------------------------------------------------------

    cccl_distance_b =
      as.numeric(
        scale(cccl_distance)
      ),

    tb_distance_b =
      as.numeric(
        scale(tb_distance)
      ),

    heated_area_b =
      as.numeric(
        scale(heated_area)
      ),

    acreage_b =
```

```
      as.numeric(
        scale(acreage)
      )
  )


# Municipality integer index for INLA


bayes_data$city_id <-
  as.integer(
    bayes_data$site_city
  )

bayesian_data_summary <- tibble(
  Metric = c(
    "Observations entering INLA",
    "Municipalities entering INLA"
  ),
  Value = c(
    nrow(bayes_data),
    nlevels(bayes_data$site_city)
  )
)

knitr::kable(
  bayesian_data_summary,
  caption = "Bayesian Modeling Dataset"
)
```

Table 4: Bayesian Modeling Dataset

Metric	Value
Observations entering INLA	245477
Municipalities entering INLA	27

5 3. Observation-count summary

```
# 3. DATA SUMMARY


eda_sample_summary <- tibble(

  Stage = c(
```

```
    "Raw input",
    "After initial filters",
    "Final complete-case analytic dataset"
  ),

  Observations = c(
    nrow(raw_data),
    nrow(df_clean),
    nrow(model_data)
  )
)

knitr::kable(
  eda_sample_summary,
  caption =
    "Observation counts through data preparation"
)
```

Table 5: Observation counts through data preparation

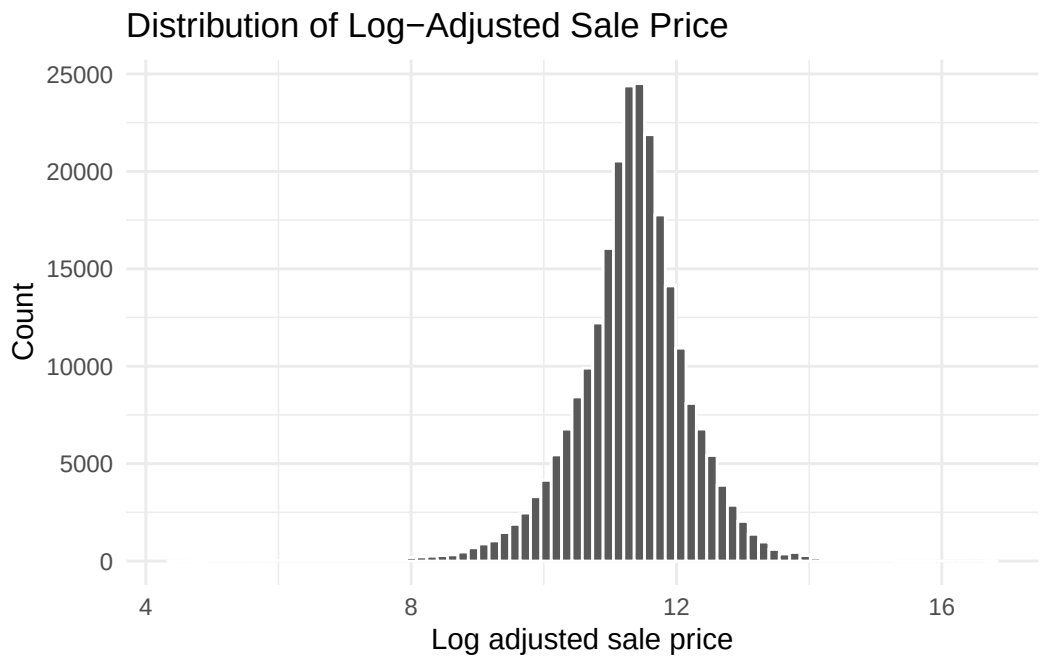
Stage	Observations
Raw input	267074
After initial filters	245477
Final complete-case analytic dataset	245477

6 4. Exploratory data analysis

6.1 4.1 Distribution of log adjusted price

```
# EDA 1: LOG PRICE

ggplot(model_data, aes(x = log_price)) +
  geom_histogram(bins = 80, color = "white") +
  labs(title = "Distribution of Log-Adjusted Sale Price", x = "Log adjusted sale price", y = "C
  theme_minimal()
```

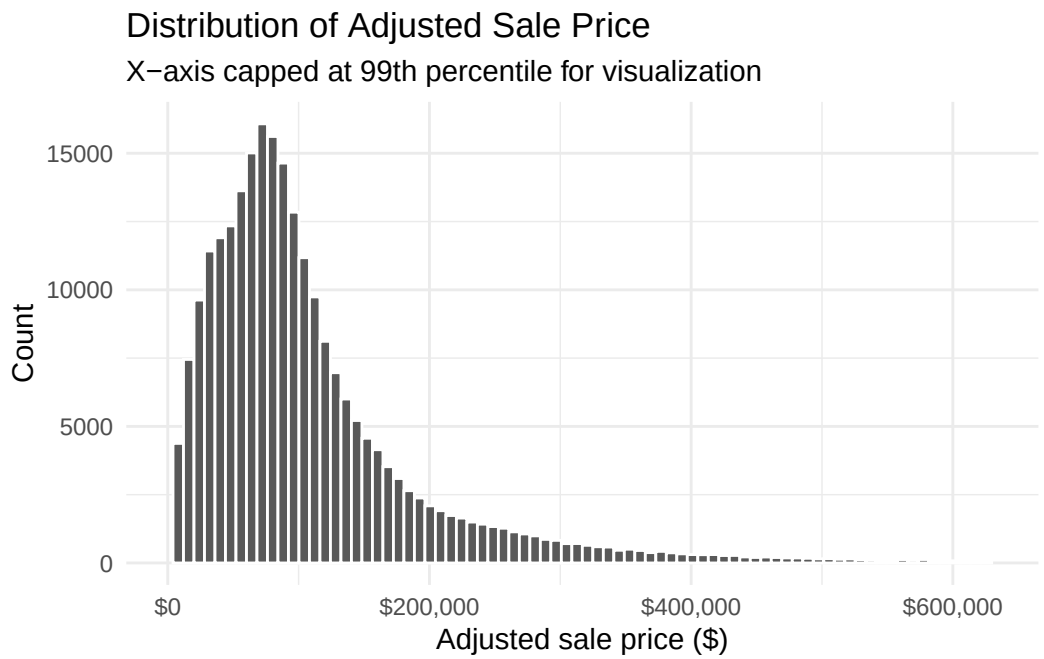


6.2 4.2 Distribution of adjusted price

```
# EDA 2: RAW PRICE

price_cap <- quantile(model_data$adj_price, 0.99, na.rm = TRUE)

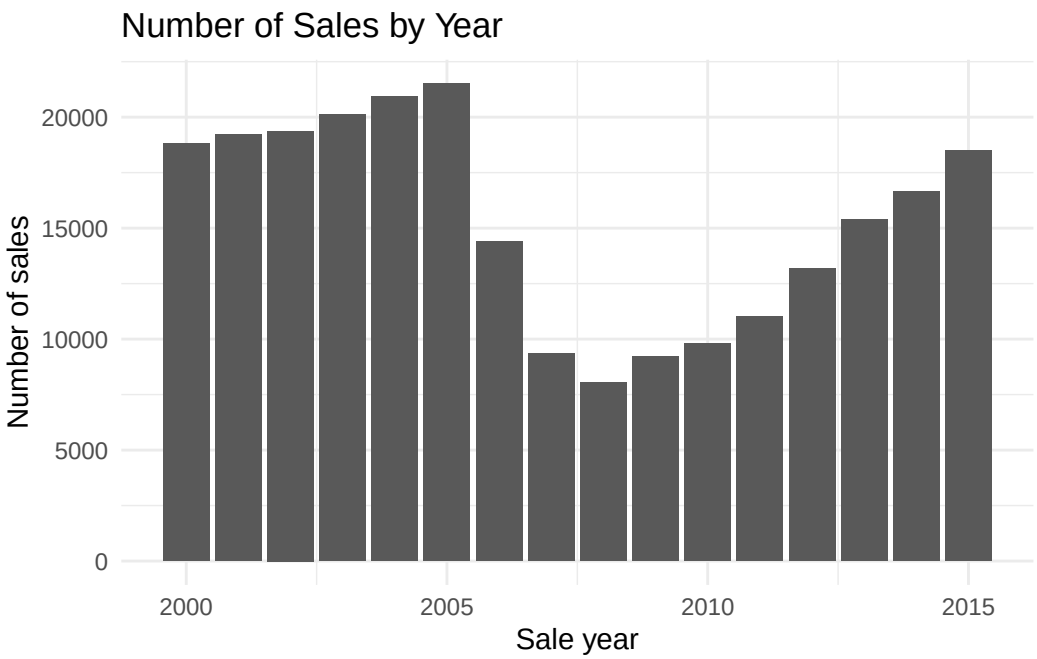
ggplot(model_data, aes(x = adj_price)) +
  geom_histogram(bins = 80, color = "white") +
  scale_x_continuous(labels = scales::dollar, limits = c(0, price_cap)) +
  labs(title = "Distribution of Adjusted Sale Price", subtitle = "X-axis capped at 99th percent
  theme_minimal()
```



6.3 4.3 Sales volume by year

```
# EDA 3: SALES BY YEAR

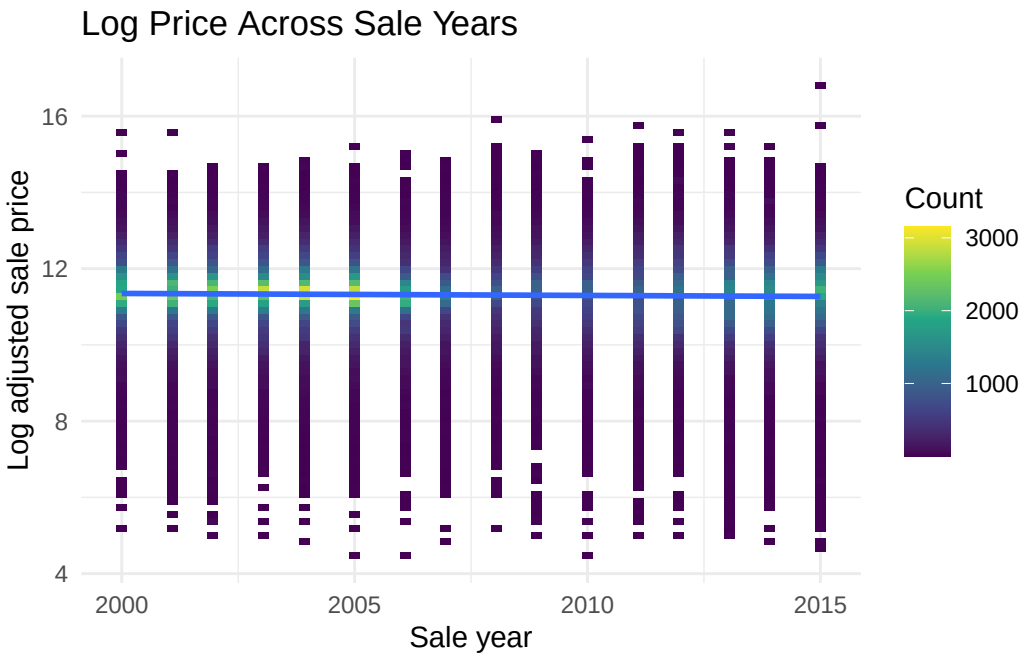
model_data %>%
  count(sale_year) %>%
  ggplot(aes(x = sale_year, y = n)) +
  geom_col() +
  labs(title = "Number of Sales by Year", x = "Sale year", y = "Number of sales") +
  theme_minimal()
```



6.4 4.4 Log price versus sale year

```
# EDA 4: YEAR VS PRICE

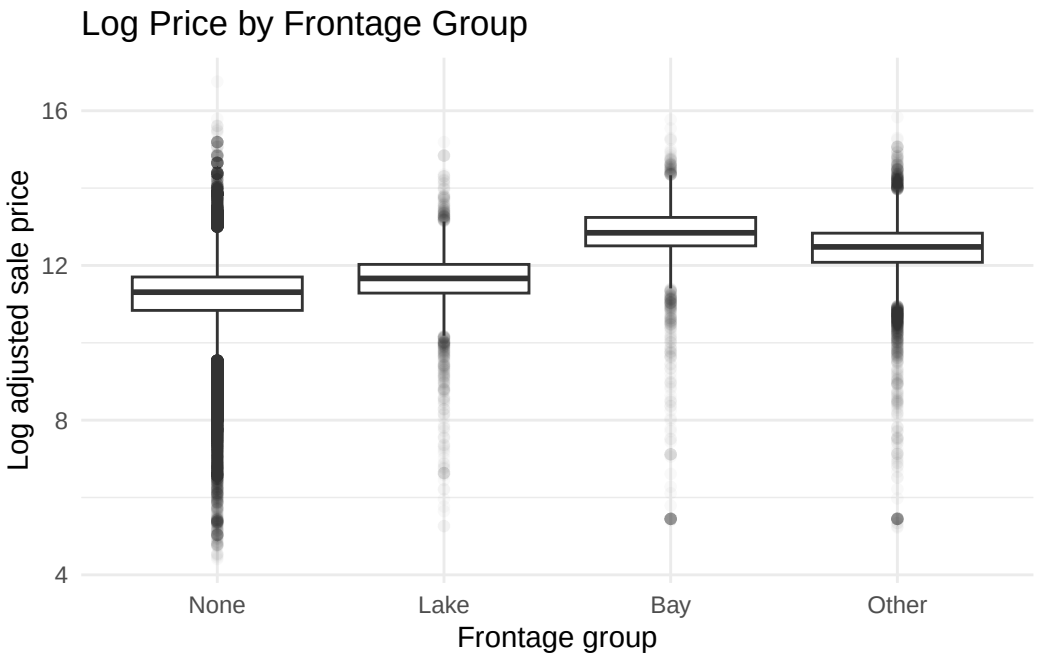
ggplot(model_data, aes(x = sale_year, y = log_price)) +
  geom_bin2d(bins = 70) +
  scale_fill_viridis_c() +
  geom_smooth(method = "lm", se = TRUE) +
  labs(title = "Log Price Across Sale Years", x = "Sale year", y = "Log adjusted sale price", 
  theme_minimal()
```



6.5 4.5 Price by frontage

```
# EDA 5: FRONTAGE

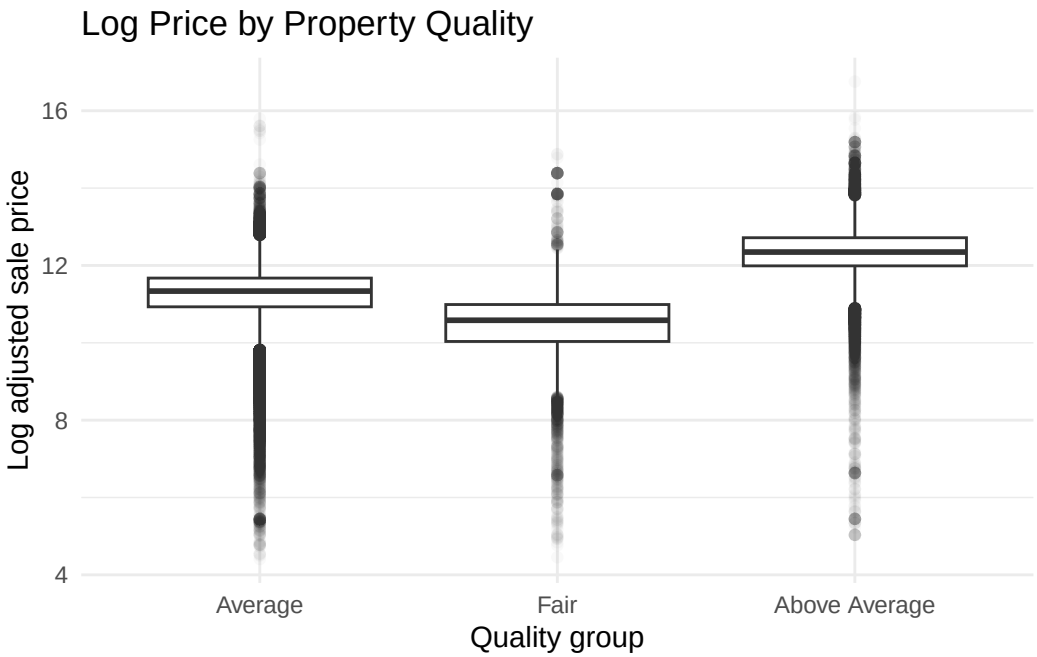
ggplot(model_data, aes(x = frontage_group, y = log_price)) +
  geom_boxplot(outlier.alpha = 0.03) +
  labs(title = "Log Price by Frontage Group", x = "Frontage group", y = "Log adjusted sale pric
  theme_minimal()
```



6.6 4.6 Price by quality

```
# EDA 6: QUALITY

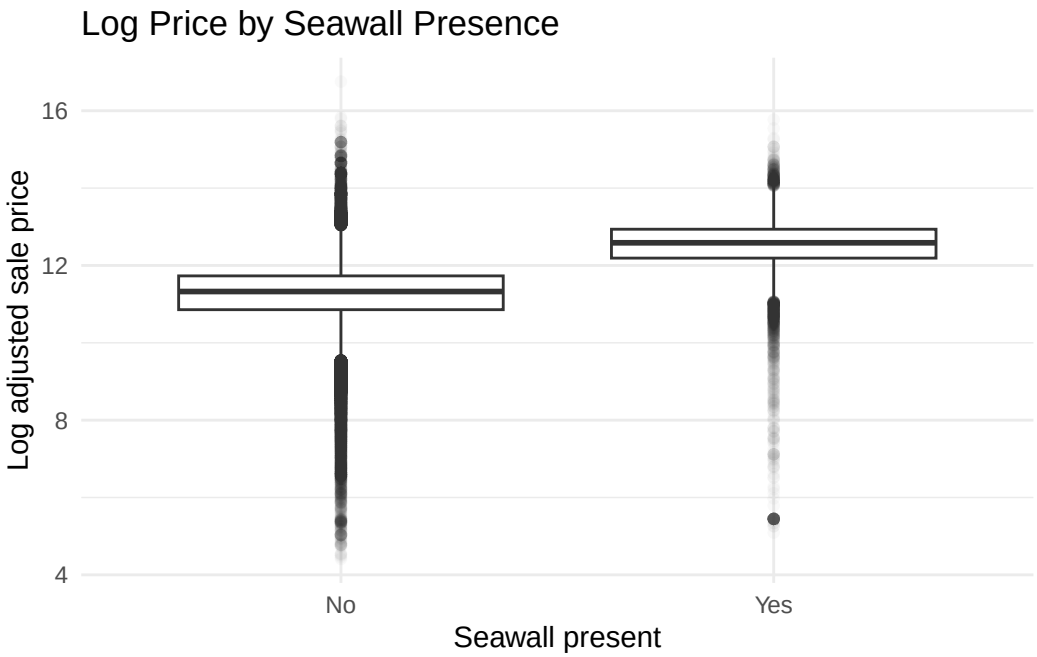
ggplot(model_data, aes(x = quality_group, y = log_price)) +
  geom_boxplot(outlier.alpha = 0.03) +
  labs(title = "Log Price by Property Quality", x = "Quality group", y = "Log adjusted sale pri
  theme_minimal()
```



6.7 4.7 Price by seawall

```
# EDA 7: SEAWALL

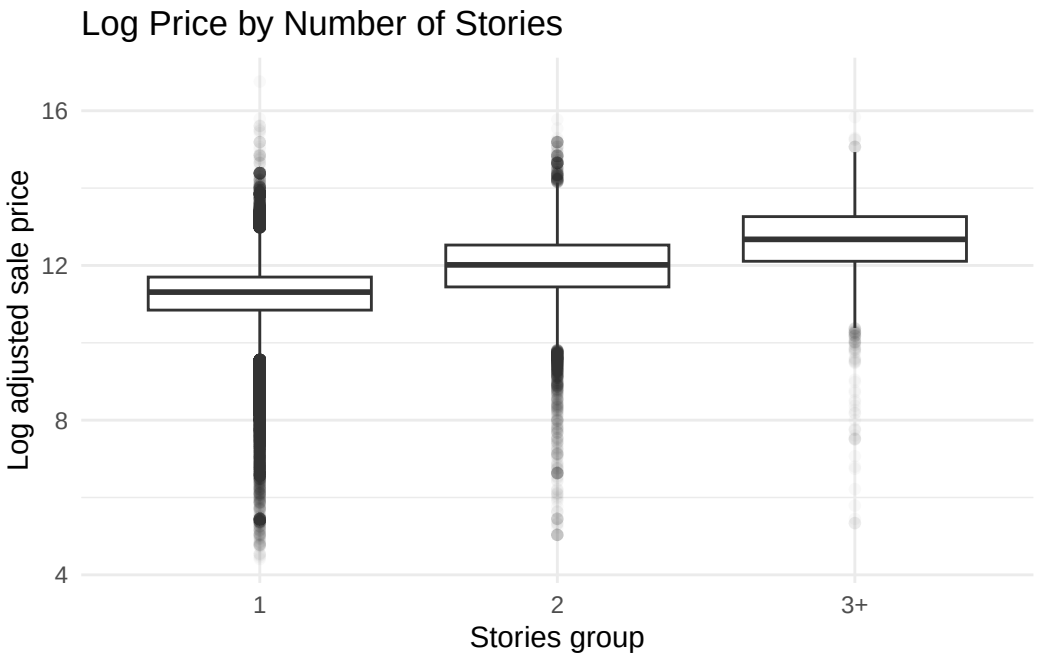
ggplot(model_data, aes(x = seawall_group, y = log_price)) +
  geom_boxplot(outlier.alpha = 0.03) +
  labs(title = "Log Price by Seawall Presence", x = "Seawall present", y = "Log adjusted sale p
  theme_minimal()
```



6.8 4.8 Price by stories

```
# EDA 8: STORIES

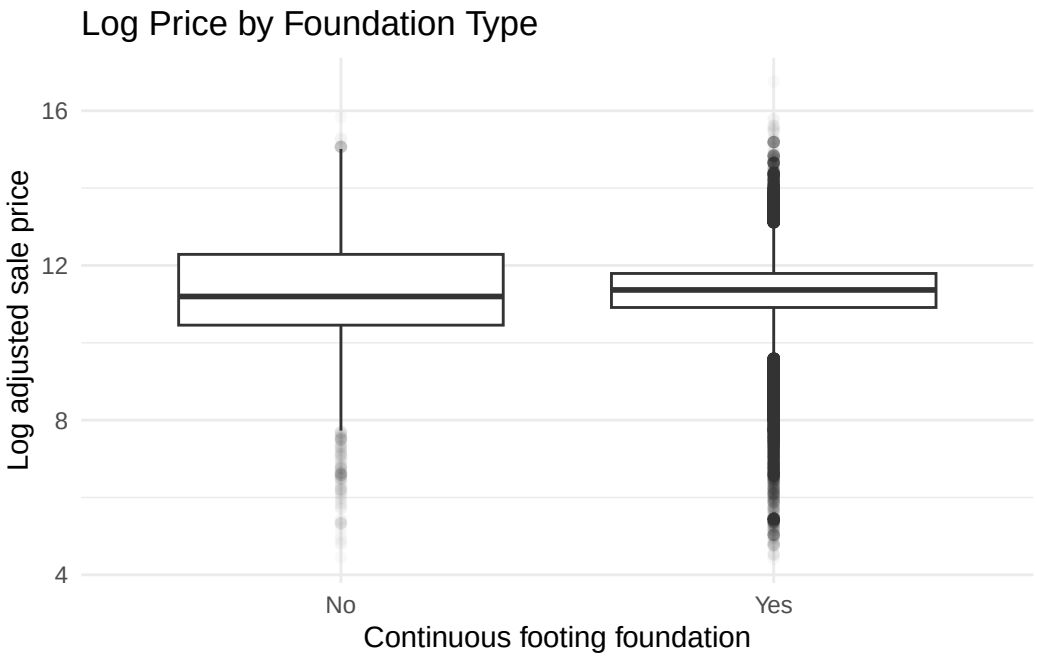
ggplot(model_data, aes(x = stories_group, y = log_price)) +
  geom_boxplot(outlier.alpha = 0.03) +
  labs(title = "Log Price by Number of Stories", x = "Stories group", y = "Log adjusted sale pr
  theme_minimal()
```



6.9 4.9 Price by foundation

```
# EDA 9: FOUNDATION

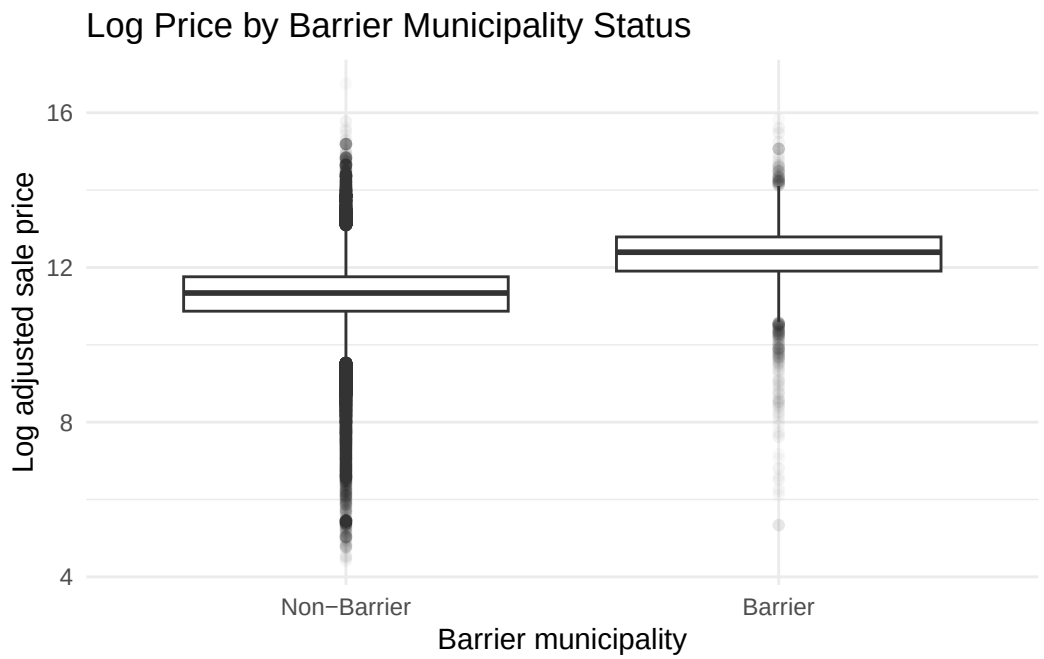
ggplot(model_data, aes(x = foundation_group, y = log_price)) +
  geom_boxplot(outlier.alpha = 0.03) +
  labs(title = "Log Price by Foundation Type", x = "Continuous footing foundation", y = "Log ad
  theme_minimal()
```



6.10 4.10 Price by barrier municipality

```
# EDA 10: BARRIER STATUS

ggplot(model_data,aes(x = barrier_municipality, y = log_price)) +
  geom_boxplot(outlier.alpha = 0.03) +
labs(title = "Log Price by Barrier Municipality Status", x = "Barrier municipality", y = "Log a
theme_minimal()
```



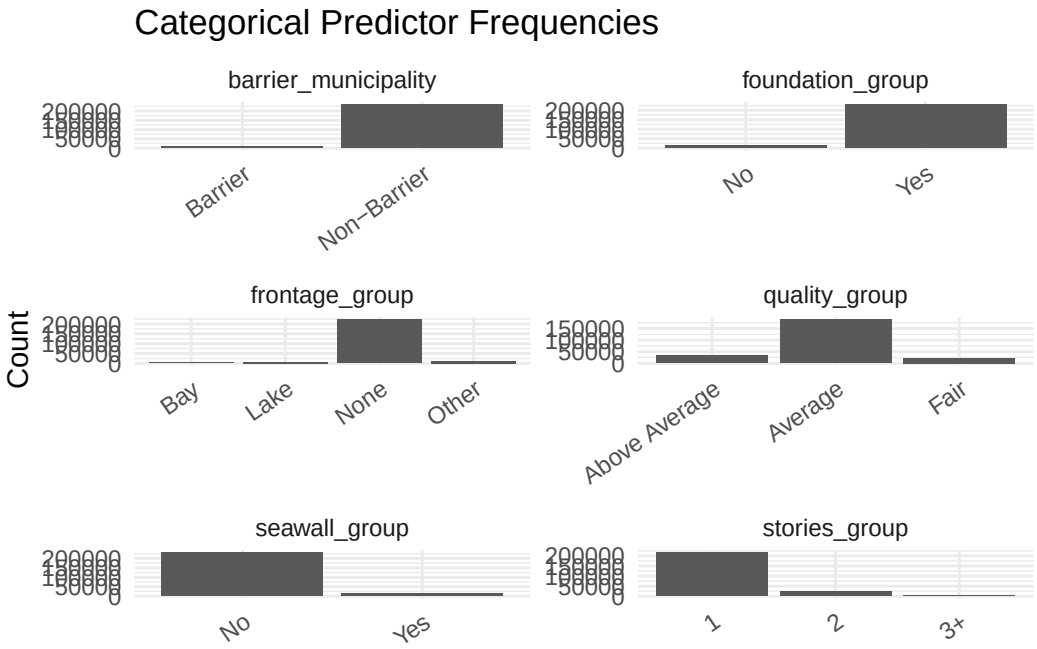
6.11 4.11 Categorical predictor frequencies

```
# EDA 11: CATEGORICAL FREQUENCIES

cat_data_long <- model_data %>%
  select(
    frontage_group,
    seawall_group,
    quality_group,
    barrier_municipality,
    stories_group,
    foundation_group
  ) %>%
  mutate(across(everything(), as.character)) %>%
  pivot_longer(
    everything(),
    names_to = "Variable",
    values_to = "Category"
  ) %>%
  count(Variable, Category)

ggplot(
  cat_data_long,
  aes(x = Category, y = n)
) +
  geom_col() +
```

```
  facet_wrap(
    ~ Variable,
    scales = "free",
    ncol = 2) +
  labs(
    title = "Categorical Predictor Frequencies",
    x = NULL,
    y = "Count"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 35,
      hjust = 1
    )
  )
```



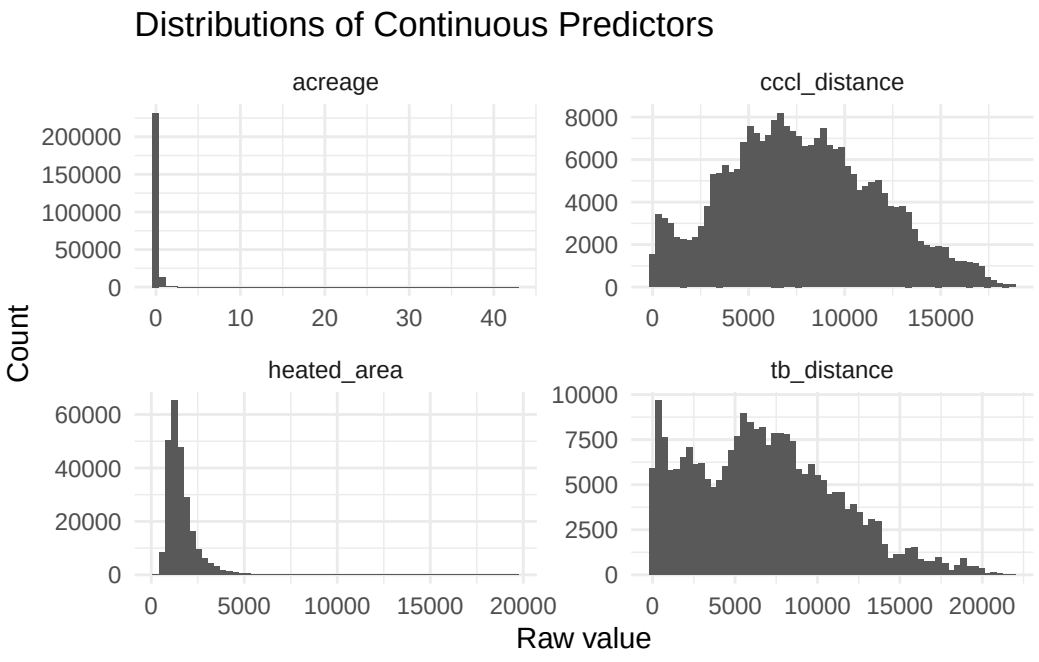
6.12 4.12 Continuous predictor distributions

```
# EDA 12: CONTINUOUS VARIABLES

cont_data_long <- model_data %>%
  select(
    cccl_distance,
    tb_distance,
```

```
    heated_area,
    acreage
  ) %>%
  pivot_longer(
    everything(),
    names_to = "Variable",
    values_to = "Value"
  )

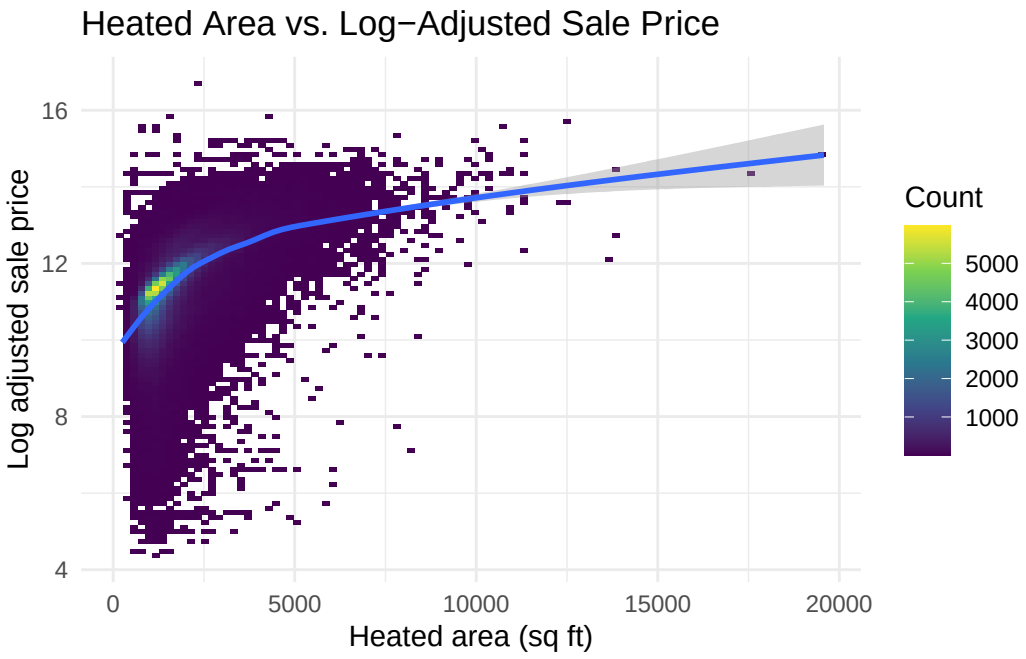
ggplot(
  cont_data_long,
  aes(x = Value)
) +
  geom_histogram(bins = 60) +
  facet_wrap(
    ~ Variable,
    scales = "free",
    ncol = 2
  ) +
  labs(
    title = "Distributions of Continuous Predictors",
    x = "Raw value",
    y = "Count"
  ) +
  theme_minimal()
```



6.13 4.13 Heated area versus price

```
# EDA 13: HEATED AREA VS PRICE

ggplot(
  model_data,
  aes(x = heated_area, y = log_price)
) +
  geom_bin2d(bins = 100) +
  scale_fill_viridis_c() +
  geom_smooth(
    method = "gam",
    se = TRUE
  ) +
  labs(
    title = "Heated Area vs. Log-Adjusted Sale Price",
    x = "Heated area (sq ft)",
    y = "Log adjusted sale price",
    fill = "Count"
  ) +
  theme_minimal()
```

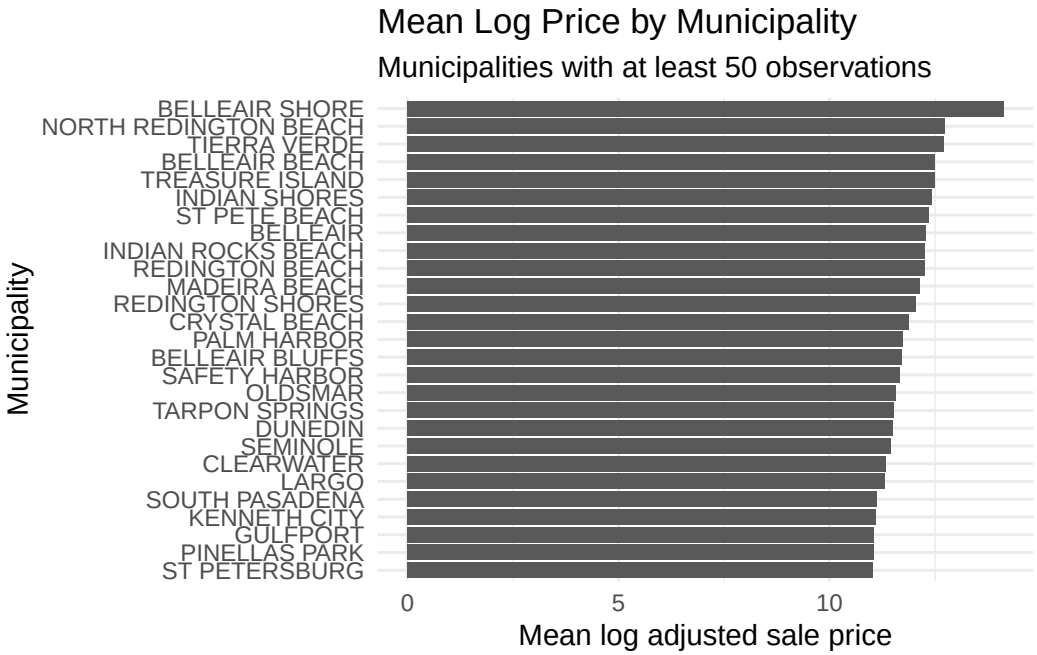


6.14 4.14 Mean price by municipality

```r
# EDA 14: MUNICIPALITY PRICE

city_summary <- model_data %>%
  group_by(site_city) %>%
  summarise(
    mean_log_price = mean(log_price, na.rm = TRUE),
    count = n(),
    .groups = "drop"
  ) %>%
  filter(count >= 50) %>%
  arrange(desc(mean_log_price))

ggplot(
  city_summary,
  aes(
    x = reorder(site_city, mean_log_price),
    y = mean_log_price
  )
) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Mean Log Price by Municipality",
    subtitle = "Municipalities with at least 50 observations",
    x = "Municipality",
    y = "Mean log adjusted sale price"
  ) +
  theme_minimal()
```



6.15 4.15 Geographic distribution

Only the visualization is sampled. The model uses all observations.

```
# EDA 15: SPATIAL DISTRIBUTION

set.seed(12345)

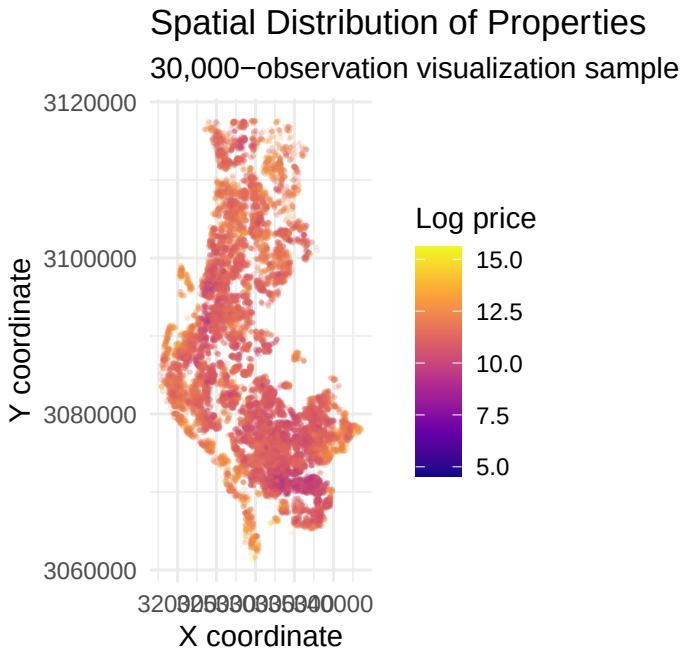
eda_map_n <- min(30000L, nrow(model_data))

eda_map_sample <- sample(
  seq_len(nrow(model_data)),
  size = eda_map_n
)

eda_map_data <- model_data[eda_map_sample, ]

ggplot(
  eda_map_data,
  aes(x = x_coord, y = y_coord, color = log_price)
) +
  geom_point(alpha = 0.15, size = 0.3) +
  scale_color_viridis_c(option = "plasma") +
  coord_equal() +
  labs(
    title = "Spatial Distribution of Properties",
    subtitle = "30,000-observation visualization sample",
```

```
    x = "X coordinate",
    y = "Y coordinate",
    color = "Log price"
  ) +
  theme_minimal()
```



7 5. Model 1 — OLS hedonic baseline

```
# 5. OLS HEDONIC BASELINE

hedonic_formula <- log_price ~
  sale_year_c +
  frontage_group +
  seawall_group +
  cccl_distance +
  tb_distance +
  quality_group +
  heated_area +
  acreage +
  barrier_municipality +
  stories_group +
  foundation_group

ols_model <- lm(
  formula = hedonic_formula,
```

```
  data = model_data
)

summary(ols_model)
```

```
Call:
lm(formula = hedonic_formula, data = model_data)

Residuals:
    Min      1Q  Median      3Q     Max
-8.1762 -0.2075  0.1620  0.3811  4.7581

Coefficients:
                             Estimate Std. Error  t value Pr(>|t|)
(Intercept)                 1.030e+01  9.386e-03 1097.083   <2e-16 ***
sale_year_c                -4.068e-03  2.844e-04  -14.306   <2e-16 ***
frontage_groupLake          1.225e-01  9.630e-03   12.719   <2e-16 ***
frontage_groupBay           2.720e-01  1.534e-02   17.732   <2e-16 ***
frontage_groupOther         3.335e-01  1.046e-02   31.886   <2e-16 ***
seawall_groupYes            2.816e-01  1.127e-02   24.989   <2e-16 ***
cccl_distance               5.371e-06  4.428e-07   12.131   <2e-16 ***
tb_distance                 1.512e-05  3.789e-07   39.908   <2e-16 ***
quality_groupFair          -4.882e-01  5.170e-03  -94.426   <2e-16 ***
quality_groupAbove Average  2.980e-01  5.656e-03   52.680   <2e-16 ***
heated_area                 4.498e-04  2.871e-06  156.646   <2e-16 ***
acreage                     6.424e-02  6.997e-03    9.181   <2e-16 ***
barrier_municipalityBarrier 4.978e-01  8.836e-03   56.338   <2e-16 ***
stories_group2             -1.160e-01  5.521e-03  -21.014   <2e-16 ***
stories_group3+            -1.662e-01  1.445e-02  -11.501   <2e-16 ***
foundation_groupYes         1.398e-01  6.949e-03   20.123   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6904 on 245461 degrees of freedom
Multiple R-squared:  0.402, Adjusted R-squared:  0.4019
F-statistic: 1.1e+04 on 15 and 245461 DF,  p-value: < 2.2e-16
```

```
ols_fit <- tibble(
  Model = "OLS hedonic baseline",
  N = nobs(ols_model),
  R_squared = summary(ols_model)$r.squared,
  Adjusted_R_squared = summary(ols_model)$adj.r.squared,
  AIC = AIC(ols_model),
  BIC = BIC(ols_model),
  RMSE = sqrt(mean(residuals(ols_model)^2)),
```

```
  MAE = mean(abs(residuals(ols_model)))
)

ols_fit
```

```
# A tibble: 1 x 8
  Model                  N R_squared Adjusted_R_squared    AIC    BIC  RMSE   MAE
  <chr>              <int>     <dbl>              <dbl>  <dbl>  <dbl> <dbl> <dbl>
1 OLS hedonic bas~ 245477     0.402              0.402 5.15e5 5.15e5 0.690 0.461
```

8 6. Model 2 — INLA hierarchical municipality model

```
# 6. INLA HIERARCHICAL MUNICIPALITY MODEL

city_formula <- log_price ~
  sale_year_c +
  frontage_group +
  seawall_group +
  cccl_distance_b +
  tb_distance_b +
  quality_group +
  heated_area_b +
  acreage_b +
  barrier_municipality +
  stories_group +
  foundation_group +
  f(city_id, model = "iid")

inla_city <- inla(
  formula = city_formula,
  family = "gaussian",
  data = bayes_data,
  control.predictor = list(
    compute = TRUE,
    link = 1
  ),
  control.compute = list(
    dic = TRUE,
    waic = TRUE,
    cpo = TRUE,
    return.marginals.predictor = TRUE
  ),
  control.inla = list(
    strategy = "simplified.laplace"
```

```
  ),
  verbose = FALSE
)

summary(inla_city)
```

```
Time used:
    Pre = 1.32, Running = 56.8, Post = 9.93, Total = 68
Fixed effects:
                              mean       sd 0.025quant 0.5quant 0.975quant
(Intercept)                 -0.201 4430.406 -10091.356   -0.146  10090.619
sale_year_c                 -0.004    0.000     -0.004   -0.004     -0.003
frontage_groupLake           0.092    0.010      0.073    0.092      0.110
frontage_groupBay            0.312    0.015      0.283    0.312      0.342
frontage_groupOther          0.318    0.010      0.297    0.318      0.338
seawall_groupYes             0.322    0.011      0.300    0.322      0.344
cccl_distance_b              0.125    0.003      0.119    0.125      0.131
tb_distance_b                0.130    0.004      0.123    0.130      0.137
quality_groupFair           -0.479    0.005     -0.489   -0.479     -0.469
quality_groupAbove Average   0.268    0.006      0.257    0.268      0.279
heated_area_b                0.315    0.002      0.311    0.315      0.320
acreage_b                    0.015    0.002      0.012    0.015      0.018
barrier_municipalityBarrier  0.000   31.623    -62.009    0.000     62.009
stories_group2              -0.071    0.005     -0.082   -0.071     -0.060
stories_group3+             -0.119    0.014     -0.147   -0.119     -0.091
foundation_groupYes          0.132    0.007      0.119    0.132      0.146
                              mode kld
(Intercept)                 -0.174   0
sale_year_c                 -0.004   0
frontage_groupLake           0.092   0
frontage_groupBay            0.312   0
frontage_groupOther          0.318   0
seawall_groupYes             0.322   0
cccl_distance_b              0.125   0
tb_distance_b                0.130   0
quality_groupFair           -0.479   0
quality_groupAbove Average   0.268   0
heated_area_b                0.315   0
acreage_b                    0.015   0
barrier_municipalityBarrier  0.000   0
stories_group2              -0.071   0
stories_group3+             -0.119   0
foundation_groupYes          0.132   0

Random effects:
  Name    Model
    city_id IID model
```

```
Model hyperparameters:
                                     mean    sd 0.025quant 0.5quant
Precision for the Gaussian observations 2.17 0.006       2.16     2.17
Precision for city_id                   0.00 0.000       0.00     0.00
                                     0.975quant mode
Precision for the Gaussian observations       2.18 2.17
Precision for city_id                         0.00 0.00

Deviance Information Criterion (DIC) ...............: 506324.93
Deviance Information Criterion (DIC, saturated) ....: 245351.07
Effective number of parameters .....................: 41.49

Watanabe-Akaike information criterion (WAIC) ...: 503207.38
Effective number of parameters .................: 56.41

Marginal log-Likelihood:  -257513.17
CPO, PIT is computed
Posterior summaries for the linear predictor and the fitted values are computed
(Posterior marginals needs also 'control.compute=list(return.marginals.predictor=TRUE)')
```

8.1 6.1 Municipality fixed effects

```
# MUNICIPALITY MODEL FIXED EFFECTS

inla_city$summary.fixed
```

```
                                 mean           sd    0.025quant
(Intercept)                -2.012591e-01 4.430406e+03 -1.009136e+04
sale_year_c                -3.831207e-03 2.796958e-04 -4.379663e-03
frontage_groupLake          9.173355e-02 9.505024e-03  7.309513e-02
frontage_groupBay           3.124940e-01 1.520002e-02  2.826882e-01
frontage_groupOther         3.177161e-01 1.033244e-02  2.974552e-01
seawall_groupYes            3.219899e-01 1.116875e-02  3.000890e-01
cccl_distance_b             1.251247e-01 2.882252e-03  1.194729e-01
tb_distance_b               1.302433e-01 3.526095e-03  1.233290e-01
quality_groupFair          -4.790986e-01 5.166070e-03 -4.892288e-01
quality_groupAbove Average  2.675840e-01 5.625405e-03  2.565531e-01
heated_area_b               3.152818e-01 2.291191e-03  3.107890e-01
acreage_b                   1.484502e-02 1.511279e-03  1.188155e-02
barrier_municipalityBarrier 1.973554e-08 3.162254e+01 -6.200867e+01
stories_group2             -7.088216e-02 5.474867e-03 -8.161784e-02
stories_group3+            -1.191706e-01 1.431889e-02 -1.472486e-01
foundation_groupYes         1.322515e-01 6.874906e-03  1.187705e-01
                               0.5quant    0.975quant          mode
```

```
(Intercept)                 -1.463958e-01  1.009062e+04 -1.742032e-01
sale_year_c                 -3.831207e-03 -3.282751e-03 -3.831207e-03
frontage_groupLake           9.173355e-02  1.103720e-01  9.173355e-02
frontage_groupBay            3.124940e-01  3.422997e-01  3.124940e-01
frontage_groupOther          3.177161e-01  3.379770e-01  3.177161e-01
seawall_groupYes             3.219899e-01  3.438907e-01  3.219899e-01
cccl_distance_b              1.251247e-01  1.307765e-01  1.251247e-01
tb_distance_b                1.302433e-01  1.371576e-01  1.302433e-01
quality_groupFair           -4.790986e-01 -4.689684e-01 -4.790986e-01
quality_groupAbove Average   2.675840e-01  2.786149e-01  2.675840e-01
heated_area_b                3.152818e-01  3.197746e-01  3.152818e-01
acreage_b                    1.484502e-02  1.780849e-02  1.484502e-02
barrier_municipalityBarrier  1.973562e-08  6.200867e+01  1.973554e-08
stories_group2              -7.088216e-02 -6.014648e-02 -7.088216e-02
stories_group3+             -1.191706e-01 -9.109268e-02 -1.191706e-01
foundation_groupYes          1.322515e-01  1.457325e-01  1.322515e-01
                                      kld
(Intercept)                 1.783395e-05
sale_year_c                 5.486798e-11
frontage_groupLake          5.485685e-11
frontage_groupBay           5.487031e-11
frontage_groupOther         5.479094e-11
seawall_groupYes            5.481946e-11
cccl_distance_b             5.491852e-11
tb_distance_b               5.483169e-11
quality_groupFair           5.433913e-11
quality_groupAbove Average  5.503680e-11
heated_area_b               5.485453e-11
acreage_b                   5.486621e-11
barrier_municipalityBarrier 5.526823e-11
stories_group2              5.484962e-11
stories_group3+             5.486366e-11
foundation_groupYes         5.485161e-11
```

8.2 6.2 Municipality random effects

```
# MUNICIPALITY RANDOM EFFECTS
inla_city$summary.random$city_id
```

```
   ID     mean       sd 0.025quant 0.5quant 0.975quant     mode          kld
1   1 11.88982 4430.406  -10078.93 11.83496   10103.04 11.86276 1.783395e-05
2   2 12.09259 4430.519  -10078.84 12.03773   10103.36 12.06553 1.783010e-05
3   3 11.76069 4430.406  -10079.06 11.70583   10102.92 11.73364 1.783395e-05
4   4 12.13279 4430.406  -10078.69 12.07793   10103.29 12.10574 1.783394e-05
5   5 11.49513 4430.406  -10079.32 11.44026   10102.65 11.46807 1.783395e-05
```

```
6   6 11.63622 4430.406  -10079.18 11.58136   10102.79 11.60917 1.783395e-05
7   7 11.54310 4430.406  -10079.28 11.48824   10102.70 11.51605 1.783395e-05
8   8 11.41338 4430.406  -10079.41 11.35852   10102.57 11.38632 1.783395e-05
9   9 11.99864 4430.519  -10078.93 11.94378   10103.26 11.97158 1.783010e-05
10 10 12.41811 4430.519  -10078.51 12.36325   10103.68 12.39106 1.783010e-05
11 11 11.27153 4430.406  -10079.55 11.21667   10102.43 11.24448 1.783395e-05
12 12 11.52331 4430.406  -10079.30 11.46845   10102.68 11.49625 1.783395e-05
13 13 11.79522 4430.519  -10079.14 11.74036   10103.06 11.76816 1.783010e-05
14 14 11.75273 4430.519  -10079.18 11.69787   10103.02 11.72567 1.783010e-05
15 15 11.40404 4430.406  -10079.42 11.34918   10102.56 11.37698 1.783395e-05
16 16 11.44885 4430.406  -10079.37 11.39398   10102.60 11.42179 1.783395e-05
17 17 11.18063 4430.406  -10079.64 11.12577   10102.34 11.15357 1.783395e-05
18 18 11.83176 4430.519  -10079.10 11.77690   10103.10 11.80471 1.783010e-05
19 19 11.70150 4430.519  -10079.23 11.64664   10102.97 11.67444 1.783010e-05
20 20 11.54992 4430.406  -10079.27 11.49506   10102.71 11.52287 1.783395e-05
21 21 11.46547 4430.406  -10079.35 11.41061   10102.62 11.43842 1.783395e-05
22 22 11.38680 4430.406  -10079.43 11.33194   10102.54 11.35975 1.783395e-05
23 23 12.17831 4430.519  -10078.75 12.12345   10103.44 12.15125 1.783010e-05
24 24 11.26137 4430.406  -10079.56 11.20650   10102.42 11.23431 1.783395e-05
25 25 11.01236 4430.406  -10079.81 10.95750   10102.17 10.98531 1.783395e-05
26 26 12.03434 4430.406  -10078.79 11.97947   10103.19 12.00728 1.783395e-05
27 27 12.02952 4430.519  -10078.90 11.97466   10103.30 12.00246 1.783010e-05
```

8.3 6.3 Municipality model fit statistics

```
# MUNICIPALITY MODEL FIT

city_model_fit <- tibble(
  Model = "INLA municipality",
  N = nrow(bayes_data),
  DIC = inla_city$dic$dic,
  WAIC = inla_city$waic$waic
)

city_model_fit
```

```
# A tibble: 1 x 4
  Model                  N     DIC    WAIC
  <chr>              <int>   <dbl>   <dbl>
1 INLA municipality 245477 506325. 503207.
```

9 7. Prepare projected spatial coordinates

Coordinates are projected UTM Zone 17N coordinates measured in meters. We are unsure whether the original datum is WGS 84 (EPSG:32617) or NAD83 (EPSG:26917).

```r
# 7. PREPARE PROJECTED SPATIAL COORDINATES
# Retain the supplied projected coordinates.
# Coordinates are consistent with UTM Zone 17N in meters.

projected_coords <-
  as.matrix(
    bayes_data[
      ,
      c("x_coord", "y_coord")
    ]
  )

colnames(projected_coords) <-
  c("x", "y")

# Verify coordinates.

stopifnot(
  nrow(projected_coords) == nrow(bayes_data),
  ncol(projected_coords) == 2,
  all(is.finite(projected_coords))
)


# Compare possible coordinate reference systems.


test_crs <- function(epsg_code, crs_name) {

  points <- st_as_sf(
    bayes_data,
    coords = c("x_coord", "y_coord"),
    crs = epsg_code
  )

  points_lonlat <- st_transform(
    points,
    4326
  )

  coords <- st_coordinates(points_lonlat)

  tibble(
    coordinate_system = crs_name,
    minimum_longitude = min(coords[, 1]),
    maximum_longitude = max(coords[, 1]),
    minimum_latitude = min(coords[, 2]),
```

```
    maximum_latitude = max(coords[, 2])
  )
}

crs_comparison <- bind_rows(

  test_crs(
    32617,
    "WGS 84 / UTM Zone 17N"
  ),

  test_crs(
    26917,
    "NAD83 / UTM Zone 17N"
  ),

  test_crs(
    2882,
    "NAD83(HARN) StatePlane Florida West (feet)"
  )

)


# Compare WGS 84 and NAD83 geographic locations.


coords_32617 <- st_coordinates(
  st_transform(
    st_as_sf(
      bayes_data,
      coords = c("x_coord", "y_coord"),
      crs = 32617
    ),
    4326
  )
)

coords_26917 <- st_coordinates(
  st_transform(
    st_as_sf(
      bayes_data,
      coords = c("x_coord", "y_coord"),
      crs = 26917
    ),
    4326
  )
```

```
)

longitude_difference <- max(
  abs(coords_32617[, 1] - coords_26917[, 1])
)

latitude_difference <- max(
  abs(coords_32617[, 2] - coords_26917[, 2])
)


# Combine results into one readable table.

coordinate_summary <-
  bind_rows(

    tibble(
      coordinate_system = "Original supplied coordinates",
      measurement = c(
        "Number of observations",
        "Minimum X",
        "Maximum X",
        "Minimum Y",
        "Maximum Y"
      ),
      value = c(
        nrow(projected_coords),
        min(projected_coords[, 1]),
        max(projected_coords[, 1]),
        min(projected_coords[, 2]),
        max(projected_coords[, 2])
      ),
      units = c(
        "observations",
        "meters",
        "meters",
        "meters",
        "meters"
      )
    ),

    crs_comparison %>%
      pivot_longer(
        cols = -coordinate_system,
        names_to = "measurement",
        values_to = "value"
      ) %>%
```

```
      mutate(
        measurement = recode(
          measurement,
          minimum_longitude = "Minimum longitude",
          maximum_longitude = "Maximum longitude",
          minimum_latitude = "Minimum latitude",
          maximum_latitude = "Maximum latitude"
        ),
        units = "decimal degrees"
      )

  )

knitr::kable(
  coordinate_summary,
  digits = 6,
  caption = "Spatial Coordinates and Coordinate Reference System Comparison",
  col.names = c(
    "Coordinate system",
    "Measurement",
    "Value",
    "Units"
  )
)
```

Table 6: Spatial Coordinates and Coordinate Reference System Comparison

Coordinate system	Measurement	Value	Units
Original supplied coordinates	Number of observations	245477.00000	observations
Original supplied coordinates	Minimum X	317784.41270	meters
Original supplied coordinates	Maximum X	343618.70940	meters
Original supplied coordinates	Minimum Y	3061037.35000	meters
Original supplied coordinates	Maximum Y	3117629.43300	meters
WGS 84 / UTM Zone 17N	Minimum longitude	-82.85109	decimal degrees
WGS 84 / UTM Zone 17N	Maximum longitude	-82.58773	decimal degrees
WGS 84 / UTM Zone 17N	Minimum latitude	27.66277	decimal degrees
WGS 84 / UTM Zone 17N	Maximum latitude	28.17327	decimal degrees
NAD83 / UTM Zone 17N	Minimum longitude	-82.85109	decimal degrees
NAD83 / UTM Zone 17N	Maximum longitude	-82.58773	decimal degrees

Coordinate system	Measurement	Value	Units
NAD83 / UTM Zone 17N	Minimum latitude	27.66277	decimal degrees
NAD83 / UTM Zone 17N	Maximum latitude	28.17327	decimal degrees
NAD83(HARN) StatePlane Florida West (feet)	Minimum longitude	-83.10140	decimal degrees
NAD83(HARN) StatePlane Florida West (feet)	Maximum longitude	-83.01708	decimal degrees
NAD83(HARN) StatePlane Florida West (feet)	Minimum latitude	32.74774	decimal degrees
NAD83(HARN) StatePlane Florida West (feet)	Maximum latitude	32.90319	decimal degrees

```
head(bayes_data[, c("x_coord", "y_coord")], 20)
```

```
# A tibble: 20 x 2
   x_coord  y_coord
     <dbl>    <dbl>
 1 328086. 3117251.
 2 328145. 3117321.
 3 328145. 3117321.
 4 328191. 3117352.
 5 328180. 3117203.
 6 328180. 3117203.
 7 328195. 3116993.
 8 328195. 3116993.
 9 328195. 3116993.
10 328195. 3117025.
11 328195. 3117025.
12 328287. 3117018.
13 328287. 3117018.
14 328287. 3117018.
15 328287. 3117018.
16 328287. 3117004.
17 328287. 3117004.
18 328287. 3116990.
19 328245. 3117208.
20 328245. 3117208.
```

10 8. Construct computationally efficient INLA SPDE mesh

```r
# 8. INLA SPDE MESH

set.seed(12345)

# Mesh construction uses a subset of coordinates.
# This does NOT sample the observations used in the model.
# All complete-case properties remain in the likelihood.
# The sample is only used to construct the mesh efficiently.

mesh_n <- min(15000L, nrow(projected_coords))

mesh_sample_id <- sample(
  seq_len(nrow(projected_coords)),
  size = mesh_n
)

mesh_coords <- projected_coords[mesh_sample_id, , drop = FALSE]
```

```r
# BUILD TRIANGULAR SPDE MESH

mesh <- inla.mesh.2d(
  loc = mesh_coords,
  # Minimum distance between mesh points (meters)
  cutoff = 150,
  # Triangle edge lengths (meters)
  # Smaller values produce a more detailed mesh
  # Larger values produce fewer latent spatial nodes
  max.edge = c(1000, 3000),
  # Buffer around observation region (meters)
  offset = c(1000, 3000)
)

mesh_summary <- tibble(
  Metric = c("Mesh vertices", "Mesh triangles", "Coordinates used for mesh construction"),
  Value = c(nrow(mesh$loc), nrow(mesh$graph$tv), nrow(mesh_coords))
)

knitr::kable(mesh_summary, caption = "SPDE Mesh Diagnostics")
```

Table 7: SPDE Mesh Diagnostics

Metric	Value
Mesh vertices	8779
Mesh triangles	17472
Coordinates used for mesh construction	15000

10.1 8.1 Plot mesh

```
# MESH VISUALIZATION

plot(mesh, asp = 1, main = "INLA SPDE Mesh - Pinellas County")
points(mesh_coords, pch = 16, cex = 0.1)
```



11 9. Define Matern SPDE model

```
# 9. MATERN SPDE

spde <- inla.spde2.pcmatern(
  mesh = mesh,
  # Matern smoothness: alpha = 2 is the common Matern SPDE construction.
  alpha = 2,
  # PC prior: P(range < 1000 m) = 0.5
  prior.range = c(1000, 0.5),
  # PC prior: P(sigma > 1) = 0.5
  prior.sigma = c(1, 0.5),
  constr = TRUE
)

spde_summary <- tibble(
  Metric = "SPDE latent nodes",
```

```
  Value = spde$n.spde
)

knitr::kable(spde_summary, caption = "SPDE Model Summary")
```

Table 8: SPDE Model Summary

Metric	Value
SPDE latent nodes	8779

12 10. Create SPDE projection matrix

```
# 10. SPDE PROJECTION MATRIX

A_spde <- inla.spde.make.A(
  mesh = mesh,
  loc = projected_coords
)

projection_summary <- tibble(
  Metric = c("Projection matrix rows", "Projection matrix columns", "SPDE latent nodes"),
  Value = c(nrow(A_spde), ncol(A_spde), spde$n.spde)
)

knitr::kable(
  projection_summary,
  caption = "SPDE Projection Matrix Dimensions"
)
```

Table 9: SPDE Projection Matrix Dimensions

Metric	Value
Projection matrix rows	245477
Projection matrix columns	8779
SPDE latent nodes	8779

13 11. Create INLA spatial index

```
# 11. SPATIAL INDEX

spatial_index <- inla.spde.make.index(name = "spatial_field", n.spde = spde$n.spde)
```

14 12. Create INLA model stack

```
# 12. INLA STACK

# Fixed effects
fixed_effects <- data.frame(
  intercept = 1,
  sale_year_c = bayes_data$sale_year_c,
  frontage_group = bayes_data$frontage_group,
  seawall_group = bayes_data$seawall_group,
  cccl_distance_b = bayes_data$cccl_distance_b,
  tb_distance_b = bayes_data$tb_distance_b,
  quality_group = bayes_data$quality_group,
  heated_area_b = bayes_data$heated_area_b,
  acreage_b = bayes_data$acreage_b,
  barrier_municipality = bayes_data$barrier_municipality,
  stories_group = bayes_data$stories_group,
  foundation_group = bayes_data$foundation_group,
  city_id = bayes_data$city_id
)

# Create stack
spatial_stack <- inla.stack(
  tag = "est",
  data = list(y = bayes_data$log_price),
  A = list(1, A_spde),
  effects = list(fixed_effects, spatial_index),
  remove.unused = FALSE
)

stack_data <- inla.stack.data(spatial_stack)

stack_summary <- tibble(
  Metric = "Stack observations",
  Value = length(stack_data$y)
)

knitr::kable(
  stack_summary,
  caption = "INLA Stack Summary"
)
```

Table 10: INLA Stack Summary

Metric	Value
Stack observations	245477

15 13. Primary INLA spatial model

```
# 13. PRIMARY INLA SPATIAL MODEL

spatial_formula <- y ~
  0 +
  intercept +
  sale_year_c +
  frontage_group +
  seawall_group +
  cccl_distance_b +
  tb_distance_b +
  quality_group +
  heated_area_b +
  acreage_b +
  barrier_municipality +
  stories_group +
  foundation_group +
  f(city_id, model = "iid") +
  f(spatial_field, model = spde)

inla_spatial <- inla(
  formula = spatial_formula,
  family = "gaussian",
  data = inla.stack.data(spatial_stack),
  control.predictor = list(
    A = inla.stack.A(spatial_stack),
    compute = TRUE,
    link = 1
  ),
  control.compute = list(
    dic = TRUE,
    waic = TRUE,
    cpo = TRUE,
    return.marginals.predictor = TRUE,
    config = TRUE
  ),
  control.inla = list(
    strategy = "simplified.laplace"
```

```
  ),
  verbose = TRUE
)

summary(inla_spatial)
```

```
Time used:
    Pre = 1.83, Running = 49.6, Post = 19.2, Total = 70.7
Fixed effects:
                              mean     sd 0.025quant 0.5quant 0.975quant   mode
intercept                    9.375 14.142    -18.356    9.375     37.106  9.375
sale_year_c                 -0.004  0.000     -0.004   -0.004     -0.003 -0.004
frontage_groupNone           2.184 14.142    -25.547    2.184     29.915  2.184
frontage_groupLake           2.233 14.142    -25.498    2.233     29.964  2.233
frontage_groupBay            2.516 14.142    -25.215    2.516     30.247  2.516
frontage_groupOther          2.442 14.142    -25.289    2.442     30.173  2.442
seawall_groupYes             0.160  0.012      0.136    0.160      0.184  0.160
cccl_distance_b              0.002  0.029     -0.055    0.002      0.059  0.002
tb_distance_b                0.071  0.031      0.009    0.071      0.132  0.071
quality_groupFair           -0.185  0.006     -0.196   -0.185     -0.174 -0.185
quality_groupAbove Average   0.158  0.006      0.146    0.158      0.171  0.158
heated_area_b                0.203  0.002      0.198    0.203      0.208  0.203
acreage_b                    0.028  0.002      0.025    0.028      0.031  0.028
barrier_municipalityBarrier  0.339  0.120      0.106    0.338      0.579  0.338
stories_group2              -0.022  0.006     -0.033   -0.022     -0.011 -0.022
stories_group3+             -0.057  0.014     -0.085   -0.057     -0.029 -0.057
foundation_groupYes          0.053  0.007      0.039    0.053      0.066  0.053
                             kld
intercept                      0
sale_year_c                    0
frontage_groupNone             0
frontage_groupLake             0
frontage_groupBay              0
frontage_groupOther            0
seawall_groupYes               0
cccl_distance_b                0
tb_distance_b                  0
quality_groupFair              0
quality_groupAbove Average     0
heated_area_b                  0
acreage_b                      0
barrier_municipalityBarrier    0
stories_group2                 0
stories_group3+                0
foundation_groupYes            0

Random effects:
```

```
  Name    Model
    city_id IID model
   spatial_field SPDE2 model

Model hyperparameters:
                                          mean     sd 0.025quant 0.5quant
Precision for the Gaussian observations    2.656  0.008      2.641    2.656
Precision for city_id                     24.190  9.986      7.865   23.100
Range for spatial_field                 1561.371 67.311   1434.955 1559.276
Stdev for spatial_field                    0.399  0.012      0.376    0.398
                                        0.975quant     mode
Precision for the Gaussian observations      2.671    2.656
Precision for city_id                       45.531   20.128
Range for spatial_field                   1699.942 1553.677
Stdev for spatial_field                      0.423    0.397

Deviance Information Criterion (DIC) ...............: 460302.91
Deviance Information Criterion (DIC, saturated) ....: 248942.55
Effective number of parameters .....................: 3464.74

Watanabe-Akaike information criterion (WAIC) ...: 457201.80
Effective number of parameters .................: 3615.38

Marginal log-Likelihood:  -232582.14
CPO, PIT is computed
Posterior summaries for the linear predictor and the fitted values are computed
(Posterior marginals needs also 'control.compute=list(return.marginals.predictor=TRUE)')
```

Bayesian hierarchical spatial regression estimated using the Integrated Nested Laplace Approximation (INLA)

15.1 Mathematical Specification of the Bayesian Spatial Hedonic Model

The primary model estimates the relationship between housing prices and structural, locational, and geographic characteristics while accounting for municipality-level variation and spatial dependence.

The response variable is the natural logarithm of the adjusted sale price:

$$Y_i = \log(\text{adj_price}_i)$$

The observation model is:

$$Y_i \mid \eta_i, \tau_\epsilon \sim \mathcal{N}(\eta_i, \tau_\epsilon^{-1})$$

The linear predictor is:

$$
\begin{aligned}
\eta_i = \beta_0 &+ \beta_1 \text{SaleYear}_i + \beta_2^\top \text{Frontage}_i + \beta_3 \text{Seawall}_i \\
&+ \beta_4 \text{CCCLDistance}_i + \beta_5 \text{TBDistance}_i + \beta_6^\top \text{Quality}_i \\
&+ \beta_7 \text{HeatedArea}_i + \beta_8 \text{Acreage}_i + \beta_9 \text{BarrierMunicipality}_i \\
&+ \beta_{10}^\top \text{Stories}_i + \beta_{11}^\top \text{Foundation}_i \\
&+ u_{c(i)} + w(\mathbf{s}_i)
\end{aligned}
$$

where:

- i indexes individual property sales.
- 0 is the overall intercept.
- The coefficients represent fixed effects associated with housing characteristics and location.
- Categorical predictors are represented by indicator variables relative to their reference categories.
- uc(i) is the municipality-specific random intercept for the municipality containing property i.
- w(si) is the spatially structured random effect at the property's geographic coordinates.
- is the observation-level error precision.

The municipality-level random effects are modeled as:

$$
u_j \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2_{\text{city}})
$$

The spatial random effect is modeled as a Matérn Gaussian random field:

$$
w(\mathbf{s}) \sim \text{Matérn}(\sigma^2_{\text{spatial}}, \rho, \nu)
$$

The Matérn spatial field is represented computationally using the Stochastic Partial Differential Equation (SPDE) approach:

$$
(\kappa^2 - \Delta)^{\alpha/2} \left(\tau_{\text{SPDE}} w(\mathbf{s})\right) = \mathcal{W}(\mathbf{s})
$$

where is the Laplacian operator, W(s) is Gaussian white noise, controls the spatial range, and SPDE is a scaling parameter. For this model, the SPDE uses:

$$
\alpha = 2, \qquad d = 2, \qquad \nu = \alpha - \frac{d}{2} = 1
$$

The approximate relationship between the Matérn range and is:

$$
\rho = \frac{\sqrt{8\nu}}{\kappa}
$$

The SPDE model uses penalized-complexity (PC) priors for the spatial range and marginal standard deviation:

$$
P(\rho < 1000) = 0.5
$$

$$P(\sigma_{\text{spatial}} > 1) = 0.5$$

These priors correspond to the INLA specification `prior.range = c(1000, 0.5)` and `prior.sigma = c(1, 0.5)`, assuming the projected coordinate system uses meters.

The complete model is estimated using Integrated Nested Laplace Approximations (INLA), which provides approximate Bayesian inference for the fixed effects, municipality-level random effects, spatial field, and model hyperparameters.

```
# MODEL OUTPUT: COEFFICIENTS, RANDOM EFFECTS, PRIORS, AND FIT

# 1. Fixed-effect coefficient estimates
fixed_effect_output <- inla_spatial$summary.fixed

print("FIXED EFFECT COEFFICIENTS")
```

```
[1] "FIXED EFFECT COEFFICIENTS"
```

```
print(fixed_effect_output)
```

```
                                mean           sd    0.025quant
intercept                  9.374969543 1.414204e+01 -18.356166503
sale_year_c               -0.003650175 2.556873e-04  -0.004151553
frontage_groupNone         2.184066590 1.414196e+01 -25.546915716
frontage_groupLake         2.233327953 1.414196e+01 -25.497657310
frontage_groupBay          2.515847486 1.414196e+01 -25.215142324
frontage_groupOther        2.442238220 1.414196e+01 -25.288745843
seawall_groupYes           0.160090511 1.235759e-02   0.135858152
cccl_distance_b            0.002087764 2.907453e-02  -0.055380018
tb_distance_b              0.070703640 3.128918e-02   0.009332911
quality_groupFair         -0.185006392 5.810751e-03  -0.196400452
quality_groupAbove Average 0.158313761 6.493042e-03   0.145580929
heated_area_b              0.203362696 2.488869e-03   0.198481825
acreage_b                  0.028038004 1.692797e-03   0.024718822
barrier_municipalityBarrier 0.338918466 1.197760e-01   0.106106348
stories_group2            -0.022264858 5.539108e-03  -0.033126349
stories_group3+           -0.057001985 1.437615e-02  -0.085192086
foundation_groupYes        0.052765655 6.853659e-03   0.039326223
                              0.5quant   0.975quant         mode          kld
intercept                  9.374969554 37.106105525  9.374969554 5.526813e-11
sale_year_c               -0.003650175 -0.003148797 -0.003650175 5.455871e-11
frontage_groupNone         2.184066604 29.915048818  2.184066604 5.526809e-11
frontage_groupLake         2.233327967 29.964313131  2.233327967 5.526809e-11
frontage_groupBay          2.515847499 30.246837219  2.515847499 5.526809e-11
frontage_groupOther        2.442238232 30.173222212  2.442238232 5.526804e-11
seawall_groupYes           0.160090623  0.184322230  0.160090624 5.437228e-11
```

```
cccl_distance_b              0.002244596  0.058669566  0.002246854 7.246814e-09
tb_distance_b                0.070694664  0.132121899  0.070693663 6.140282e-10
quality_groupFair           -0.185006482 -0.173611822 -0.185006482 5.481847e-11
quality_groupAbove Average   0.158313975  0.171045377  0.158313976 5.484799e-11
heated_area_b                0.203362863  0.208242623  0.203362865 6.048902e-11
acreage_b                    0.028037920  0.031357662  0.028037920 5.306403e-11
barrier_municipalityBarrier  0.337574020  0.579073847  0.337530798 6.666129e-08
stories_group2              -0.022264914 -0.011403045 -0.022264915 5.461657e-11
stories_group3+             -0.057002030 -0.028811629 -0.057002031 5.454029e-11
foundation_groupYes          0.052765681  0.066204939  0.052765681 5.455288e-11
```

```
# 2. Municipality random effects
municipality_output <- inla_spatial$summary.random$city_id

print("MUNICIPALITY RANDOM EFFECTS")
```

```
[1] "MUNICIPALITY RANDOM EFFECTS"
```

```
print(municipality_output)
```

```
   ID        mean          sd   0.025quant    0.5quant   0.975quant        mode
1   1  0.32510755 0.08768483  0.155658621  0.32402765  0.500498032  0.32397999
2   2 -0.28482684 0.13906186 -0.556521752 -0.28584282 -0.007420051 -0.28583516
3   3  0.10061855 0.07759381 -0.051300707  0.10026364  0.254481957  0.10024760
4   4  0.56489961 0.14999296  0.284189813  0.56005813  0.870414069  0.55921740
5   5 -0.08705562 0.06331019 -0.214046206 -0.08646788  0.036736775 -0.08645911
6   6  0.06039011 0.10508078 -0.147960034  0.06099361  0.265364102  0.06100612
7   7 -0.06404468 0.07132613 -0.206905944 -0.06325632  0.074473512 -0.06323665
8   8  0.01403855 0.07692749 -0.140189983  0.01501684  0.162885974  0.01505037
9   9  0.13216717 0.12643546 -0.113599043  0.13093675  0.384715673  0.13089924
10 10 -0.05508030 0.14532122 -0.345904982 -0.05343596  0.226729391 -0.05337208
11 11 -0.20198837 0.07985751 -0.362906066 -0.20059571 -0.048675789 -0.20052863
12 12  0.02887120 0.06296981 -0.097265438  0.02937254  0.152277145  0.02937787
13 13 -0.10486461 0.13678237 -0.377708476 -0.10365242  0.161329983 -0.10360591
14 14  0.06657161 0.13500996 -0.201529492  0.06720401  0.331129083  0.06720541
15 15 -0.03048082 0.10031780 -0.229989854 -0.02966877  0.164512769 -0.02964496
16 16 -0.08312578 0.07534067 -0.235886562 -0.08151526  0.060908302 -0.08143385
17 17 -0.24232429 0.07599891 -0.395625608 -0.24096755 -0.096422088 -0.24090456
18 18  0.21346247 0.13448278 -0.046849787  0.21177678  0.482957876  0.21169265
19 19 -0.25217333 0.13965761 -0.535775604 -0.24885955  0.013655926 -0.24862396
20 20  0.13830308 0.09281140 -0.043786419  0.13813028  0.321341461  0.13812118
21 21 -0.14185923 0.06682610 -0.276162421 -0.14103205 -0.012070772 -0.14101041
22 22 -0.10029489 0.08687209 -0.274095983 -0.09923207  0.067642794 -0.09919202
23 23  0.20356938 0.12626375 -0.039647804  0.20154104  0.457830998  0.20144677
24 24 -0.26843189 0.06626691 -0.403301168 -0.26694668 -0.141512835 -0.26687288
25 25 -0.32798109 0.09695067 -0.523775244 -0.32591891 -0.143295574 -0.32576020
```

```
26 26  0.31581391 0.16328746  0.008957997  0.31091124  0.647885583  0.31013229
27 27  0.08119510 0.12747557 -0.167336016  0.08018576  0.335268281  0.08015134
            kld
1  4.129890e-08
2  6.047656e-08
3  3.603300e-08
4  2.273359e-07
5  7.953305e-08
6  1.996722e-08
7  5.812477e-08
8  5.494767e-08
9  4.845792e-08
10 5.155722e-08
11 7.036973e-08
12 8.080094e-08
13 4.469877e-08
14 4.763669e-08
15 2.473671e-08
16 9.804144e-08
17 7.420431e-08
18 5.488665e-08
19 1.134414e-07
20 1.648539e-08
21 7.318700e-08
22 4.259576e-08
23 7.146277e-08
24 1.113650e-07
25 9.354466e-08
26 1.864474e-07
27 4.709523e-08
```

```
# 3. Spatial random field summary
spatial_output <- inla_spatial$summary.random$spatial_field

print("SPATIAL RANDOM FIELD")
```

```
[1] "SPATIAL RANDOM FIELD"
```

```
print(spatial_output)
```

```
       ID          mean         sd    0.025quant      0.5quant    0.975quant
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2       1  3.498048e-03 0.40126719 -0.7838151520  3.456801e-03  7.910466e-01
3       2  1.106917e-03 0.40676237 -0.7969305188  1.094722e-03  7.992139e-01
4       3 -2.227825e-05 0.41310932 -0.8104605874 -2.215488e-05  8.104153e-01
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177	176	2.484324e-03	0.40627208	-0.7945671475	2.460255e-03	7.996731e-01
178	177	-2.234801e-05	0.39829103	-0.7813859490	-2.222438e-05	7.813405e-01
179	178	6.303190e-03	0.42116261	-0.8198499241	6.253741e-03	8.327382e-01
180	179	-1.299461e-04	0.41730703	-0.8188995977	-1.287838e-04	8.186331e-01
181	180	-2.288938e-05	0.41877029	-0.8216431801	-2.273960e-05	8.215965e-01
182	181	1.094950e-01	0.41226452	-0.6981865131	1.090887e-01	9.194886e-01
183	182	-2.253457e-05	0.39754084	-0.7799603842	-2.239871e-05	7.799145e-01
184	183	1.740702e-01	0.38807060	-0.5863254513	1.737895e-01	9.360629e-01
185	184	1.417918e-04	0.40102742	-0.7866014394	1.398122e-04	7.868963e-01
186	185	1.906293e-02	0.41992767	-0.8044377128	1.893050e-02	8.433182e-01
187	186	3.827787e-03	0.41482928	-0.8099456772	3.798150e-03	8.177702e-01
188	187	6.926015e-02	0.39615308	-0.7071862246	6.900197e-02	8.471764e-01
189	188	1.273319e-02	0.42050307	-0.8119860938	1.263587e-02	8.380072e-01
190	189	6.382708e-02	0.33915275	-0.6015947600	6.391607e-02	7.287396e-01
191	190	-5.111969e-06	0.39858118	-0.7819457071	-5.233961e-06	7.819362e-01
192	191	1.621257e-01	0.37933132	-0.5810731584	1.618151e-01	9.070928e-01
193	192	6.491682e-05	0.40800495	-0.8005050109	6.372396e-05	8.006417e-01
194	193	1.675310e-02	0.40872241	-0.7847717347	1.664654e-02	8.188852e-01
195	194	-1.902957e-05	0.38963542	-0.7643799356	-1.895592e-05	7.643415e-01
196	195	9.031468e-03	0.41357336	-0.8021898640	8.971560e-03	8.205943e-01
197	196	1.145969e-03	0.41539487	-0.8139110908	1.130848e-03	8.162893e-01

198	197	8.034902e-03	0.39151170	-0.7598774214	7.990064e-03	7.762028e-01
199	198	3.772435e-04	0.41181206	-0.8076107609	3.718104e-04	8.083963e-01
200	199	4.070526e-03	0.40946923	-0.7993168251	4.030493e-03	8.076861e-01
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202	201	4.807560e-03	0.40430073	-0.7882432660	4.778900e-03	7.980216e-01
203	202	1.504578e-03	0.41345275	-0.8095764638	1.491401e-03	8.126608e-01
204	203	8.343712e-02	0.39858419	-0.6976943181	8.317505e-02	8.660593e-01
205	204	1.375817e-01	0.40683424	-0.6593321863	1.371675e-01	9.368516e-01
206	205	-7.350397e-05	0.39316422	-0.7713529569	-7.284424e-05	7.712022e-01
207	206	2.154238e-05	0.41779984	-0.8196838178	2.143453e-05	8.197275e-01
208	207	8.720065e-03	0.41918658	-0.8135130170	8.651824e-03	8.313421e-01
209	208	-1.633090e-05	0.39774074	-0.7802999065	-1.629851e-05	7.802671e-01
210	209	1.596134e-04	0.41076052	-0.8057822253	1.569924e-04	8.061164e-01
211	210	-2.270344e-05	0.40414508	-0.7928509502	-2.257863e-05	7.928048e-01
212	211	1.245689e-03	0.40497845	-0.7931938124	1.236086e-03	7.957399e-01
213	212	5.543244e-02	0.40442772	-0.7372384676	5.517994e-02	8.495414e-01
214	213	-2.295814e-05	0.42124298	-0.8264743382	-2.281220e-05	8.264276e-01
215	214	-1.515608e-03	0.41788210	-0.8214178177	-1.504624e-03	8.183240e-01
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217	216	-2.031238e-05	0.41610171	-0.8165032936	-2.019332e-05	8.164620e-01
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222	221	9.440997e-06	0.41948606	-0.8230372966	9.172959e-06	8.230577e-01
223	222	-2.304252e-05	0.42136976	-0.8267496698	-2.289237e-05	8.267027e-01
224	223	2.499611e-03	0.39823453	-0.7787304009	2.476506e-03	7.838614e-01
225	224	-9.627729e-04	0.40210975	-0.7898148806	-9.562509e-04	7.878522e-01
226	225	1.690857e-03	0.40673541	-0.7961742847	1.674949e-03	7.996467e-01
227	226	3.835165e-03	0.40959609	-0.7997181959	3.794519e-03	8.076204e-01
228	227	-2.775282e-03	0.39154122	-0.7707401602	-2.804517e-03	7.653561e-01
229	228	1.900486e-03	0.39793242	-0.7787093745	1.881497e-03	7.826187e-01
230	229	3.682438e-02	0.39174903	-0.7311548725	3.663888e-02	8.058604e-01
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235	234	-1.412363e-05	0.39976421	-0.7842341531	-1.411107e-05	7.842058e-01
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238	237	1.599748e-02	0.08532699	-0.1514215754	1.602883e-02	1.832380e-01
239	238	-1.725360e-02	0.10365018	-0.2205020262	-1.725519e-02	1.860038e-01
240	239	3.069731e-01	0.11961394	0.0725678201	3.069150e-01	5.417098e-01
241	240	2.501385e-01	0.10890002	0.0363405676	2.502242e-01	4.634490e-01
242	241	-2.233895e-01	0.09891724	-0.4173707659	-2.233871e-01	-2.942203e-02
243	242	-1.382630e+00	0.11909036	-1.6162982800	-1.382581e+00	-1.149242e+00
244	243	2.878487e-01	0.10732734	0.0773532566	2.878609e-01	4.982747e-01
245	244	1.297911e-01	0.14104046	-0.1467985967	1.297940e-01	4.063646e-01

246	245	-2.339784e-01	0.10064588	-0.4313078149	-2.339898e-01	-3.658406e-02
247	246	-3.303553e-01	0.12107650	-0.5677122435	-3.303796e-01	-9.286039e-02
248	247	-1.669030e-01	0.09489419	-0.3529265484	-1.669251e-01	1.924608e-02
249	248	2.663742e-01	0.11939751	0.0322796986	2.663577e-01	5.005628e-01
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252	251	1.626148e-01	0.12798041	-0.0883567554	1.626177e-01	4.135692e-01
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257	256	-3.641859e-01	0.10268106	-0.5655972177	-3.641660e-01	-1.628882e-01
258	257	9.773739e-01	0.10437246	0.7726880251	9.773775e-01	1.182039e+00
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264	263	9.481513e-02	0.10202639	-0.1050833585	9.474904e-02	2.950903e-01
265	264	-4.416822e-01	0.09563366	-0.6293540860	-4.416338e-01	-2.542857e-01
266	265	-6.605280e-01	0.09126358	-0.8394955939	-6.605265e-01	-4.815695e-01
267	266	-1.096737e-01	0.09372030	-0.2935279135	-1.096474e-01	7.403096e-02
268	267	-4.859493e-01	0.10813455	-0.6981069896	-4.859103e-01	-2.740134e-01
269	268	-2.598916e-01	0.08933074	-0.4350784062	-2.598868e-01	-8.473242e-02
270	269	-5.815668e-02	0.09756791	-0.2494704018	-5.816071e-02	1.331799e-01
271	270	-9.221407e-01	0.11836406	-1.1543357148	-9.221085e-01	-6.901286e-01
272	271	-5.057861e-01	0.09547944	-0.6931792919	-5.057276e-01	-3.187253e-01
273	272	-1.022544e+00	0.10049099	-1.2196942479	-1.022512e+00	-8.255771e-01
274	273	-1.344826e-01	0.11418320	-0.3584455238	-1.344629e-01	8.936836e-02
275	274	-2.818585e-01	0.09066329	-0.4596882413	-2.818428e-01	-1.041184e-01
276	275	-2.132096e-01	0.09276497	-0.3953343077	-2.131347e-01	-3.151154e-02
277	276	2.909552e-01	0.11947785	0.0565468070	2.909965e-01	5.251289e-01
278	277	5.506306e-02	0.11191069	-0.1644402537	5.507769e-02	2.744831e-01
279	278	-1.126291e-01	0.09444224	-0.2978973586	-1.126042e-01	7.249725e-02
280	279	-2.964195e-01	0.10617907	-0.5046999669	-2.963950e-01	-8.827852e-02
281	280	-7.377170e-01	0.08876782	-0.9119170992	-7.376702e-01	-5.637835e-01
282	281	-2.678156e-01	0.08874026	-0.4418609915	-2.678046e-01	-9.383302e-02
283	282	-1.863582e-01	0.10669943	-0.3957486900	-1.863036e-01	2.272192e-02
284	283	7.801104e-02	0.11270299	-0.1429433750	7.799264e-02	2.990700e-01
285	284	4.418543e-01	0.16766484	0.1130559587	4.418583e-01	7.706301e-01
286	285	-9.283122e-01	0.09303492	-1.1108828216	-9.282666e-01	-7.460008e-01
287	286	6.318406e-01	0.11425544	0.4078108866	6.318318e-01	8.559206e-01
288	287	-3.420046e-01	0.09322252	-0.5249740698	-3.419468e-01	-1.593634e-01
289	288	-4.670234e-01	0.10210013	-0.6673965930	-4.669671e-01	-2.669704e-01
290	289	-3.415006e-02	0.15423644	-0.3373549546	-3.389445e-02	2.676128e-01
291	290	-4.044570e-02	0.09932486	-0.2352707168	-4.042653e-02	1.542702e-01
292	291	7.355531e-02	0.10122944	-0.1249225018	7.354566e-02	2.720881e-01
293	292	-4.203862e-01	0.11489573	-0.6461190880	-4.202410e-01	-1.954780e-01

294	293	-2.662241e-01	0.11238502	-0.4865232917	-2.662526e-01	-4.576370e-02
295	294	-4.885965e-01	0.12683722	-0.7373271604	-4.885933e-01	-2.398836e-01
296	295	-2.880752e-01	0.13837338	-0.5603826460	-2.877458e-01	-1.763066e-02
297	296	-8.996958e-02	0.17024910	-0.4217017679	-9.082855e-02	2.465597e-01
298	297	8.877257e-01	0.10866846	0.6746273338	8.877258e-01	1.100824e+00
299	298	-5.376901e-01	0.10415484	-0.7420643250	-5.376445e-01	-3.335756e-01
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303	302	-1.562137e-01	0.10241118	-0.3571624415	-1.561704e-01	4.448896e-02
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305	304	-2.061062e-01	0.11948832	-0.4405437903	-2.060732e-01	2.814327e-02
306	305	2.863136e-01	0.10920070	0.0721263286	2.863314e-01	5.003996e-01
307	306	-9.433034e-01	0.11464299	-1.1682066509	-9.432699e-01	-7.185905e-01
308	307	-1.462170e-01	0.15251455	-0.4453185722	-1.462065e-01	1.528246e-01
309	308	-1.114230e-01	0.09919243	-0.3059992463	-1.114003e-01	8.302353e-02
310	309	2.486418e-02	0.12897530	-0.2283622615	2.497301e-02	2.774721e-01
311	310	-8.408905e-01	0.10097648	-1.0390788464	-8.408270e-01	-6.430635e-01
312	311	-6.269181e-02	0.10206458	-0.2628417822	-6.269030e-02	1.374495e-01
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316	315	1.549501e-01	0.10856494	-0.0581714126	1.550305e-01	3.676144e-01
317	316	-3.744442e-02	0.11647843	-0.2661012044	-3.735826e-02	1.907225e-01
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319	318	2.243644e-01	0.12715892	-0.0249530015	2.243500e-01	4.737638e-01
320	319	-7.577999e-01	0.09389878	-0.9421190714	-7.577327e-01	-5.738624e-01
321	320	-7.409821e-01	0.09779066	-0.9326854703	-7.410031e-01	-5.491603e-01
322	321	1.118509e-01	0.10419889	-0.0924588734	1.118379e-01	3.162342e-01
323	322	-4.480985e-01	0.08516183	-0.6151818721	-4.480705e-01	-2.811747e-01
324	323	-3.764246e-01	0.11521592	-0.6024779472	-3.763826e-01	-1.506101e-01
325	324	-4.058987e-01	0.18714837	-0.7753852439	-4.050321e-01	-4.127287e-02
326	325	-4.535203e-02	0.10159961	-0.2445966890	-4.534806e-02	1.538700e-01
327	326	-4.098212e-01	0.10494175	-0.6155631692	-4.098377e-01	-2.039855e-01
328	327	3.274507e-01	0.11031517	0.1111588337	3.274399e-01	5.438043e-01
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330	329	-8.335822e-02	0.09175371	-0.2633432613	-8.333662e-02	9.650386e-02
331	330	-6.125266e-01	0.09492026	-0.7987023612	-6.125110e-01	-4.264395e-01
332	331	4.730465e-02	0.14854186	-0.2439629798	4.729805e-02	3.386098e-01
333	332	6.898392e-01	0.10474243	0.4844418758	6.898377e-01	8.952456e-01
334	333	3.430809e-01	0.18987813	-0.0348740760	3.451566e-01	7.096689e-01
335	334	1.398087e-01	0.10571394	-0.0674855704	1.398048e-01	3.471255e-01
336	335	-5.055773e-01	0.12022241	-0.7412894477	-5.055990e-01	-2.697415e-01
337	336	-7.224383e-01	0.09319168	-0.9052308786	-7.224210e-01	-5.397445e-01
338	337	1.476315e-01	0.13920448	-0.1253124782	1.476220e-01	4.206295e-01
339	338	6.443167e-03	0.12359423	-0.2359428320	6.451742e-03	2.487804e-01
340	339	3.253758e-01	0.11719942	0.0954352662	3.254167e-01	5.550830e-01
341	340	-5.863969e-01	0.12539863	-0.8324591505	-5.863420e-01	-3.406471e-01

342	341	-1.754738e-01	0.11695268	-0.4049581240	-1.754352e-01	5.379116e-02
343	342	-5.905905e-01	0.08686082	-0.7608610298	-5.906128e-01	-4.201941e-01
344	343	-4.889269e-01	0.10524356	-0.6954712742	-4.888686e-01	-2.827146e-01
345	344	5.842400e-01	0.13079300	0.3276472472	5.842797e-01	8.406061e-01
346	345	-3.319598e-02	0.11306666	-0.2549014306	-3.320243e-02	1.885462e-01
347	346	1.339230e-01	0.10867158	-0.0792056594	1.339334e-01	3.469922e-01
348	347	-3.514753e-01	0.10735474	-0.5622599377	-3.513822e-01	-1.412196e-01
349	348	2.607474e-01	0.12060160	0.0241137256	2.607960e-01	4.971050e-01
350	349	-3.897167e-01	0.09280703	-0.5717184714	-3.897125e-01	-2.077386e-01
351	350	3.575113e-02	0.11054064	-0.1812183131	3.582149e-02	2.523205e-01
352	351	-1.534428e-01	0.10519326	-0.3596834151	-1.534658e-01	5.292882e-02
353	352	-1.638634e-01	0.08753089	-0.3356158267	-1.638244e-01	7.667279e-03
354	353	5.121959e-03	0.12464071	-0.2392511439	5.107574e-03	2.495769e-01
355	354	-1.325036e-01	0.09922728	-0.3271507254	-1.324808e-01	6.201388e-02
356	355	-3.112349e-01	0.13133023	-0.5688440049	-3.112073e-01	-5.378304e-02
357	356	-6.071099e-01	0.09285656	-0.7893532734	-6.070556e-01	-4.251756e-01
358	357	2.309149e-02	0.12713462	-0.2265910291	2.322329e-02	2.720249e-01
359	358	-8.626468e-01	0.08167829	-1.0230495707	-8.625632e-01	-7.027195e-01
360	359	-4.496751e-02	0.11915225	-0.2790053852	-4.483437e-02	1.883149e-01
361	360	-6.219173e-01	0.08965789	-0.7978448109	-6.218767e-01	-4.462213e-01
362	361	-4.046348e-01	0.09788064	-0.5967079901	-4.045897e-01	-2.128187e-01
363	362	2.766945e-01	0.11411323	0.0528511076	2.767200e-01	5.003926e-01
364	363	-4.197039e-01	0.10387453	-0.6233961139	-4.197042e-01	-2.160102e-01
365	364	2.629603e-01	0.13468876	-0.0012037417	2.629753e-01	5.270387e-01
366	365	-1.890869e-01	0.11014909	-0.4051697705	-1.890561e-01	2.682012e-02
367	366	-2.642488e-01	0.16303816	-0.5839533929	-2.642515e-01	5.547133e-02
368	367	-3.709111e-01	0.10109119	-0.5692094210	-3.708884e-01	-1.727422e-01
369	368	-9.557791e-02	0.10650986	-0.3048176438	-9.544503e-02	1.129075e-01
370	369	-4.000361e-01	0.13205503	-0.6591007540	-4.000071e-01	-1.411364e-01
371	370	2.530868e-01	0.13503335	-0.0117408930	2.530980e-01	5.178513e-01
372	371	-5.069533e-02	0.09810106	-0.2430823586	-5.069425e-02	1.416855e-01
373	372	4.087364e-02	0.10370429	-0.1629049994	4.101964e-02	2.438233e-01
374	373	2.155977e-01	0.09269842	0.0333060898	2.157785e-01	3.968638e-01
375	374	-2.748749e-01	0.11866996	-0.5075513659	-2.748856e-01	-4.213814e-02
376	375	-9.894505e-02	0.10220386	-0.2993580775	-9.894589e-02	1.014726e-01
377	376	1.863577e-01	0.10788645	-0.0253921020	1.864235e-01	3.977329e-01
378	377	1.150883e-01	0.10643417	-0.0937261370	1.151239e-01	3.237000e-01
379	378	-1.547727e-01	0.12410187	-0.3981936247	-1.547498e-01	8.851710e-02
380	379	-8.069502e-02	0.10175255	-0.2802242405	-8.069934e-02	1.188587e-01
381	380	7.788852e-01	0.09801330	0.5866380932	7.788996e-01	9.710504e-01
382	381	1.193571e-01	0.10843544	-0.0932400243	1.193410e-01	3.320458e-01
383	382	-2.215573e-01	0.09700666	-0.4119500907	-2.214990e-01	-3.149557e-02
384	383	-5.380050e-02	0.10675263	-0.2631589875	-5.379285e-02	1.555141e-01
385	384	1.945576e-01	0.09517077	0.0079220952	1.945614e-01	3.811708e-01
386	385	3.061910e-01	0.10448953	0.1012962976	3.061893e-01	5.110952e-01
387	386	-4.666094e-01	0.08471179	-0.6330628276	-4.664901e-01	-3.008341e-01
388	387	6.316445e-01	0.10968950	0.4166072883	6.316267e-01	8.467823e-01
389	388	-3.368060e-03	0.11215719	-0.2233966550	-3.335034e-03	2.164725e-01

390	389	-4.471559e-01	0.09481621	-0.6332658027	-4.470938e-01	-2.613991e-01
391	390	-4.347183e-01	0.10077592	-0.6324632126	-4.346725e-01	-2.372339e-01
392	391	-5.469516e-01	0.09452779	-0.7324773513	-5.468952e-01	-3.617465e-01
393	392	1.611327e-01	0.10306353	-0.0409951973	1.611422e-01	3.632064e-01
394	393	-7.181028e-01	0.10622553	-0.9264762358	-7.180802e-01	-5.098582e-01
395	394	-2.743077e-01	0.11248803	-0.4949716890	-2.742796e-01	-5.380412e-02
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397	396	2.023064e-01	0.11405918	-0.0217303978	2.024357e-01	4.256079e-01
398	397	-1.537625e-01	0.10481535	-0.3593475270	-1.537460e-01	5.172863e-02
399	398	-8.402792e-01	0.09230104	-1.0213739273	-8.402463e-01	-6.593720e-01
400	399	-2.705294e-02	0.14902859	-0.3193948015	-2.701675e-02	2.650829e-01
401	400	-2.781421e-01	0.11189997	-0.4975994526	-2.781327e-01	-5.873854e-02
402	401	-1.638654e-01	0.10696931	-0.3736328077	-1.638644e-01	4.589662e-02
403	402	-6.405155e-01	0.09031921	-0.8177947105	-6.404557e-01	-4.635768e-01
404	403	-5.433187e-01	0.09802302	-0.7355355441	-5.433187e-01	-3.511020e-01
405	404	-8.911846e-02	0.09649586	-0.2784740467	-8.907176e-02	9.997174e-02
406	405	2.812300e-01	0.10751504	0.0703506456	2.812467e-01	4.920140e-01
407	406	-7.863458e-01	0.10803504	-0.9982678746	-7.863217e-01	-5.745611e-01
408	407	2.248050e-03	0.14726179	-0.2864285754	2.211952e-03	2.911303e-01
409	408	-5.968583e-01	0.09363488	-0.7805040722	-5.968474e-01	-4.132748e-01
410	409	-2.156296e-01	0.09444227	-0.4010819827	-2.155399e-01	-3.068711e-02
411	410	1.919856e-01	0.12265587	-0.0485452343	1.919874e-01	4.325061e-01
412	411	-3.041322e-01	0.10070701	-0.5015312018	-3.041617e-01	-1.065653e-01
413	412	-6.088862e-02	0.11298900	-0.2825690222	-6.084912e-02	1.605669e-01
414	413	-6.248870e-01	0.09595557	-0.8132563963	-6.248173e-01	-4.369145e-01
415	414	-8.683541e-02	0.16669273	-0.4116179858	-8.768400e-02	2.426868e-01
416	415	-4.404243e-01	0.10228033	-0.6410572730	-4.404011e-01	-2.399231e-01
417	416	-3.319766e-01	0.13213605	-0.5913301551	-3.318919e-01	-7.310341e-02
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423	422	3.435629e-01	0.13330260	0.0818634061	3.436635e-01	6.046904e-01
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428	427	-8.605226e-01	0.10096603	-1.0585546489	-8.605071e-01	-6.625793e-01
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430	429	-1.961131e-01	0.09699179	-0.3863148576	-1.961106e-01	-5.925930e-03
431	430	-1.465070e-01	0.10538299	-0.3531729149	-1.465016e-01	6.012808e-02
432	431	6.608570e-02	0.13385808	-0.1966472926	6.617141e-02	3.283324e-01
433	432	1.381797e-01	0.15557056	-0.1666491132	1.380891e-01	4.435245e-01
434	433	-1.083656e+00	0.09985127	-1.2795771224	-1.083616e+00	-8.879678e-01
435	434	-2.424112e-01	0.09363935	-0.4261238594	-2.423787e-01	-5.888349e-02
436	435	8.441656e-03	0.09303499	-0.1740808000	8.466078e-03	1.908251e-01
437	436	-1.743590e-01	0.10320874	-0.3769302330	-1.742952e-01	2.784944e-02

438	437	-3.863520e-01	0.09159497	-0.5661774753	-3.862774e-01	-2.069512e-01
439	438	-2.213866e-01	0.09148999	-0.4009871783	-2.213205e-01	-4.216263e-02
440	439	-4.313755e-01	0.09228644	-0.6124078740	-4.313527e-01	-2.504725e-01
441	440	1.837246e-01	0.11299809	-0.0376187755	1.836105e-01	4.057187e-01
442	441	-9.502710e-02	0.10422313	-0.2993994682	-9.502813e-02	1.093511e-01
443	442	1.629026e-01	0.11585944	-0.0643247992	1.629099e-01	3.900888e-01
444	443	-1.045343e+00	0.09338813	-1.2286005002	-1.045298e+00	-8.623407e-01
445	444	-9.965259e-01	0.09545924	-1.1838746685	-9.964723e-01	-8.094817e-01
446	445	-6.958434e-01	0.17157109	-1.0323160450	-6.958335e-01	-3.594272e-01
447	446	-7.058664e-01	0.08547881	-0.8736646553	-7.058036e-01	-5.384254e-01
448	447	-6.073534e-02	0.08905604	-0.2354887939	-6.069542e-02	1.137909e-01
449	448	-7.062089e-03	0.08797808	-0.1796673137	-7.032433e-03	1.653743e-01
450	449	-1.968169e-01	0.11511411	-0.4224578512	-1.968487e-01	2.900425e-02
451	450	6.701480e-01	0.14164652	0.3924416319	6.701278e-01	9.479688e-01
452	451	7.259789e-01	0.13039190	0.4703861558	7.259398e-01	9.817943e-01
453	452	-6.796176e-02	0.09550289	-0.2552634848	-6.795496e-02	1.193012e-01
454	453	-2.347989e-01	0.11041143	-0.4513160598	-2.348055e-01	-1.824364e-02
455	454	1.107518e+00	0.09858358	0.9141479312	1.107533e+00	1.300801e+00
456	455	-4.009736e-01	0.10587720	-0.6086341254	-4.009600e-01	-1.933907e-01
457	456	6.970063e-02	0.09389455	-0.1144156914	6.969762e-02	2.538341e-01
458	457	4.730520e-01	0.11265403	0.2521631914	4.730428e-01	6.939933e-01
459	458	1.756615e-01	0.13304386	-0.0852203974	1.756563e-01	4.365727e-01
460	459	-3.839904e-01	0.12275514	-0.6248200260	-3.839511e-01	-1.433846e-01
461	460	8.562052e-01	0.10974709	0.6409949888	8.562025e-01	1.071431e+00
462	461	1.850205e-01	0.11776822	-0.0460479617	1.850659e-01	4.158309e-01
463	462	-2.430180e-01	0.10026984	-0.4396635653	-2.430106e-01	-4.641466e-02
464	463	-1.803227e-01	0.11958467	-0.4152136756	-1.801902e-01	5.381552e-02
465	464	-4.346783e-02	0.10378348	-0.2470019653	-4.346067e-02	1.600254e-01
466	465	2.553166e-01	0.11309858	0.0335593481	2.553087e-01	4.771190e-01
467	466	7.442295e-01	0.09890667	0.5502404380	7.442400e-01	9.381590e-01
468	467	-2.393883e-01	0.09271022	-0.4212815933	-2.393546e-01	-5.768631e-02
469	468	8.181281e-02	0.10873630	-0.1315751961	8.186881e-02	2.948826e-01
470	469	-6.094549e-01	0.10384018	-0.8130553975	-6.094647e-01	-4.057982e-01
471	470	-2.807943e-01	0.12128484	-0.5186974263	-2.807698e-01	-4.303038e-02
472	471	-4.211854e-01	0.11710245	-0.6508188215	-4.211846e-01	-1.915572e-01
473	472	-3.996666e-01	0.09904764	-0.5940622594	-3.996089e-01	-2.055995e-01
474	473	-7.398688e-02	0.11026406	-0.2902054068	-7.398824e-02	1.422395e-01
475	474	-6.731649e-01	0.10567805	-0.8804856431	-6.731341e-01	-4.660194e-01
476	475	-9.350549e-01	0.08570281	-1.1032826354	-9.349953e-01	-7.671662e-01
477	476	7.852918e-01	0.14037938	0.5100665749	7.852630e-01	1.060681e+00
478	477	-2.886300e-01	0.12234282	-0.5285483140	-2.886268e-01	-4.872947e-02
479	478	-2.441574e-01	0.09764868	-0.4356746321	-2.441515e-01	-5.267366e-02
480	479	-1.750602e-01	0.09930258	-0.3698099485	-1.750548e-01	1.965856e-02
481	480	-8.055143e-01	0.09094824	-0.9839556065	-8.054798e-01	-6.272696e-01
482	481	9.579399e-02	0.13189683	-0.1630698637	9.586993e-02	3.542261e-01
483	482	-2.575032e-01	0.10362871	-0.4608305151	-2.574620e-01	-5.441064e-02
484	483	1.444342e-01	0.11167260	-0.0746768236	1.444779e-01	3.632969e-01
485	484	5.101145e-01	0.11669935	0.2813020058	5.101004e-01	7.390072e-01

486	485	-2.275821e-01	0.13421662	-0.4911149608	-2.274662e-01	3.529199e-02
487	486	-3.716652e-01	0.10347691	-0.5747643651	-3.715998e-01	-1.689380e-01
488	487	-3.828648e-01	0.09979830	-0.5787076801	-3.828150e-01	-1.873058e-01
489	488	2.923381e-01	0.13843600	0.0210216055	2.922836e-01	5.639650e-01
490	489	5.634082e-01	0.09894633	0.3693283873	5.634232e-01	7.574026e-01
491	490	6.424193e-01	0.11855702	0.4097785647	6.424735e-01	8.747516e-01
492	491	-4.350621e-01	0.09717881	-0.6256706252	-4.350464e-01	-2.445435e-01
493	492	-1.167524e-01	0.12576371	-0.3633847160	-1.167473e-01	1.298514e-01
494	493	5.666389e-01	0.09443470	0.3813195171	5.666871e-01	7.516841e-01
495	494	-1.175989e-01	0.10192161	-0.3176171510	-1.175446e-01	8.211014e-02
496	495	-2.982645e-01	0.10146806	-0.4972582811	-2.982578e-01	-9.930913e-02
497	496	-3.696769e-01	0.10627381	-0.5781492178	-3.696506e-01	-1.613543e-01
498	497	-2.465818e-01	0.09520663	-0.4333121163	-2.465706e-01	-5.991536e-02
499	498	-4.803629e-01	0.11311772	-0.7023262672	-4.803130e-01	-2.586830e-01
500	499	-1.854304e-01	0.15971167	-0.4998878839	-1.849896e-01	1.265405e-01
501	500	-6.632934e-01	0.10226278	-0.8639223862	-6.632594e-01	-4.628581e-01
502	501	-1.503542e-01	0.12578482	-0.3973858737	-1.502276e-01	9.595807e-02
503	502	3.201737e-01	0.11436068	0.0959147697	3.201744e-01	5.444282e-01
504	503	-6.375712e-01	0.10821975	-0.8498862382	-6.375355e-01	-4.254595e-01
505	504	-3.161573e-01	0.10826465	-0.5287045499	-3.160719e-01	-1.040955e-01
506	505	-4.277036e-02	0.10454307	-0.2478772413	-4.273480e-02	1.621342e-01
507	506	1.110947e-01	0.10858521	-0.1016922607	1.110305e-01	3.242483e-01
508	507	-9.538705e-02	0.09658474	-0.2846542085	-9.543677e-02	9.416291e-02
509	508	-7.528521e-01	0.09903205	-0.9471003777	-7.528353e-01	-5.586998e-01
510	509	1.048313e-02	0.13758902	-0.2592778984	1.045780e-02	2.803887e-01
511	510	9.540810e-01	0.10743288	0.7433848595	9.540878e-01	1.164739e+00
512	511	3.346646e-01	0.11638706	0.1064454010	3.346618e-01	5.628998e-01
513	512	4.679098e-01	0.12734986	0.2181223334	4.679316e-01	7.175729e-01
514	513	7.743305e-01	0.09634885	0.5853636222	7.743381e-01	9.632541e-01
515	514	1.430668e-02	0.11180502	-0.2052116110	1.440218e-02	2.332820e-01
516	515	-5.075181e-01	0.15367638	-0.8090340125	-5.074634e-01	-2.063135e-01
517	516	2.319210e-01	0.14989703	-0.0618871463	2.318705e-01	5.260166e-01
518	517	-5.450151e-01	0.09834378	-0.7380230038	-5.449596e-01	-3.523229e-01
519	518	-6.519300e-01	0.10036638	-0.8488883149	-6.518794e-01	-4.552601e-01
520	519	-4.414263e-01	0.08837433	-0.6148027168	-4.413979e-01	-2.682113e-01
521	520	2.686365e-01	0.14525476	-0.0158136064	2.684882e-01	5.539310e-01
522	521	4.070010e-01	0.14145842	0.1296881397	4.069701e-01	6.844901e-01
523	522	1.616942e-01	0.10179718	-0.0379074877	1.616885e-01	3.613284e-01
524	523	-2.988615e-01	0.10527912	-0.5055180740	-2.987882e-01	-9.262170e-02
525	524	6.018684e-02	0.09674024	-0.1295252977	6.018694e-02	2.498984e-01
526	525	-3.429359e-02	0.10644614	-0.2431790115	-3.424064e-02	1.742903e-01
527	526	-1.386730e-01	0.09397120	-0.3229784407	-1.386656e-01	4.558989e-02
528	527	-1.033117e-01	0.10554206	-0.3104226483	-1.032599e-01	1.035049e-01
529	528	5.885082e-01	0.11674944	0.3596345442	5.884812e-01	8.175355e-01
530	529	-4.878155e-02	0.09802100	-0.2409540821	-4.879703e-02	1.434792e-01
531	530	-8.944247e-02	0.10279481	-0.2909732172	-8.945881e-02	1.121812e-01
532	531	-4.031644e-01	0.09059317	-0.5810561108	-4.030779e-01	-2.257653e-01
533	532	-7.361487e-02	0.11606362	-0.3012143563	-7.361463e-02	1.539832e-01

534	533	-2.531683e-01	0.11427543	-0.4772437173	-2.531721e-01	-2.907131e-02
535	534	1.542116e-01	0.11092804	-0.0630737921	1.541221e-01	3.720056e-01
536	535	4.907083e-02	0.10768341	-0.1624386609	4.919179e-02	2.598930e-01
537	536	-4.305449e-01	0.11579764	-0.6575492108	-4.305687e-01	-2.034051e-01
538	537	-2.154874e-01	0.09516851	-0.4020860241	-2.154953e-01	-2.884384e-02
539	538	-2.934782e-01	0.09707300	-0.4838349924	-2.934776e-01	-1.031247e-01
540	539	2.693230e-01	0.12101066	0.0320301796	2.693222e-01	5.066202e-01
541	540	-6.935735e-01	0.09479568	-0.8795181828	-6.935542e-01	-5.077383e-01
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543	542	-1.349919e-01	0.12517505	-0.3808050713	-1.348706e-01	1.101329e-01
544	543	7.571955e-02	0.11998073	-0.1598802274	7.583236e-02	3.106802e-01
545	544	4.208600e-01	0.11606281	0.1933388729	4.208304e-01	6.485500e-01
546	545	-3.766328e-01	0.07699047	-0.5277400598	-3.765890e-01	-2.257749e-01
547	546	-1.663745e-01	0.10344381	-0.3691991609	-1.663831e-01	3.649917e-02
548	547	-5.892998e-01	0.07920415	-0.7447502828	-5.892514e-01	-4.341240e-01
549	548	-5.075705e-01	0.10406909	-0.7118288865	-5.075057e-01	-3.036807e-01
550	549	4.187385e-01	0.13871926	0.1467798384	4.187051e-01	6.908877e-01
551	550	-4.249209e-01	0.11680967	-0.6543781809	-4.247859e-01	-1.962306e-01
552	551	-2.591380e-01	0.09838130	-0.4520742276	-2.591344e-01	-6.622190e-02
553	552	-1.720932e-01	0.09598038	-0.3604078883	-1.720568e-01	1.601427e-02
554	553	6.524769e-02	0.11155013	-0.1535714128	6.527426e-02	2.839155e-01
555	554	-1.902846e-01	0.09945155	-0.3852740163	-1.902950e-01	4.763705e-03
556	555	3.434356e-01	0.10963714	0.1284040013	3.434482e-01	5.583954e-01
557	556	1.627859e-01	0.11654308	-0.0657799238	1.627971e-01	3.912878e-01
558	557	-7.570815e-01	0.08933242	-0.9324718024	-7.570053e-01	-5.821248e-01
559	558	3.822877e-01	0.16872417	0.0495765803	3.828616e-01	7.117678e-01
560	559	2.931929e-01	0.10349784	0.0902331586	2.931941e-01	4.961458e-01
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562	561	1.716007e-01	0.12712825	-0.0780066953	1.717078e-01	4.205995e-01
563	562	-2.333141e-01	0.10357708	-0.4365152080	-2.332827e-01	-3.029217e-02
564	563	-2.192746e-02	0.11048095	-0.2386959310	-2.188561e-02	1.946028e-01
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566	565	-1.277218e-01	0.18880769	-0.5032693549	-1.257680e-01	2.370973e-01
567	566	5.761258e-01	0.11439814	0.3517720791	5.761330e-01	8.004391e-01
568	567	1.185323e-02	0.10827639	-0.2005775921	1.189008e-02	2.240743e-01
569	568	-2.639592e-01	0.11215643	-0.4839335455	-2.639444e-01	-4.406901e-02
570	569	2.426872e-02	0.11292975	-0.1972998182	2.430975e-02	2.456037e-01
571	570	-7.303872e-01	0.09669703	-0.9201319311	-7.303442e-01	-5.408875e-01
572	571	-2.788222e-01	0.11832824	-0.5108938625	-2.788105e-01	-4.681709e-02
573	572	-6.817626e-02	0.11027609	-0.2844916617	-6.816272e-02	1.480626e-01
574	573	2.007598e-01	0.11330545	-0.0216255347	2.008287e-01	4.227532e-01
575	574	-4.073791e-01	0.09599575	-0.5956470983	-4.073707e-01	-2.191585e-01
576	575	-2.045787e-01	0.12178640	-0.4434322886	-2.045676e-01	3.421192e-02
577	576	-1.098318e+00	0.09114788	-1.2771846030	-1.098273e+00	-9.197099e-01
578	577	-3.558499e-01	0.10176178	-0.5554473880	-3.558332e-01	-1.563474e-01
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587	586	-2.866062e-01	0.09220760	-0.4674564242	-2.865958e-01	-1.058148e-01
588	587	6.807029e-01	0.12362770	0.4382417383	6.807144e-01	9.230984e-01
589	588	-1.062572e+00	0.10676903	-1.2720598104	-1.062531e+00	-8.533183e-01
590	589	4.011466e-02	0.13312251	-0.2209991475	4.013831e-02	3.010939e-01
591	590	6.000674e-02	0.11467117	-0.1649520821	6.003914e-02	2.847812e-01
592	591	-1.103268e+00	0.08282341	-1.2657869841	-1.103231e+00	-9.409610e-01
593	592	-5.007932e-02	0.10025717	-0.2467156359	-5.006603e-02	1.464813e-01
594	593	1.804047e-01	0.10846074	-0.0324743394	1.804721e-01	3.929005e-01
595	594	9.797349e-03	0.12398088	-0.2332044605	9.755784e-03	2.530355e-01
596	595	-1.058146e+00	0.08852673	-1.2318505090	-1.058108e+00	-8.846582e-01
597	596	-5.587968e-01	0.11318705	-0.7808582009	-5.587588e-01	-3.369517e-01
598	597	-4.071775e-01	0.08052704	-0.5650309297	-4.071990e-01	-2.492024e-01
599	598	-5.931639e-01	0.10123050	-0.7917301038	-5.931431e-01	-3.947158e-01
600	599	-2.480211e-01	0.09960140	-0.4433038061	-2.480320e-01	-5.267641e-02
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602	601	1.338956e-01	0.10531493	-0.0726863465	1.339129e-01	3.403794e-01
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606	605	-1.085453e-01	0.10517808	-0.3149470677	-1.084915e-01	9.755099e-02
607	606	1.586275e-01	0.10201006	-0.0413614597	1.586105e-01	3.587128e-01
608	607	-1.319370e-02	0.09721847	-0.2037441559	-1.323696e-02	1.776029e-01
609	608	-5.371030e-03	0.10071468	-0.2028449492	-5.378778e-03	1.921469e-01
610	609	5.872296e-02	0.11276807	-0.1623878486	5.871600e-02	2.798733e-01
611	610	-1.604049e-01	0.11696272	-0.3898481848	-1.603746e-01	6.886557e-02
612	611	-5.275176e-01	0.08761691	-0.6993774201	-5.275013e-01	-3.557510e-01
613	612	6.953850e-02	0.12986104	-0.1852530552	6.958779e-02	3.240496e-01
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616	615	-5.244368e-01	0.11180015	-0.7438303671	-5.243801e-01	-3.053653e-01
617	616	6.397553e-01	0.12092683	0.4026743534	6.397348e-01	8.769527e-01
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621	620	-4.623931e-02	0.10535709	-0.2528488860	-4.623753e-02	1.603601e-01
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623	622	5.393229e-02	0.11784302	-0.1771558925	5.393341e-02	2.850141e-01
624	623	6.692103e-03	0.09645631	-0.1828405767	6.827879e-03	1.954539e-01
625	624	-3.598979e-01	0.10848594	-0.5727592016	-3.598529e-01	-1.472929e-01
626	625	6.866500e-02	0.10500968	-0.1373721832	6.870725e-02	2.744620e-01
627	626	-3.311204e-01	0.09920768	-0.5257224845	-3.310987e-01	-1.366413e-01
628	627	3.504720e-01	0.13771870	0.0804001985	3.504771e-01	6.205152e-01
629	628	-3.061682e-01	0.10966091	-0.5211045086	-3.062054e-01	-9.102125e-02

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631	630	5.590850e-01	0.10615331	0.3506935063	5.591661e-01	7.670145e-01
632	631	1.835871e-01	0.11305718	-0.0381419911	1.835979e-01	4.052547e-01
633	632	7.256079e-02	0.17252145	-0.2635823948	7.169519e-02	4.135415e-01
634	633	-1.192858e-01	0.11879557	-0.3522863263	-1.192694e-01	1.136215e-01
635	634	-9.417051e-01	0.09053513	-1.1193431670	-9.416701e-01	-7.642662e-01
636	635	2.167903e-01	0.11243901	-0.0037237118	2.167981e-01	4.372599e-01
637	636	-3.791211e-01	0.10413547	-0.5833363505	-3.791177e-01	-1.749249e-01
638	637	-8.660966e-01	0.09027755	-1.0432166621	-8.660646e-01	-6.891590e-01
639	638	-1.130288e-01	0.09764254	-0.3045515667	-1.130137e-01	7.840805e-02
640	639	2.053416e-01	0.13961019	-0.0683804533	2.053246e-01	4.791600e-01
641	640	-3.175815e-02	0.11347455	-0.2543036967	-3.174782e-02	1.907285e-01
642	641	-1.778631e-01	0.11081883	-0.3951663689	-1.778663e-01	3.945812e-02
643	642	-5.609626e-01	0.10350283	-0.7640963257	-5.609023e-01	-3.581721e-01
644	643	-3.215072e-01	0.08771193	-0.4935325224	-3.214986e-01	-1.495314e-01
645	644	-8.988597e-02	0.09746654	-0.2811841108	-8.982571e-02	1.010694e-01
646	645	-4.681529e-02	0.10869137	-0.2599714010	-4.680902e-02	1.663051e-01
647	646	-7.647163e-02	0.10426974	-0.2809007615	-7.648589e-02	1.280387e-01
648	647	-2.919046e-01	0.10108917	-0.4903420821	-2.918311e-01	-9.388514e-02
649	648	-5.661308e-01	0.10736139	-0.7767066557	-5.661145e-01	-3.556479e-01
650	649	-1.840700e-01	0.09687431	-0.3740255703	-1.840748e-01	5.912333e-03
651	650	2.647205e-01	0.11156745	0.0458586629	2.647494e-01	4.834180e-01
652	651	6.348256e-02	0.13173001	-0.1947703121	6.345409e-02	3.218979e-01
653	652	2.263968e-01	0.10813628	0.0143824511	2.263843e-01	4.384826e-01
654	653	-3.987349e-01	0.08796500	-0.5712646500	-3.987225e-01	-2.262756e-01
655	654	-2.597501e-01	0.10764773	-0.4708270194	-2.597557e-01	-4.864202e-02
656	655	1.871219e-02	0.12290390	-0.2223339167	1.872479e-02	2.596866e-01
657	656	-1.513674e-01	0.13307311	-0.4124498688	-1.513294e-01	1.094991e-01
658	657	-6.464596e-01	0.09414231	-0.8311292289	-6.464377e-01	-4.619147e-01
659	658	4.150057e-01	0.17107821	0.0782365172	4.153769e-01	7.496780e-01
660	659	-6.358050e-01	0.10337060	-0.8385780645	-6.357811e-01	-4.331683e-01
661	660	-4.869058e-01	0.10403115	-0.6909537933	-4.868886e-01	-2.829553e-01
662	661	-4.235240e-01	0.09472762	-0.6094164193	-4.234760e-01	-2.379044e-01
663	662	-3.034471e-01	0.11231537	-0.5236706637	-3.034548e-01	-8.318019e-02
664	663	-3.974302e-01	0.09648163	-0.5867151788	-3.973983e-01	-2.083266e-01
665	664	3.050311e-01	0.10286340	0.1032947143	3.050394e-01	5.067203e-01
666	665	-6.291963e-01	0.10110716	-0.8277902235	-6.290809e-01	-4.312584e-01
667	666	5.487482e-01	0.11344583	0.3261295252	5.488025e-01	7.710575e-01
668	667	-4.134091e-02	0.17319163	-0.3808923655	-4.136539e-02	2.983498e-01
669	668	-5.459900e-02	0.16972747	-0.3881561074	-5.435134e-02	2.775552e-01
670	669	2.051006e-02	0.09686980	-0.1696771069	2.059077e-02	2.102383e-01
671	670	7.029830e-02	0.15697857	-0.2383006041	7.055665e-02	3.774356e-01
672	671	-5.509871e-01	0.11529925	-0.7771206788	-5.509733e-01	-3.249323e-01
673	672	1.881677e-01	0.13899769	-0.0842823995	1.881210e-01	4.608840e-01
674	673	1.808768e-01	0.11852144	-0.0516392745	1.809119e-01	4.131931e-01
675	674	-1.538922e-01	0.11189604	-0.3733148966	-1.538935e-01	6.553789e-02
676	675	-1.018477e-01	0.11152348	-0.3205265723	-1.018545e-01	1.168700e-01
677	676	1.280579e-01	0.12987144	-0.1269144344	1.281605e-01	3.824478e-01

678	677	3.625572e-01	0.17209253	0.0212807266	3.638974e-01	6.963955e-01
679	678	-4.075796e-01	0.09912058	-0.6020860736	-4.075314e-01	-2.133471e-01
680	679	-4.124158e-01	0.10868239	-0.6257675233	-4.123361e-01	-1.995174e-01
681	680	-2.849758e-01	0.09287839	-0.4673050458	-2.849057e-01	-1.030452e-01
682	681	1.521361e-01	0.14286731	-0.1278467977	1.520714e-01	4.324875e-01
683	682	3.953788e-02	0.09734225	-0.1513015302	3.951292e-02	2.305192e-01
684	683	-5.416920e-01	0.13933171	-0.8150271498	-5.416529e-01	-2.685793e-01
685	684	-7.214250e-01	0.09622479	-0.9102740653	-7.213689e-01	-5.328956e-01
686	685	-2.546594e-01	0.10595057	-0.4627247098	-2.545544e-01	-4.719082e-02
687	686	2.414559e-01	0.16343570	-0.0788872001	2.414038e-01	5.620957e-01
688	687	2.643769e-01	0.11397545	0.0408778500	2.643764e-01	4.878785e-01
689	688	1.473860e-01	0.10848423	-0.0654570623	1.474235e-01	3.600152e-01
690	689	-3.200085e-01	0.10982158	-0.5354480655	-3.199785e-01	-1.047400e-01
691	690	-7.786822e-01	0.09293603	-0.9610118956	-7.786514e-01	-5.965289e-01
692	691	1.136917e-01	0.13239020	-0.1462256303	1.137969e-01	3.730115e-01
693	692	1.201912e-02	0.12827411	-0.2394500042	1.199274e-02	2.636384e-01
694	693	-5.097450e-01	0.10014611	-0.7061099636	-5.097507e-01	-3.133474e-01
695	694	-1.157833e-01	0.12816842	-0.3672315603	-1.157517e-01	1.354848e-01
696	695	9.855395e-02	0.11164934	-0.1206174158	9.863544e-02	3.172623e-01
697	696	5.353171e-02	0.10737531	-0.1570349179	5.353223e-02	2.640954e-01
698	697	-7.911996e-02	0.08520140	-0.2462936106	-7.908889e-02	8.787679e-02
699	698	-3.278707e-02	0.10454751	-0.2378084352	-3.278494e-02	1.722221e-01
700	699	-2.042194e-01	0.10158122	-0.4034333923	-2.042128e-01	-5.043342e-03
701	700	-3.782006e-01	0.11712533	-0.6079169384	-3.781878e-01	-1.485572e-01
702	701	-5.968824e-02	0.11177791	-0.2789005776	-5.968139e-02	1.594851e-01
703	702	2.280172e-01	0.09999797	0.0319435189	2.280107e-01	4.241280e-01
704	703	4.442384e-01	0.11530874	0.2180936275	4.442462e-01	6.703385e-01
705	704	9.680049e-02	0.11744475	-0.1334783535	9.679188e-02	3.271278e-01
706	705	-7.750965e-01	0.08939021	-0.9506759487	-7.749938e-01	-6.001003e-01
707	706	2.616698e-01	0.09255762	0.0800932610	2.616932e-01	4.431136e-01
708	707	1.900066e-01	0.12088768	-0.0470607979	1.900107e-01	4.270504e-01
709	708	-3.818641e-01	0.11114374	-0.5997803179	-3.818835e-01	-1.638374e-01
710	709	-1.761255e-01	0.10565554	-0.3834182743	-1.760878e-01	3.095297e-02
711	710	-3.643755e-01	0.09807877	-0.5566920915	-3.643809e-01	-1.720283e-01
712	711	-1.959343e-01	0.10462627	-0.4011613591	-1.959134e-01	9.173873e-03
713	712	-2.292546e-02	0.09759096	-0.2143158434	-2.292138e-02	1.684416e-01
714	713	-1.957607e-01	0.10697350	-0.4055907968	-1.957398e-01	1.394996e-02
715	714	-5.171643e-01	0.12339798	-0.7590466964	-5.172000e-01	-2.750787e-01
716	715	-2.564922e-01	0.10808959	-0.4684268854	-2.565158e-01	-4.442266e-02
717	716	-2.914884e-02	0.10698273	-0.2389492675	-2.914423e-02	1.806253e-01
718	717	-1.049608e-01	0.13954143	-0.3786629561	-1.049374e-01	1.686079e-01
719	718	7.735537e-01	0.12049845	0.5370880900	7.736154e-01	1.009668e+00
720	719	-4.434831e-01	0.08867319	-0.6173437222	-4.434918e-01	-2.695734e-01
721	720	-3.006173e-01	0.11163579	-0.5195890185	-3.005957e-01	-8.176885e-02
722	721	-2.577796e-01	0.11005765	-0.4736531163	-2.577599e-01	-4.201797e-02
723	722	1.066951e-01	0.11537226	-0.1195421096	1.066945e-01	3.329357e-01
724	723	-2.389573e-01	0.08611830	-0.4079197081	-2.389257e-01	-7.017464e-02
725	724	-6.958003e-01	0.09122135	-0.8748639353	-6.957354e-01	-5.171067e-01

726	725	1.265637e-01	0.12265732	-0.1142526445	1.266649e-01	3.668052e-01
727	726	-1.873948e-01	0.11780995	-0.4184466597	-1.873830e-01	4.359035e-02
728	727	-6.883567e-01	0.11915889	-0.9219722266	-6.883814e-01	-4.545995e-01
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730	729	-7.570539e-01	0.10520998	-0.9635274175	-7.569978e-01	-5.508994e-01
731	730	-8.556815e-01	0.09938402	-1.0508005501	-8.555991e-01	-6.610308e-01
732	731	-3.375796e-02	0.09948503	-0.2291491398	-3.365146e-02	1.610287e-01
733	732	3.721213e-02	0.11269510	-0.1837361734	3.719768e-02	2.582426e-01
734	733	-1.010744e+00	0.08315469	-1.1739902548	-1.010680e+00	-8.478583e-01
735	734	1.228394e-01	0.11016656	-0.0931252874	1.228105e-01	3.389683e-01
736	735	9.535315e-02	0.12074581	-0.1414131150	9.534890e-02	3.321436e-01
737	736	-8.565069e-01	0.13395025	-1.1193660098	-8.564402e-01	-5.940275e-01
738	737	2.361070e-01	0.13164031	-0.0220789000	2.361239e-01	4.941963e-01
739	738	4.949652e-02	0.13029349	-0.2058628185	4.944695e-02	3.051379e-01
740	739	-2.355171e-01	0.10327847	-0.4380255320	-2.355236e-01	-3.297175e-02
741	740	1.479690e-01	0.11971443	-0.0865858140	1.478946e-01	3.829470e-01
742	741	-2.311073e-01	0.14318125	-0.5119127020	-2.310983e-01	4.964668e-02
743	742	-2.053343e-01	0.12355626	-0.4479379174	-2.052251e-01	3.664895e-02
744	743	-2.994624e-01	0.09517279	-0.4862673992	-2.994025e-01	-1.129983e-01
745	744	2.876027e-02	0.09694957	-0.1613998743	2.877521e-02	2.188354e-01
746	745	-4.796724e-01	0.09832582	-0.6724126058	-4.796989e-01	-2.867818e-01
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751	750	1.074523e-01	0.14732092	-0.1815008546	1.074747e-01	3.962779e-01
752	751	-4.254143e-01	0.10841962	-0.6380882732	-4.253894e-01	-2.128816e-01
753	752	5.156314e-01	0.11131092	0.2973951200	5.156150e-01	7.339607e-01
754	753	2.207038e-01	0.11547426	-0.0056048556	2.206541e-01	4.472959e-01
755	754	-1.936942e-01	0.11081641	-0.4109100593	-1.937262e-01	2.370344e-02
756	755	-3.818365e-01	0.12140547	-0.6203272074	-3.816935e-01	-1.441583e-01
757	756	-3.226052e-01	0.08650373	-0.4923036764	-3.225823e-01	-1.530367e-01
758	757	2.706532e-01	0.11759164	0.0399822527	2.706815e-01	5.011627e-01
759	758	-2.781697e-01	0.09573567	-0.4659182954	-2.781639e-01	-9.045431e-02
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762	761	-3.989771e-01	0.09787469	-0.5910485396	-3.989256e-01	-2.071987e-01
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765	764	1.892011e-01	0.09508160	0.0026636176	1.892298e-01	3.755747e-01
766	765	-1.630516e-01	0.10337992	-0.3658475562	-1.630262e-01	3.959950e-02
767	766	4.893830e-02	0.09987904	-0.1469555339	4.894959e-02	2.447679e-01
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771	770	1.539761e-01	0.10613022	-0.0541145402	1.539674e-01	3.621162e-01
772	771	1.701577e-01	0.12038469	-0.0658092730	1.701209e-01	4.063336e-01
773	772	-3.526813e-01	0.10009217	-0.5490803294	-3.526398e-01	-1.565187e-01

774	773	1.077343e-02	0.09143049	-0.1685874315	1.079598e-02	1.900059e-01
775	774	6.315653e-01	0.14436664	0.3476595899	6.318387e-01	9.139223e-01
776	775	-1.066665e+00	0.12836805	-1.3182649792	-1.066710e+00	-8.148097e-01
777	776	9.283041e-01	0.15089186	0.6325427557	9.282493e-01	1.224378e+00
778	777	-6.089588e-01	0.09510166	-0.7955931185	-6.089098e-01	-4.226038e-01
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780	779	3.993233e-01	0.10573119	0.1919615704	3.993333e-01	6.066281e-01
781	780	3.190121e-01	0.20757127	-0.0884452125	3.191559e-01	7.256533e-01
782	781	-2.635226e-02	0.11786899	-0.2578868042	-2.621518e-02	2.044041e-01
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784	783	2.152001e-01	0.12828822	-0.0364320299	2.152230e-01	4.667024e-01
785	784	2.336859e-01	0.10846395	0.0210600475	2.336569e-01	4.464767e-01
786	785	-4.413970e-01	0.11632992	-0.6694778145	-4.414181e-01	-2.131957e-01
787	786	-7.552842e-01	0.10159757	-0.9545285441	-7.552781e-01	-5.560743e-01
788	787	4.322102e-01	0.13665897	0.1642501263	4.322010e-01	7.002226e-01
789	788	-4.398101e-02	0.09809798	-0.2363572081	-4.397829e-02	1.483797e-01
790	789	-4.016307e-01	0.08641980	-0.5712215270	-4.015870e-01	-2.322886e-01
791	790	-2.889300e-01	0.11220002	-0.5092471371	-2.888289e-01	-6.918805e-02
792	791	-2.020295e-02	0.09498906	-0.2064835508	-2.019833e-02	1.660513e-01
793	792	2.652311e-01	0.15392927	-0.0372352284	2.654308e-01	5.665709e-01
794	793	-3.811117e-03	0.14591839	-0.2901200039	-3.752454e-03	2.821641e-01
795	794	-8.501092e-02	0.11183983	-0.3043126615	-8.501469e-02	1.343123e-01
796	795	-2.743816e-01	0.09796277	-0.4667149439	-2.743010e-01	-8.250700e-02
797	796	-5.715667e-01	0.11333397	-0.7939702027	-5.715127e-01	-3.494702e-01
798	797	-7.373954e-02	0.09469609	-0.2594374477	-7.373826e-02	1.119511e-01
799	798	-1.661436e-01	0.11471505	-0.3910983660	-1.661423e-01	5.880366e-02
800	799	-2.163481e-01	0.09908840	-0.4108216383	-2.162908e-01	-2.220114e-02
801	800	8.340413e-03	0.09673181	-0.1812506404	8.305981e-03	1.981274e-01
802	801	-1.372422e+00	0.13482311	-1.6369805952	-1.372358e+00	-1.108224e+00
803	802	9.032335e-02	0.11583882	-0.1370334789	9.039358e-02	3.172806e-01
804	803	5.190326e-02	0.09831490	-0.1412377353	5.202640e-02	2.443449e-01
805	804	-1.377020e-01	0.16747846	-0.4640540603	-1.385375e-01	1.933175e-01
806	805	-3.348430e-01	0.09121511	-0.5137489161	-3.348328e-01	-1.559953e-01
807	806	-5.764665e-02	0.11947993	-0.2921416444	-5.757648e-02	1.764494e-01
808	807	3.214328e-01	0.10961061	0.1064569007	3.214438e-01	5.363462e-01
809	808	-6.524081e-01	0.09972314	-0.8480186847	-6.523880e-01	-4.569120e-01
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812	811	2.409017e-02	0.11616035	-0.2037022512	2.409322e-02	2.518652e-01
813	812	-8.023067e-01	0.09607964	-0.9908151496	-8.022718e-01	-6.139975e-01
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815	814	-7.707018e-01	0.11039791	-0.9872093234	-7.706935e-01	-5.542421e-01
816	815	3.243802e-01	0.11048992	0.1076611933	3.243996e-01	5.409884e-01
817	816	-6.017731e-01	0.08697311	-0.7723919968	-6.017480e-01	-4.312968e-01
818	817	-5.715023e-01	0.10578079	-0.7789831189	-5.714849e-01	-3.641206e-01
819	818	-9.782809e-02	0.13516427	-0.3628616481	-9.783822e-02	1.672633e-01
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824	823	-4.025165e-01	0.09260102	-0.5840784298	-4.025250e-01	-2.209067e-01
825	824	4.819380e-01	0.14378071	0.1999873130	4.819394e-01	7.638811e-01
826	825	-1.017891e+00	0.09437703	-1.2030656442	-1.017853e+00	-8.329324e-01
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828	827	-3.130117e-01	0.10209810	-0.5132337465	-3.130067e-01	-1.128182e-01
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845	844	2.171933e-01	0.12422139	-0.0263716367	2.171826e-01	4.608192e-01
846	845	-4.086948e-01	0.10961532	-0.6237108075	-4.086724e-01	-1.938062e-01
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848	847	7.826918e-01	0.12731292	0.5330720534	7.826783e-01	1.032388e+00
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851	850	1.071862e-01	0.11435908	-0.1170302705	1.071737e-01	3.314737e-01
852	851	-2.511764e-01	0.11184516	-0.4706066733	-2.511383e-01	-3.196291e-02
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855	854	5.452320e-02	0.13472342	-0.2098933785	5.460079e-02	3.184992e-01
856	855	-9.189651e-01	0.11039975	-1.1354795626	-9.189630e-01	-7.024620e-01
857	856	5.956911e-01	0.11176388	0.3761673077	5.958171e-01	8.144984e-01
858	857	4.415089e-01	0.11041127	0.2248413806	4.415639e-01	6.578622e-01
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7359	7358	-5.215981e-03	0.42915624	-0.8471328689	-5.205067e-03	8.366386e-01
7360	7359	9.188844e-02	0.41672712	-0.7246799194	9.156006e-02	9.103263e-01
7361	7360	-2.458395e-05	0.41759051	-0.8192173851	-2.443887e-05	8.191674e-01
7362	7361	1.362846e-04	0.41534326	-0.8146374919	1.348311e-04	8.149184e-01
7363	7362	3.520672e-02	0.42676773	-0.8015301488	3.503863e-02	8.729011e-01
7364	7363	-2.285635e-01	0.39816012	-1.0116196028	-2.278483e-01	5.504246e-01
7365	7364	9.799330e-02	0.42339140	-0.7317942765	9.773500e-02	9.292507e-01
7366	7365	3.475437e-03	0.42498191	-0.8301893445	3.458615e-03	8.372360e-01
7367	7366	1.920975e-02	0.42168501	-0.8077746892	1.912497e-02	8.466770e-01
7368	7367	7.142135e-02	0.42475848	-0.7609128671	7.110023e-02	9.055845e-01
7369	7368	3.697135e-04	0.42859412	-0.8404196175	3.662601e-04	8.411787e-01
7370	7369	7.529970e-05	0.42979342	-0.8430669388	7.443486e-05	8.432225e-01
7371	7370	1.451239e-04	0.42870841	-0.8409084884	1.435670e-04	8.412076e-01
7372	7371	5.133459e-06	0.41672141	-0.8174601327	4.936641e-06	8.174715e-01
7373	7372	5.356478e-06	0.42606653	-0.8358761752	4.230205e-06	8.358933e-01
7374	7373	1.594312e-02	0.41360590	-0.7951723598	1.585186e-02	8.275786e-01
7375	7374	5.870819e-06	0.42770430	-0.8390597974	5.641103e-06	8.390729e-01
7376	7375	1.373067e-01	0.43021490	-0.7051988671	1.367772e-01	9.828260e-01
7377	7376	1.635991e-02	0.42462952	-0.8166316999	1.637081e-02	8.492888e-01
7378	7377	3.443993e-03	0.42697241	-0.8341160298	3.418790e-03	8.411477e-01
7379	7378	6.845662e-02	0.42748761	-0.7693693682	6.817148e-02	9.079062e-01
7380	7379	1.661362e-02	0.42122674	-0.8095351598	1.655456e-02	8.430983e-01
7381	7380	4.231853e-03	0.43030487	-0.8398933262	4.194902e-03	8.485677e-01
7382	7381	1.451109e-03	0.43112577	-0.8443691786	1.435777e-03	8.473588e-01
7383	7382	5.024368e-03	0.42825663	-0.8350462805	4.982684e-03	8.453327e-01
7384	7383	1.301460e-02	0.42831252	-0.8270982727	1.293265e-02	8.535943e-01
7385	7384	6.932299e-02	0.43265936	-0.7784837849	6.896227e-02	9.191841e-01
7386	7385	-1.549386e-02	0.34342325	-0.6890458905	-1.548468e-02	6.580054e-01
7387	7386	2.471876e-02	0.41608666	-0.7911579700	2.458441e-02	8.413609e-01
7388	7387	-1.370730e-06	0.42636675	-0.8364208247	-1.495929e-06	8.364188e-01
7389	7388	1.227291e-03	0.43141747	-0.8451739079	1.215971e-03	8.476930e-01
7390	7389	8.327667e-04	0.43201569	-0.8467176713	8.245707e-04	8.484299e-01
7391	7390	3.329012e-05	0.42791568	-0.8394987439	3.270650e-05	8.395687e-01
7392	7391	3.114007e-03	0.41427760	-0.8095132966	3.097450e-03	8.158355e-01
7393	7392	4.730817e-04	0.43414028	-0.8512830463	4.686517e-04	8.522545e-01
7394	7393	1.309569e-02	0.42875496	-0.8278899760	1.303976e-02	8.543998e-01
7395	7394	-2.346689e-05	0.43374819	-0.8509934438	-2.342270e-05	8.509463e-01
7396	7395	1.416067e-01	0.42157812	-0.6843411110	1.412453e-01	9.696111e-01
7397	7396	1.182250e-01	0.42605073	-0.7163590164	1.178074e-01	9.551859e-01

7398	7397	3.519639e-03	0.42371810	-0.8276705013	3.502630e-03	8.348066e-01
7399	7398	1.433574e-03	0.42117437	-0.8247502565	1.421550e-03	8.276860e-01
7400	7399	2.744933e-01	0.40175644	-0.5120253466	2.739560e-01	1.064067e+00
7401	7400	4.175417e-03	0.41986161	-0.8193773422	4.143479e-03	8.279103e-01
7402	7401	4.667220e-03	0.42872983	-0.8363237649	4.628991e-03	8.458761e-01
7403	7402	6.721274e-02	0.41541258	-0.7470239271	6.697872e-02	8.827817e-01
7404	7403	4.137504e-04	0.42879946	-0.8408130635	4.108823e-04	8.416569e-01
7405	7404	1.903153e-03	0.42927766	-0.8402072863	1.887523e-03	8.441027e-01
7406	7405	1.133535e-02	0.42359996	-0.8194486438	1.125656e-02	8.425683e-01
7407	7406	4.839635e-04	0.43223982	-0.8475834968	4.906684e-04	8.485130e-01
7408	7407	4.533880e-04	0.42857249	-0.8403110251	4.489771e-04	8.412430e-01
7409	7408	1.783561e-03	0.42381415	-0.8296361230	1.768452e-03	8.332894e-01
7410	7409	-1.835524e-02	0.42703623	-0.8564059658	-1.825402e-02	8.191187e-01
7411	7410	-4.377556e-02	0.42506664	-0.8781598154	-4.358435e-02	7.895195e-01
7412	7411	1.384699e-01	0.42390051	-0.6919724141	1.380707e-01	9.711837e-01
7413	7412	-1.562387e-06	0.42861107	-0.8408691907	-1.734150e-06	8.408670e-01
7414	7413	4.189901e-03	0.43071990	-0.8407417842	4.166259e-03	8.492563e-01
7415	7414	-4.226381e-06	0.42237593	-0.8285964917	-4.346231e-06	8.285887e-01
7416	7415	-1.607021e-05	0.41968416	-0.8233292363	-1.603895e-05	8.232969e-01
7417	7416	4.125616e-01	0.42779381	-0.4235379906	4.114564e-01	1.254939e+00
7418	7417	-2.063062e-05	0.42411554	-0.8320363695	-2.053397e-05	8.319946e-01
7419	7418	9.469250e-03	0.42990134	-0.8338416092	9.433330e-03	8.529845e-01
7420	7419	-2.206045e-03	0.42883340	-0.8435570925	-2.191763e-03	8.390636e-01
7421	7420	1.521026e-01	0.42602016	-0.6822703239	1.516252e-01	9.891923e-01
7422	7421	4.421698e-02	0.43149760	-0.8016721005	4.398101e-02	8.914500e-01
7423	7422	-1.720171e-04	0.43252455	-0.8487492478	-1.708581e-04	8.483986e-01
7424	7423	7.279849e-03	0.43020720	-0.8366306207	7.240543e-03	8.514142e-01
7425	7424	3.751114e-01	0.41635210	-0.4391498747	3.742502e-01	1.194266e+00
7426	7425	3.880562e-02	0.42206959	-0.7888505528	3.868518e-02	8.671474e-01
7427	7426	8.467880e-06	0.43124858	-0.8460966907	8.173179e-06	8.461153e-01
7428	7427	6.587824e-02	0.41493159	-0.7478028659	6.579170e-02	8.800514e-01
7429	7428	2.877820e-01	0.42557447	-0.5448248769	2.869969e-01	1.124853e+00
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7437	7436	1.436289e-02	0.42764885	-0.8244262983	1.430204e-02	8.534985e-01
7438	7437	-2.946509e-01	0.32424877	-0.9310647551	-2.944766e-01	3.407711e-01
7439	7438	-6.016008e-03	0.42425707	-0.8383967736	-5.984149e-03	8.261833e-01
7440	7439	8.436099e-02	0.43076750	-0.7597108633	8.399717e-02	9.305037e-01
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7448	7447	7.223905e-02	0.43231260	-0.7749564991	7.190786e-02	9.213206e-01
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7453	7452	6.888436e-05	0.42800806	-0.8395916805	6.797254e-05	8.397347e-01
7454	7453	-1.247113e-05	0.43430669	-0.8520839060	-1.248314e-05	8.520590e-01
7455	7454	5.602685e-01	0.38626332	-0.1956208510	5.596585e-01	1.319625e+00
7456	7455	6.209194e-03	0.42320904	-0.8238967104	6.169073e-03	8.365437e-01
7457	7456	3.019159e-02	0.43107041	-0.8150024202	3.001335e-02	8.764010e-01
7458	7457	9.102367e-03	0.43221691	-0.8386666621	9.034470e-03	8.572584e-01
7459	7458	1.842802e-01	0.42753012	-0.6528104284	1.837246e-01	1.024532e+00
7460	7459	5.523437e-05	0.42658491	-0.8368577750	5.439254e-05	8.369730e-01
7461	7460	1.405986e-01	0.41930767	-0.6812195451	1.403524e-01	9.638164e-01
7462	7461	6.221661e-03	0.42074940	-0.8190772026	6.183599e-03	8.317374e-01
7463	7462	1.837449e-01	0.41952202	-0.6382569665	1.834167e-01	1.007613e+00
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7465	7464	-7.422160e-02	0.42905105	-0.9165082730	-7.401522e-02	7.668899e-01
7466	7465	-2.956257e-04	0.43039457	-0.8446488769	-2.927244e-04	8.440411e-01
7467	7466	1.723929e-01	0.41061379	-0.6325188938	1.722062e-01	9.783670e-01
7468	7467	3.743168e-04	0.42336320	-0.8301567838	3.700629e-04	8.309297e-01
7469	7468	4.090112e-02	0.43444754	-0.8107417003	4.066051e-02	8.939146e-01
7470	7469	5.203997e-04	0.43435291	-0.8516145267	5.158040e-04	8.526815e-01
7471	7470	5.841635e-02	0.43300262	-0.7902341232	5.811920e-02	9.087592e-01
7472	7471	7.675066e-02	0.43312632	-0.7718902846	7.635078e-02	9.276686e-01
7473	7472	6.259610e-02	0.43655931	-0.7930868729	6.232243e-02	9.198372e-01
7474	7473	1.721919e-01	0.40970989	-0.6301366789	1.717143e-01	9.772383e-01
7475	7474	1.879436e-03	0.43441643	-0.8503888556	1.861238e-03	8.542515e-01
7476	7475	5.311244e-02	0.43345017	-0.7965800619	5.285804e-02	9.042535e-01
7477	7476	-9.163334e-04	0.43555801	-0.8554469399	-9.094517e-04	8.535750e-01
7478	7477	3.834420e-02	0.42690830	-0.7986315667	3.816679e-02	8.763305e-01
7479	7478	1.139200e-03	0.42733499	-0.8371744858	1.126984e-03	8.395226e-01
7480	7479	1.401458e-01	0.43041106	-0.7027922727	1.396269e-01	9.860368e-01
7481	7480	3.504690e-01	0.41269048	-0.4576662183	3.499915e-01	1.161319e+00
7482	7481	-1.826233e-02	0.43513530	-0.8721492666	-1.820003e-02	8.352699e-01
7483	7482	2.748100e-02	0.43530308	-0.8260817448	2.731569e-02	8.819854e-01
7484	7483	-1.998980e-05	0.43158078	-0.8467184676	-1.988250e-05	8.466779e-01
7485	7484	6.495320e-04	0.43314570	-0.8491178399	6.424315e-04	8.504574e-01
7486	7485	-2.506945e-05	0.43416094	-0.8517955461	-2.490092e-05	8.517444e-01
7487	7486	2.105410e-05	0.43359799	-0.8506535646	2.063699e-05	8.506981e-01
7488	7487	-1.773622e-05	0.42475960	-0.8332834480	-1.767680e-05	8.332476e-01
7489	7488	1.889586e-04	0.42455828	-0.8327155758	1.868262e-04	8.331057e-01
7490	7489	-2.019335e-05	0.42843700	-0.8405122038	-2.010815e-05	8.404713e-01
7491	7490	4.290450e-04	0.43078913	-0.8447069204	4.246284e-04	8.455902e-01
7492	7491	2.805374e-02	0.43256998	-0.8201325764	2.789040e-02	8.771706e-01
7493	7492	-7.597823e-04	0.42931512	-0.8430174889	-7.561321e-04	8.414771e-01

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7495	7494	1.058148e-03	0.43613914	-0.8545585202	1.049171e-03	8.567259e-01
7496	7495	-2.041478e-03	0.43172806	-0.8490699036	-2.025535e-03	8.448960e-01
7497	7496	8.477490e-02	0.42675501	-0.7516828347	8.452637e-02	9.226474e-01
7498	7497	1.190460e-01	0.42583217	-0.7151981239	1.186451e-01	9.555718e-01
7499	7498	7.400975e-02	0.42918183	-0.7671454326	7.373669e-02	9.167196e-01
7500	7499	8.575355e-02	0.42911457	-0.7554352290	8.553727e-02	9.281731e-01
7501	7500	3.674018e-02	0.32347108	-0.5973981003	3.665511e-02	6.713625e-01
7502	7501	1.193311e-03	0.41323840	-0.8094337702	1.182425e-03	8.118825e-01
7503	7502	8.578779e-04	0.42647993	-0.8357991320	8.497319e-04	8.375613e-01
7504	7503	-3.324641e-05	0.43252443	-0.8485723456	-3.308850e-05	8.485050e-01
7505	7504	9.328562e-02	0.42922906	-0.7475391092	9.285078e-02	9.365859e-01
7506	7505	1.136349e-01	0.43559262	-0.7393846932	1.130897e-01	9.697582e-01
7507	7506	7.988132e-05	0.42895160	-0.8414330593	7.873168e-05	8.415994e-01
7508	7507	-3.508603e-01	0.37086682	-1.0795808235	-3.503943e-01	3.752073e-01
7509	7508	7.219475e-03	0.42702523	-0.8303897905	7.170117e-03	8.451100e-01
7510	7509	4.928401e-03	0.41934174	-0.8175938511	4.891175e-03	8.276629e-01
7511	7510	7.737184e-02	0.43464732	-0.7742005790	7.697950e-02	9.311786e-01
7512	7511	-2.435891e-05	0.43652674	-0.8564280252	-2.420758e-05	8.563784e-01
7513	7512	8.094580e-04	0.43070938	-0.8441444255	8.009351e-04	8.458120e-01
7514	7513	3.287405e-02	0.43051518	-0.8113836826	3.276975e-02	8.777248e-01
7515	7514	5.388585e-02	0.43501575	-0.7989979606	5.369841e-02	9.078365e-01
7516	7515	-2.114164e-05	0.43649539	-0.8563935422	-2.106767e-05	8.563508e-01
7517	7516	1.921616e-02	0.42244033	-0.8093960567	1.919288e-02	8.479604e-01
7518	7517	2.557593e-01	0.40956982	-0.5468326913	2.554890e-01	1.059889e+00
7519	7518	1.363035e-01	0.41721661	-0.6812883728	1.360186e-01	9.555155e-01
7520	7519	2.102741e-02	0.43954973	-0.8408840112	2.087118e-02	8.838293e-01
7521	7520	-5.657105e-02	0.42944213	-0.8995413728	-5.640136e-02	7.854330e-01
7522	7521	6.970066e-03	0.42140318	-0.8195616272	6.914926e-03	8.338161e-01
7523	7522	-3.851729e-04	0.41134520	-0.8073611790	-3.552077e-04	8.064199e-01
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8722  7.673758e-03 2.531442e-09
8723  2.266876e-02 2.592317e-09
8724  1.138834e-03 2.852240e-09
8725  4.079977e-02 2.305780e-09
8726  5.566843e-03 3.025321e-09
8727  8.877970e-04 3.095577e-09
8728  1.183159e-03 2.274587e-09
8729  1.283365e-05 2.355875e-09
8730  3.669823e-04 3.405921e-09
8731  3.318722e-03 3.380239e-09
8732 -1.912997e-05 2.676001e-10
8733 -2.261422e-05 9.133284e-10
8734  8.781802e-03 8.283754e-10
8735  9.755812e-06 8.719596e-10
8736  2.078688e-02 4.334968e-10
8737  8.823281e-03 2.671692e-09
8738  1.168554e-01 6.185962e-10
8739 -1.989553e-05 4.553782e-10
8740 -1.478484e-03 4.311112e-10
8741  3.843129e-04 2.011074e-09
8742  1.726605e-03 4.754840e-10
8743 -2.082560e-05 1.042322e-09
8744  5.010849e-04 4.061254e-10
8745  3.151365e-04 1.198786e-09
```

```
8746  1.722098e-05 1.453125e-09
8747  4.124299e-03 5.555190e-10
8748  3.913141e-02 8.107147e-10
8749  4.465613e-03 5.056539e-10
8750  1.781041e-03 6.114613e-10
8751  1.088974e-02 5.654742e-10
8752 -2.146050e-05 2.515519e-10
8753 -2.246477e-05 1.447079e-09
8754  2.503699e-04 6.373824e-10
8755 -2.398834e-04 6.377368e-10
8756  3.706145e-03 6.392398e-10
8757  2.522137e-02 6.737605e-10
8758 -2.268967e-05 6.379801e-10
8759 -2.282529e-05 6.382498e-10
8760  1.723307e-02 6.723730e-10
8761  3.960487e-02 7.184514e-10
8762  8.894487e-04 6.662049e-10
8763 -2.077309e-05 6.645701e-10
8764  1.788775e-02 6.706718e-10
8765  2.826310e-03 6.648511e-10
8766  5.892766e-03 6.759735e-10
8767  7.080671e-04 6.708297e-10
8768  4.946802e-03 6.702556e-10
8769 -2.276452e-05 6.637769e-10
8770 -2.703680e-05 6.646131e-10
8771  1.288164e-01 1.040614e-09
8772  6.122986e-02 7.641344e-10
8773  5.853419e-02 6.611912e-10
8774 -2.255068e-05 6.355303e-10
8775 -2.264334e-05 6.374904e-10
8776 -2.276666e-05 6.327421e-10
8777 -9.877955e-06 6.278745e-10
8778 -2.293245e-05 6.335203e-10
8779  1.156595e-02 6.497243e-10
```

```
# 4. Hyperparameter estimates
hyperparameter_output <- inla_spatial$summary.hyperpar

print("MODEL HYPERPARAMETERS")
```

```
[1] "MODEL HYPERPARAMETERS"
```

```
print(hyperparameter_output)
```

```
                                             mean           sd   0.025quant
Precision for the Gaussian observations 2.6561274 0.007689995    2.6409486
```

```
Precision for city_id                      24.1902947  9.986313201    7.8646341
Range for spatial_field                  1561.3708820 67.311226620 1434.9550958
Stdev for spatial_field                     0.3985649  0.012168285    0.3755018
                                             0.5quant  0.975quant         mode
Precision for the Gaussian observations     2.6561409    2.671227    2.6562210
Precision for city_id                      23.1002097   45.531210   20.1279007
Range for spatial_field                  1559.2758387 1699.941882 1553.6770229
Stdev for spatial_field                     0.3982585    0.423409    0.3973791
```

```
# 5. Model fit statistics
model_fit_output <- data.frame(
  DIC = inla_spatial$dic$dic,
  WAIC = inla_spatial$waic$waic,
  log_marginal_likelihood = inla_spatial$mlik[1]
)

print("MODEL FIT STATISTICS")
```

```
[1] "MODEL FIT STATISTICS"
```

```
print(model_fit_output)
```

```
       DIC     WAIC log_marginal_likelihood
1 460302.9 457201.8                -232584
```

```
# 6. Save all model output tables
write.csv(
  fixed_effect_output,
  "fixed_effect_coefficients.csv"
)

write.csv(
  municipality_output,
  "municipality_random_effects.csv"
)

write.csv(
  spatial_output,
  "spatial_random_effects.csv"
)

write.csv(
  hyperparameter_output,
  "model_hyperparameters.csv"
)
```

```
write.csv(
  model_fit_output,
  "model_fit_statistics.csv",
  row.names = FALSE
)
```

16 14. Primary model fixed effects

```
# 14. FIXED EFFECTS
spatial_fixed_effects <- inla_spatial$summary.fixed
spatial_fixed_effects
```

```
                                 mean           sd    0.025quant
intercept                   9.374969543 1.414204e+01 -18.356166503
sale_year_c                -0.003650175 2.556873e-04  -0.004151553
frontage_groupNone          2.184066590 1.414196e+01 -25.546915716
frontage_groupLake          2.233327953 1.414196e+01 -25.497657310
frontage_groupBay           2.515847486 1.414196e+01 -25.215142324
frontage_groupOther         2.442238220 1.414196e+01 -25.288745843
seawall_groupYes            0.160090511 1.235759e-02   0.135858152
cccl_distance_b             0.002087764 2.907453e-02  -0.055380018
tb_distance_b               0.070703640 3.128918e-02   0.009332911
quality_groupFair          -0.185006392 5.810751e-03  -0.196400452
quality_groupAbove Average  0.158313761 6.493042e-03   0.145580929
heated_area_b               0.203362696 2.488869e-03   0.198481825
acreage_b                   0.028038004 1.692797e-03   0.024718822
barrier_municipalityBarrier 0.338918466 1.197760e-01   0.106106348
stories_group2             -0.022264858 5.539108e-03  -0.033126349
stories_group3+            -0.057001985 1.437615e-02  -0.085192086
foundation_groupYes         0.052765655 6.853659e-03   0.039326223
                               0.5quant   0.975quant         mode          kld
intercept                   9.374969554 37.106105525  9.374969554 5.526813e-11
sale_year_c                -0.003650175 -0.003148797 -0.003650175 5.455871e-11
frontage_groupNone          2.184066604 29.915048818  2.184066604 5.526809e-11
frontage_groupLake          2.233327967 29.964313131  2.233327967 5.526809e-11
frontage_groupBay           2.515847499 30.246837219  2.515847499 5.526809e-11
frontage_groupOther         2.442238232 30.173222212  2.442238232 5.526804e-11
seawall_groupYes            0.160090623  0.184322230  0.160090624 5.437228e-11
cccl_distance_b             0.002244596  0.058669566  0.002246854 7.246814e-09
tb_distance_b               0.070694664  0.132121899  0.070693663 6.140282e-10
quality_groupFair          -0.185006482 -0.173611822 -0.185006482 5.481847e-11
quality_groupAbove Average  0.158313975  0.171045377  0.158313976 5.484799e-11
heated_area_b               0.203362863  0.208242623  0.203362865 6.048902e-11
acreage_b                   0.028037920  0.031357662  0.028037920 5.306403e-11
```

```
barrier_municipalityBarrier  0.337574020  0.579073847  0.337530798 6.666129e-08
stories_group2              -0.022264914 -0.011403045 -0.022264915 5.461657e-11
stories_group3+             -0.057002030 -0.028811629 -0.057002031 5.454029e-11
foundation_groupYes          0.052765681  0.066204939  0.052765681 5.455288e-11
```

```
# FIXED EFFECTS: CLEAN TABLE

spatial_fixed_effects_clean <-
  spatial_fixed_effects %>%

  rownames_to_column(
    var = "Term"
  ) %>%

  select(
    Term,
    mean,
    sd,
    `0.025quant`,
    `0.5quant`,
    `0.975quant`
  )

spatial_fixed_effects_clean
```

```
                         Term         mean           sd    0.025quant
1                   intercept  9.374969543 1.414204e+01 -18.356166503
2                 sale_year_c -0.003650175 2.556873e-04  -0.004151553
3           frontage_groupNone  2.184066590 1.414196e+01 -25.546915716
4           frontage_groupLake  2.233327953 1.414196e+01 -25.497657310
5            frontage_groupBay  2.515847486 1.414196e+01 -25.215142324
6          frontage_groupOther  2.442238220 1.414196e+01 -25.288745843
7             seawall_groupYes  0.160090511 1.235759e-02   0.135858152
8              cccl_distance_b  0.002087764 2.907453e-02  -0.055380018
9                tb_distance_b  0.070703640 3.128918e-02   0.009332911
10           quality_groupFair -0.185006392 5.810751e-03  -0.196400452
11  quality_groupAbove Average  0.158313761 6.493042e-03   0.145580929
12               heated_area_b  0.203362696 2.488869e-03   0.198481825
13                   acreage_b  0.028038004 1.692797e-03   0.024718822
14 barrier_municipalityBarrier  0.338918466 1.197760e-01   0.106106348
15              stories_group2 -0.022264858 5.539108e-03  -0.033126349
16             stories_group3+ -0.057001985 1.437615e-02  -0.085192086
17         foundation_groupYes  0.052765655 6.853659e-03   0.039326223
       0.5quant   0.975quant
1   9.374969554 37.106105525
2  -0.003650175 -0.003148797
3   2.184066604 29.915048818
```

```
4   2.233327967 29.964313131
5   2.515847499 30.246837219
6   2.442238232 30.173222212
7   0.160090623  0.184322230
8   0.002244596  0.058669566
9   0.070694664  0.132121899
10 -0.185006482 -0.173611822
11  0.158313975  0.171045377
12  0.203362863  0.208242623
13  0.028037920  0.031357662
14  0.337574020  0.579073847
15 -0.022264914 -0.011403045
16 -0.057002030 -0.028811629
17  0.052765681  0.066204939
```

17 15. Municipality random effects

```
# 15. MUNICIPALITY RANDOM EFFECTS
municipality_effects <-inla_spatial$summary.random$city_id
municipality_effects
```

```
   ID        mean         sd   0.025quant    0.5quant   0.975quant        mode
1   1  0.32510755 0.08768483  0.155658621  0.32402765  0.500498032  0.32397999
2   2 -0.28482684 0.13906186 -0.556521752 -0.28584282 -0.007420051 -0.28583516
3   3  0.10061855 0.07759381 -0.051300707  0.10026364  0.254481957  0.10024760
4   4  0.56489961 0.14999296  0.284189813  0.56005813  0.870414069  0.55921740
5   5 -0.08705562 0.06331019 -0.214046206 -0.08646788  0.036736775 -0.08645911
6   6  0.06039011 0.10508078 -0.147960034  0.06099361  0.265364102  0.06100612
7   7 -0.06404468 0.07132613 -0.206905944 -0.06325632  0.074473512 -0.06323665
8   8  0.01403855 0.07692749 -0.140189983  0.01501684  0.162885974  0.01505037
9   9  0.13216717 0.12643546 -0.113599043  0.13093675  0.384715673  0.13089924
10 10 -0.05508030 0.14532122 -0.345904982 -0.05343596  0.226729391 -0.05337208
11 11 -0.20198837 0.07985751 -0.362906066 -0.20059571 -0.048675789 -0.20052863
12 12  0.02887120 0.06296981 -0.097265438  0.02937254  0.152277145  0.02937787
13 13 -0.10486461 0.13678237 -0.377708476 -0.10365242  0.161329983 -0.10360591
14 14  0.06657161 0.13500996 -0.201529492  0.06720401  0.331129083  0.06720541
15 15 -0.03048082 0.10031780 -0.229989854 -0.02966877  0.164512769 -0.02964496
16 16 -0.08312578 0.07534067 -0.235886562 -0.08151526  0.060908302 -0.08143385
17 17 -0.24232429 0.07599891 -0.395625608 -0.24096755 -0.096422088 -0.24090456
18 18  0.21346247 0.13448278 -0.046849787  0.21177678  0.482957876  0.21169265
19 19 -0.25217333 0.13965761 -0.535775604 -0.24885955  0.013655926 -0.24862396
20 20  0.13830308 0.09281140 -0.043786419  0.13813028  0.321341461  0.13812118
21 21 -0.14185923 0.06682610 -0.276162421 -0.14103205 -0.012070772 -0.14101041
22 22 -0.10029489 0.08687209 -0.274095983 -0.09923207  0.067642794 -0.09919202
23 23  0.20356938 0.12626375 -0.039647804  0.20154104  0.457830998  0.20144677
```

```
24 24 -0.26843189 0.06626691 -0.403301168 -0.26694668 -0.141512835 -0.26687288
25 25 -0.32798109 0.09695067 -0.523775244 -0.32591891 -0.143295574 -0.32576020
26 26  0.31581391 0.16328746  0.008957997  0.31091124  0.647885583  0.31013229
27 27  0.08119510 0.12747557 -0.167336016  0.08018576  0.335268281  0.08015134
            kld
1  4.129890e-08
2  6.047656e-08
3  3.603300e-08
4  2.273359e-07
5  7.953305e-08
6  1.996722e-08
7  5.812477e-08
8  5.494767e-08
9  4.845792e-08
10 5.155722e-08
11 7.036973e-08
12 8.080094e-08
13 4.469877e-08
14 4.763669e-08
15 2.473671e-08
16 9.804144e-08
17 7.420431e-08
18 5.488665e-08
19 1.134414e-07
20 1.648539e-08
21 7.318700e-08
22 4.259576e-08
23 7.146277e-08
24 1.113650e-07
25 9.354466e-08
26 1.864474e-07
27 4.709523e-08
```

18 16. Spatial random effects

```r
# 16. SPATIAL RANDOM FIELD
spatial_field_summary <- inla_spatial$summary.random$spatial_field
head(spatial_field_summary)
```

```
  ID          mean        sd 0.025quant      0.5quant 0.975quant          mode
1  0  2.096946e-02 0.3603352 -0.6858497  2.098531e-02  0.7276936  2.098407e-02
2  1  3.498048e-03 0.4012672 -0.7838152  3.456801e-03  0.7910466  3.456783e-03
3  2  1.106917e-03 0.4067624 -0.7969305  1.094722e-03  0.7992139  1.094710e-03
4  3 -2.227825e-05 0.4131093 -0.8104606 -2.215488e-05  0.8104153 -2.215454e-05
5  4  3.711483e-02 0.3967916 -0.7410017  3.697902e-02  0.8160041  3.697832e-02
```

```
6  5  5.247173e-02 0.4001765 -0.7321755  5.233394e-02  0.8379037  5.233341e-02
           kld
1 3.645780e-11
2 8.298945e-10
3 4.161954e-10
4 2.332455e-10
5 4.263288e-10
6 2.138909e-10
```

19 17. Spatial hyperparameters

```r
# 17. SPATIAL HYPERPARAMETERS

spatial_hyperparameters <- inla_spatial$summary.hyperpar
spatial_hyperparameters
```

```
                                                mean            sd   0.025quant
Precision for the Gaussian observations    2.6561274  0.007689995    2.6409486
Precision for city_id                     24.1902947  9.986313201    7.8646341
Range for spatial_field                 1561.3708820 67.311226620 1434.9550958
Stdev for spatial_field                    0.3985649  0.012168285    0.3755018
                                            0.5quant  0.975quant         mode
Precision for the Gaussian observations    2.6561409    2.671227    2.6562210
Precision for city_id                     23.1002097   45.531210   20.1279007
Range for spatial_field                 1559.2758387 1699.941882 1553.6770229
Stdev for spatial_field                    0.3982585    0.423409    0.3973791
```

20 18. Model fit statistics

```r
# 18. MODEL FIT STATISTICS
spatial_model_fit <- tibble(Model = "INLA spatial SPDE", N = nrow(bayes_data), DIC = inla_spati
spatial_model_fit
```

```
# A tibble: 1 x 4
  Model                  N     DIC    WAIC
  <chr>              <int>   <dbl>   <dbl>
1 INLA spatial SPDE 245477 460303. 457202.
```

21 19. Compare OLS, municipality, and spatial models

```
# 19. MODEL COMPARISON

model_comparison <- tibble(
  Model = c("OLS hedonic baseline", "INLA municipality", "INLA spatial SPDE"),
  N = c(nobs(ols_model), nrow(bayes_data), nrow(bayes_data)),
  DIC = c(NA_real_, inla_city$dic$dic, inla_spatial$dic$dic),
  WAIC = c(NA_real_, inla_city$waic$waic, inla_spatial$waic$waic),
  R_squared = c(summary(ols_model)$r.squared, NA_real_, NA_real_)
)

model_comparison
```

```
# A tibble: 3 x 5
  Model                     N     DIC    WAIC R_squared
  <chr>                 <int>   <dbl>   <dbl>     <dbl>
1 OLS hedonic baseline 245477     NA      NA      0.402
2 INLA municipality    245477 506325. 503207.    NA
3 INLA spatial SPDE    245477 460303. 457202.    NA
```

22 20. Extract fitted values

```
# 20. FITTED VALUES

# Extract fitted values from INLA
fitted_summary <- inla_spatial$summary.fitted.values

# Identify observed-data rows
estimation_index <- inla.stack.index(spatial_stack, tag = "est")$data

# Extract fitted values for observed observations only
observed_fitted <- fitted_summary$mean[estimation_index]

# Verify dimensions
if (length(observed_fitted) != nrow(bayes_data)) {
  stop(paste0(
    "ERROR: Fitted values do not match bayes_data. ",
    "Fitted values: ", length(observed_fitted),
    "; Observations: ", nrow(bayes_data)
  ))
}
```

```r
# Assign fitted values
bayes_data$inla_fitted <- observed_fitted

# Calculate residuals
bayes_data$inla_residual <- bayes_data$log_price - bayes_data$inla_fitted

# Summary statistics
fitted_value_summary <- tibble(
  Statistic = c("Total INLA fitted values", "Observed-data stack rows", "Bayesian dataset rows"
  Value = c(length(fitted_summary$mean), length(estimation_index), nrow(bayes_data))
)

print(fitted_value_summary)
```

```
# A tibble: 3 x 2
  Statistic                 Value
  <chr>                     <int>
1 Total INLA fitted values 480659
2 Observed-data stack rows 245477
3 Bayesian dataset rows    245477
```

```r
model_value_summary <- tibble(
  Variable = c("Fitted values", "Residuals"),
  Minimum = c(min(bayes_data$inla_fitted), min(bayes_data$inla_residual)),
  Mean = c(mean(bayes_data$inla_fitted), mean(bayes_data$inla_residual)),
  Median = c(median(bayes_data$inla_fitted), median(bayes_data$inla_residual)),
  Maximum = c(max(bayes_data$inla_fitted), max(bayes_data$inla_residual))
)

print(model_value_summary)
```

```
# A tibble: 2 x 5
  Variable      Minimum          Mean Median Maximum
  <chr>           <dbl>         <dbl>  <dbl>   <dbl>
1 Fitted values    8.96 11.3          11.2     17.7
2 Residuals       -7.73  0.0000000122  0.118    4.67
```

23 21. INLA prediction accuracy

```r
# 21. PREDICTION ACCURACY


inla_rmse <-
```

```
  sqrt(
    mean(
      bayes_data$inla_residual^2,
      na.rm = TRUE
    )
  )

inla_mae <-
  mean(
    abs(
      bayes_data$inla_residual
    ),
    na.rm = TRUE
  )

inla_r2 <-
  1 -

  sum(
    bayes_data$inla_residual^2,
    na.rm = TRUE
  ) /

  sum(
    (
      bayes_data$log_price -
        mean(
          bayes_data$log_price,
          na.rm = TRUE
        )
    )^2,

    na.rm = TRUE
  )

inla_accuracy <-
  tibble(

    Model =
      "INLA spatial SPDE",

    N =
      nrow(bayes_data),

    RMSE =
      inla_rmse,
```

```
    MAE =
      inla_mae,

    R_squared =
      inla_r2
  )

inla_accuracy
```

```
# A tibble: 1 x 5
  Model                   N  RMSE   MAE R_squared
  <chr>               <int> <dbl> <dbl>     <dbl>
1 INLA spatial SPDE 245477 0.609 0.384     0.534
```

24 22. Residual distribution

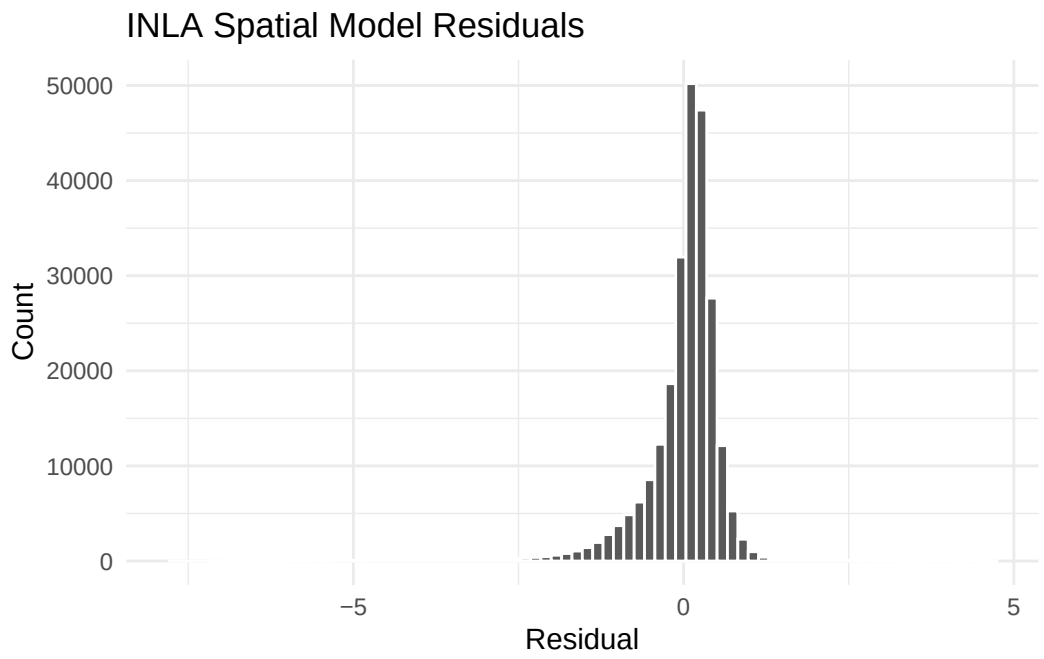
```
# 22. RESIDUAL DISTRIBUTION


ggplot(
  bayes_data,
  aes(
    x = inla_residual
  )
) +

  geom_histogram(
    bins = 80,
    color = "white"
  ) +

  labs(
    title =
      "INLA Spatial Model Residuals",
    x =
      "Residual",
    y =
      "Count"
  ) +

  theme_minimal()
```



25 23. Residuals versus fitted values

```
# 23. RESIDUALS VS FITTED


ggplot(
  bayes_data,
  aes(
    x = inla_fitted,
    y = inla_residual
  )
) +

  geom_bin2d(
    bins = 80
  ) +

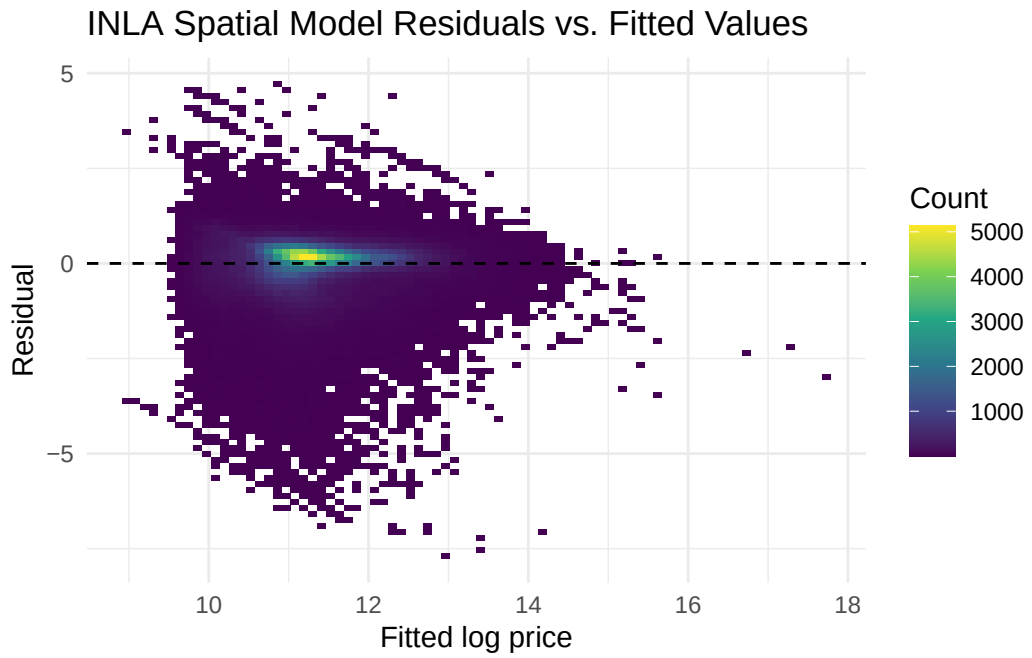
  scale_fill_viridis_c() +

  geom_hline(
    yintercept = 0,
    linetype = "dashed"
  ) +

  labs(
```

```
    title =
      "INLA Spatial Model Residuals vs. Fitted Values",
    x =
      "Fitted log price",
    y =
      "Residual",
    fill =
      "Count"
  ) +

  theme_minimal()
```



26 24. Spatial residual map

```
# 24. SPATIAL RESIDUAL MAP


set.seed(12345)

residual_map_n <-
  min(
    30000L,
    nrow(bayes_data)
  )
```

```
residual_map_id <-
  sample(
    seq_len(
      nrow(bayes_data)
    ),
    size =
      residual_map_n
  )

residual_map_data <-
  bayes_data[
    residual_map_id,
  ]

ggplot(
  residual_map_data,
  aes(
    x = x_coord,
    y = y_coord,
    color = inla_residual
  )
) +

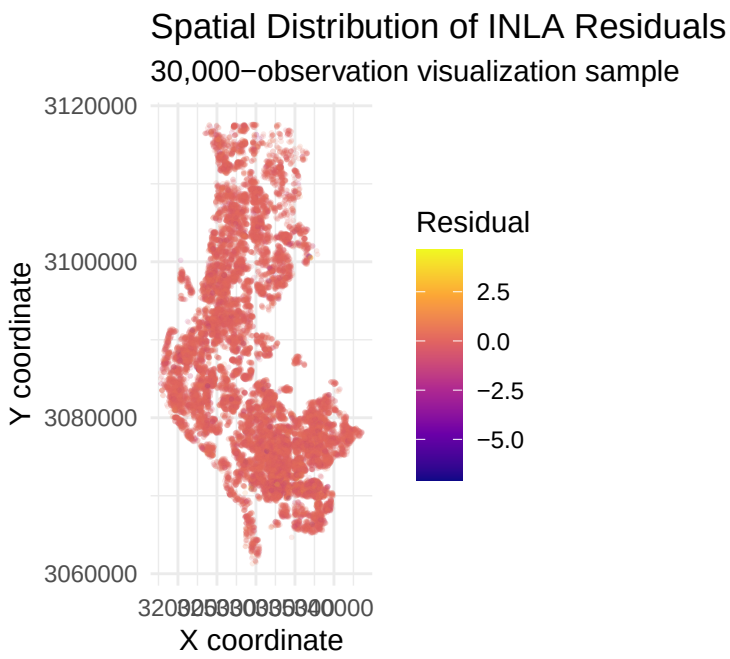
  geom_point(
    alpha = 0.15,
    size = 0.3
  ) +

  scale_color_viridis_c(
    option = "plasma"
  ) +

  coord_equal() +

  labs(
    title =
      "Spatial Distribution of INLA Residuals",
    subtitle =
      "30,000-observation visualization sample",
    x =
      "X coordinate",
    y =
      "Y coordinate",
    color =
      "Residual"
  ) +
```

```
  theme_minimal()
```



27 25. Spatial field map

```
# 25. SPATIAL RANDOM FIELD MAP



# The SPDE random effect is defined at mesh vertices.


mesh_field <-
  tibble(

    x =
      mesh$loc[, 1],

    y =
      mesh$loc[, 2],

    spatial_effect =
      spatial_field_summary$mean,

    spatial_lower =
      spatial_field_summary$`0.025quant`,
```

```
    spatial_upper =
      spatial_field_summary$`0.975quant`
  )

ggplot(
  mesh_field,
  aes(
    x = x,
    y = y,
    color = spatial_effect
  )
) +

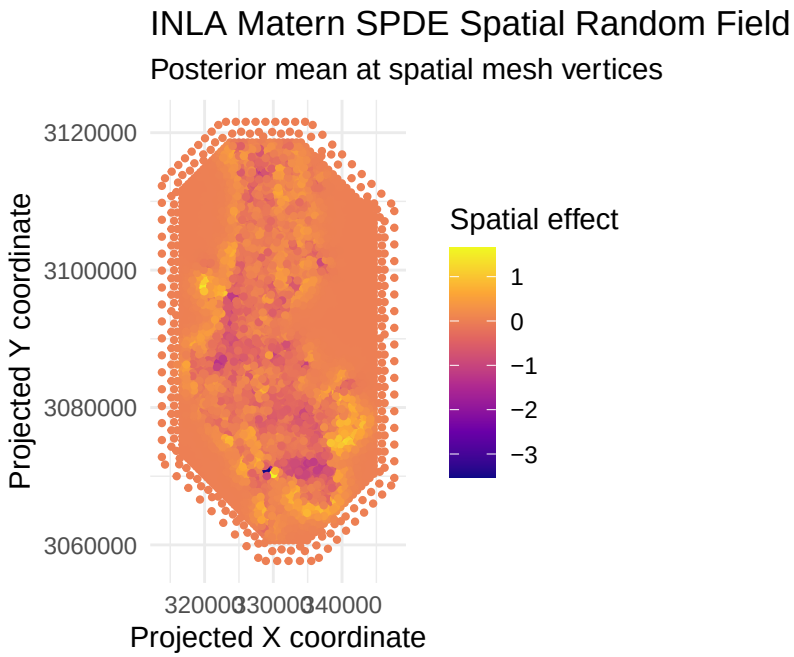
  geom_point(
    size = 0.8
  ) +

  scale_color_viridis_c(
    option = "plasma"
  ) +

  coord_equal() +

  labs(
    title =
      "INLA Matern SPDE Spatial Random Field",
    subtitle =
      "Posterior mean at spatial mesh vertices",
    x =
      "Projected X coordinate",
    y =
      "Projected Y coordinate",
    color =
      "Spatial effect"
  ) +

  theme_minimal()
```



28 26. Spatial uncertainty map

```
# 26. SPATIAL EFFECT UNCERTAINTY


mesh_field <-
  mesh_field %>%

  mutate(
    spatial_sd =
      spatial_upper -
      spatial_lower
  )

ggplot(
  mesh_field,
  aes(
    x = x,
    y = y,
    color = spatial_sd
  )
) +

  geom_point(
    size = 0.8
```

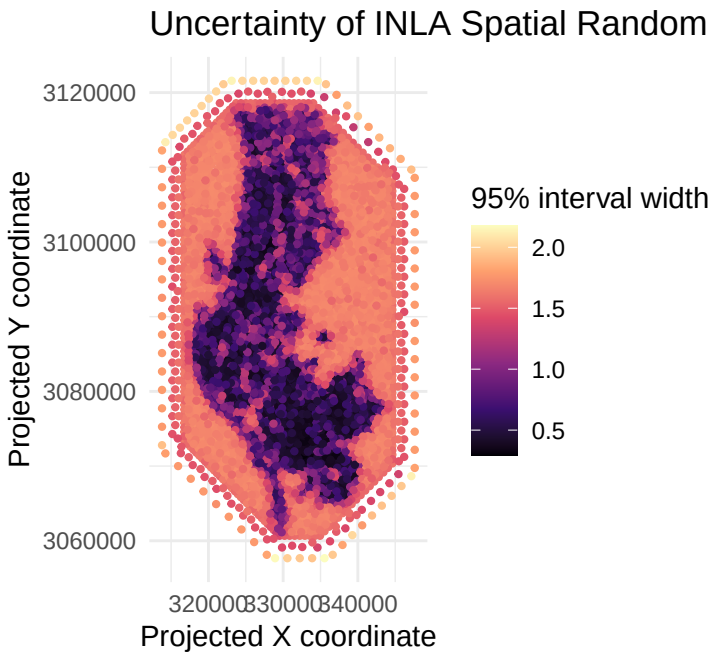
```
  ) +

  scale_color_viridis_c(
    option = "magma"
  ) +

  coord_equal() +

  labs(
    title =
      "Uncertainty of INLA Spatial Random Field",
    x =
      "Projected X coordinate",
    y =
      "Projected Y coordinate",
    color =
      "95% interval width"
  ) +

  theme_minimal()
```



29 27. Observed versus fitted values

```
# 27. OBSERVED VS FITTED
```

```r
set.seed(12345)

fit_plot_n <-
  min(
    30000L,
    nrow(bayes_data)
  )

fit_plot_id <-
  sample(
    seq_len(
      nrow(bayes_data)
    ),
    size =
      fit_plot_n
  )

fit_plot_data <-
  bayes_data[
    fit_plot_id,
  ]

ggplot(
  fit_plot_data,
  aes(
    x = inla_fitted,
    y = log_price
  )
) +

  geom_bin2d(
    bins = 80
  ) +

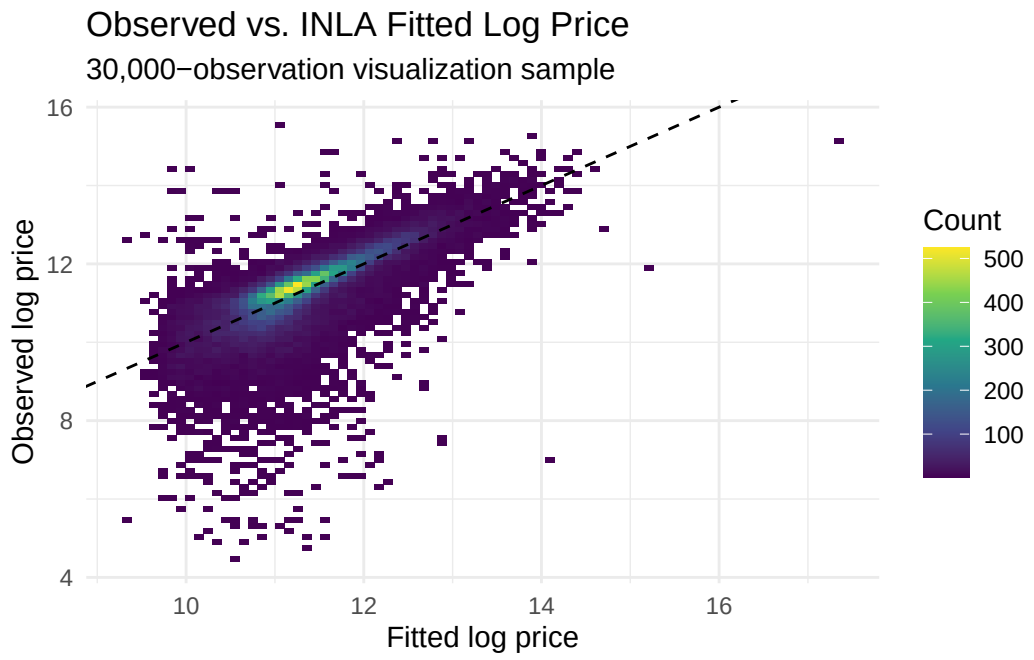
  scale_fill_viridis_c() +

  geom_abline(
    slope = 1,
    intercept = 0,
    linetype = "dashed"
  ) +

  labs(
    title =
      "Observed vs. INLA Fitted Log Price",
    subtitle =
      "30,000-observation visualization sample",
```

```
    x =
      "Fitted log price",
    y =
      "Observed log price",
    fill =
      "Count"
  ) +

  theme_minimal()
```



30 28. Municipality effects plot

```
# 28. MUNICIPALITY EFFECT PLOT


# Extract municipality random effects
city_effect_plot <-
  inla_spatial$summary.random$city_id %>%

  # INLA stores municipality IDs in the ID column
  mutate(
    city_id = as.integer(ID)
  ) %>%

  # Join municipality names
```

```
  left_join(
    bayes_data %>%
      select(
        city_id,
        site_city
      ) %>%
      distinct(),
    by = "city_id"
  ) %>%

  # Arrange by estimated municipality effect
  arrange(
    mean
  )

print(
  head(city_effect_plot)
)
```

```
  ID       mean         sd 0.025quant   0.5quant   0.975quant       mode
1 25 -0.3279811 0.09695067 -0.5237752 -0.3259189 -0.143295574 -0.3257602
2  2 -0.2848268 0.13906186 -0.5565218 -0.2858428 -0.007420051 -0.2858352
3 24 -0.2684319 0.06626691 -0.4033012 -0.2669467 -0.141512835 -0.2668729
4 19 -0.2521733 0.13965761 -0.5357756 -0.2488596  0.013655926 -0.2486240
5 17 -0.2423243 0.07599891 -0.3956256 -0.2409675 -0.096422088 -0.2409046
6 11 -0.2019884 0.07985751 -0.3629061 -0.2005957 -0.048675789 -0.2005286
           kld city_id        site_city
1 9.354466e-08      25   TARPON SPRINGS
2 6.047656e-08       2   BELLEAIR BEACH
3 1.113650e-07      24    ST PETERSBURG
4 1.134414e-07      19 REDINGTON SHORES
5 7.420431e-08      17    PINELLAS PARK
6 7.036973e-08      11     KENNETH CITY
```

```
# Check for unmatched municipality names
if (any(is.na(city_effect_plot$site_city))) {
  warning(
    "Some municipality IDs did not match municipality names."
  )
}

# Plot municipality random effects
ggplot(
  city_effect_plot,
  aes(
    x = reorder(site_city, mean),
    y = mean
```

```
  )
) +

  geom_point(
    size = 2,
    color = "steelblue"
  ) +

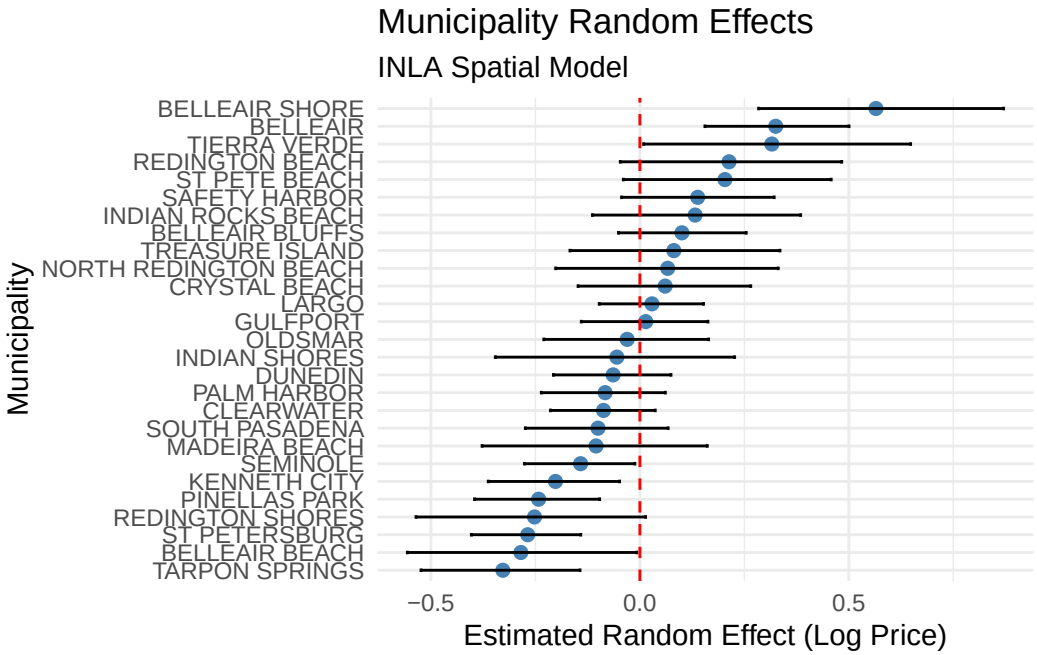
  geom_errorbar(
    aes(
      ymin = `0.025quant`,
      ymax = `0.975quant`
    ),
    width = 0.2
  ) +

  coord_flip() +

  geom_hline(
    yintercept = 0,
    linetype = "dashed",
    color = "red"
  ) +

  labs(
    title = "Municipality Random Effects",
    subtitle = "INLA Spatial Model",
    x = "Municipality",
    y = "Estimated Random Effect (Log Price)"
  ) +

  theme_minimal()
```



31 29. Save fitted models and objects

```
# 29. SAVE MODELS AND OBJECTS
#saveRDS(ols_model, "pinellas_ols_inla_v1.rds")
#saveRDS(inla_city, "pinellas_inla_city_v1.rds")
#saveRDS(inla_spatial, "pinellas_inla_spatial_spde_v1.rds")
#saveRDS(mesh,"pinellas_inla_mesh_v1.rds")
#saveRDS(spde,"pinellas_inla_spde_v1.rds")
#saveRDS(model_data, "pinellas_model_data_v1.rds")
#saveRDS(bayes_data, "pinellas_bayes_data_v1.rds")
#saveRDS(model_comparison, "pinellas_model_comparison_v1.rds")
```

32 30. Final model summary table

```
# 30. FINAL SUMMARY TABLE

final_summary <- tibble(Model = c("OLS Hedonic Baseline", "INLA Municipality", "INLA Spatial SP

    Observations = c(
      nobs(ols_model),
      nrow(bayes_data),
      nrow(bayes_data)
    ),
```

```
  R_squared = c(
    summary(
      ols_model
    )$r.squared,

    NA_real_,

    inla_r2
  ),

  RMSE = c(
    sqrt(
      mean(
        residuals(ols_model)^2
      )
    ),

    NA_real_,

    inla_rmse
  ),

  MAE = c(
    mean(
      abs(
        residuals(ols_model)
      )
    ),

    NA_real_,

    inla_mae
  ),

  DIC = c(
    NA_real_,
    inla_city$dic$dic,
    inla_spatial$dic$dic
  ),

  WAIC = c(
    NA_real_,
    inla_city$waic$waic,
    inla_spatial$waic$waic
  )
)
```

```
final_summary
```

```
# A tibble: 3 x 7
  Model                Observations R_squared   RMSE    MAE     DIC    WAIC
  <chr>                       <int>     <dbl>  <dbl>  <dbl>   <dbl>   <dbl>
1 OLS Hedonic Baseline       245477     0.402  0.690  0.461      NA      NA
2 INLA Municipality          245477    NA     NA     NA     506325. 503207.
3 INLA Spatial SPDE          245477     0.534  0.609  0.384 460303. 457202.
```